Rigid Analytic Geometry

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1 Introduction - Spectral Spaces and Spaces of Valuation Spectra

1.1 Introduction

Definition 1.1. *Let $\Omega \subseteq \mathbb{C}$ be a domain. We define a sheaf of holomorphic functions as,*

$$\Omega \supseteq V \mapsto \mathscr{O}(V) = \{f : V \to \mathbb{C} \mid f \textit{ is holomorphic}\}.$$

One can equip $\mathscr{O}(V)$ this with the semi-norm such that,

$$\|f\|_{V,A} = \max\{|f(z)| \mid z \in A\},$$

where $A \subseteq \Omega$ is a closed and bounded in $\mathbb{C}$, namely a compact set A. With respect to this semi-norm, we define basis elements of the topology in the following way.

Proposition 1.1. *The semi-norm induces the topology on $\mathscr{O}(V)$ whose basis elements of the neighbourhood of 0 are of the following form. For $f \in \mathscr{O}(V)$ and $\varepsilon > 0$,*

$$\{f \in \mathscr{O}(V) \mid \|f\|_{V,A} < \varepsilon\}.$$

Definition 1.2. *A ring is said to be Bézout if every finitely generated ideal is principal.*

Remark 1.1. *By Weierstrass product theorem, $\mathscr{O}(V)$ is not Noetherian but it is Bézout.*

Proposition 1.2. *The stalks of $\mathscr{O}(V)$ which is a ring of germs of holomorphic functions are Noetherian. This holds for holomorphic functions in several complex variables as well.*

The goal of the course is to generalize these results to p-adic cases.
We will allow ground field K with an absolute value $|\cdot| : K \to [0, \infty)$ satisfying,

$$|0| = 0, |1| = 1, |ab| = |a||b|, |a + b| \leq \max\{|a|, |b|\}.$$

Definition 1.3. *Let K be a field which is complete with respect to the map $|\cdot|_K : K \to [0, \infty)$ is an absolute value of K if we have the following conditions.*

i). $|1|_K = 1$,

ii). $|0|_K = 0$,

iii). $\forall a, b \in K, |ab|_K = |a|_K|b|_K$.

Furthermore, it is called ultra-metric if we have,

$$\forall a, b, |a + b|_K \leq \max\{|a|_K, |b|_K\}.$$

Proposition 1.3. *Suppose we have an ultra-metric* $\mathrm{abs} \cdot_K$ *on a field K. If K is complete with respect to the topology induced by the metric $d(x, y) = |x - y|_K$, then for a finite extension L/K, there is an extension $|\cdot|_L : L \to [0, \infty)$ of $|\cdot|_K$ such that L is complete with respect to the canonical metric obtained from $|\cdot|_L$.*

Proposition 1.4. *Let K be a field with an ultrametric $|\cdot|_K$. The completion $\widehat{\overline{K}}$ of the algebraic closure $\overline{K}$ of K with respect to $|\cdot|_K$ is again algebraically closed.*

Example 1.1. *In Proposition 1.4, we can take for example $\widehat{\overline{K}}$ to be $\mathbb{C}_p$ where $K = \mathbb{Q}_p$.*

We will now give the definition of our basic objects.

Definition 1.4. *Let $(K, |\cdot|_K)$ be a field which is complete with respect to the absolute value $|\cdot|_K$. We define the Tate algebra over K as*

$$\mathbb{T}_m(K) := \left\{ f = \sum_{\alpha \in \mathbb{N}^m} f_\alpha X^\alpha \;\middle|\; \substack{\text{for all } \varepsilon > 0, \text{ there are only} \\ \text{finitely many } \alpha \text{ such that } |f_\alpha| > \varepsilon} \right\}$$

We define,

$$\|f|_{\mathbb{T}_m}\| := \max\{|f_\alpha|_K \mid \alpha \in \mathbb{N}^m\}.$$

Proposition 1.5. *Suppose K is algebraically closed in Definition 1.4. Then we have,*

$$\|f|_{\mathbb{T}_m}\| = \max\{|f(z)| \mid z = (z_i)_{i=1,\cdots,m} \in K^m, |z| \le 1\},$$

and such maximal is actually attained.

Remark 1.2. *In p-adic settings, K^m is totally disconneted so assertion like the "identity theorem" for holomorphic functions becomes problematic.*

In order to resolve the above obstacle Grothendieck introduced Étale topology.

1.2 Main Theorems of the Course

Definition 1.5. *An element a is power-bounded with respect to a norm $\|\cdot\|$ if there is $M > 0$ such that for all $n \in \mathbb{N}$, $\|a\|^n \le M$.*

Definition 1.6. *Let A be a topological ring. An element $a \in A$ is topologically nilpotent if*

$$\lim_{n \to \infty} a^n = 0.$$

Definition 1.7. *We define the following subset of a Tate algebra $\mathbb{T}_m$.*

$$\mathbb{T}_m^\circ := \{f \in \mathbb{T}_m \mid f \text{ is power bounded}\},$$
$$\mathbb{T}_m^{\circ\circ} := \{f \in \mathbb{T}_m \mid f \text{ is topologicaly nilpotent } \lim_{k \to \infty} f^k = 0\}.$$

Proposition 1.6. *$\|\cdot|_{\mathbb{T}_m}\|$ is multiplicative. (i.e. $\|fg\| = \|f\|\|g\|$ and $\|1\| = 1$). In particular,*

$$\mathbb{T}_m^\circ = \{f \in \mathbb{T}_m(K) \mid \|f|_{\mathbb{T}_m}\| \le 1\}.$$

and

$$\mathbb{T}_m^{\circ\circ} = \{f \in \mathbb{T}_m(K) \mid \|f|_{\mathbb{T}_m}\| < 1\}.$$

From the definition of ultrametric, it is clear that $\mathbb{T}_m^\circ$ is a subring and $\mathbb{T}_m^{\circ\circ}$ is an ideal.

Proposition 1.7. *$\mathbb{T}_m(K)$ is Noetherian and all ideals are closed in it. If $\mathfrak{m}$ is a maximal ideal, its residue field $k(\mathfrak{m}) = \mathbb{T}_m/\mathfrak{m}$ is a finite field extension of K (thus $|\cdot|_K$ extends uniquely to $k(\mathfrak{m})$).*

Proposition 1.8. *There is a topologically nilpotent unit, thus there are no proper open ideals.*

Proposition 1.9. *$\mathbb{T}_m(K)$ is regular of Krull dimension m.*

Definition 1.8. *A ring A is Jacobson if every prime ideal is an intersection of some maximal ideals.*

Proposition 1.10. *$\mathbb{T}_m(K)$ is a Jacobson ring. (The closed points of $\operatorname{Spec}\mathbb{T}_m$ are dense in every closed subsets)*

Notation 1.1. *For a ring A, we denote*

$$\operatorname{Sp} A := \{\mathfrak{m} \in \operatorname{Spec} A \mid \mathfrak{m} \text{ is a maximal ideal}\}.$$

Notation 1.2. *Let $f \in \mathbb{T}_m(K)$ and $x \in \operatorname{Sp}\mathbb{T}_m(K)$. We denote,*

$$f(x) := f \bmod x \in k(x).$$

Proposition 1.11.
$$\|f|_{\mathbb{T}_m}\| = \max\{|f(x)| \mid x \in \operatorname{Sp}\mathbb{T}_m\}.$$

We define a basis element of the topology in $\mathbb{T}_m(K)$ as follows.

Definition 1.9 (Rational Open Sets)**.** *Suppose $\mathbb{T}_m(K) = \langle f_0, \cdots, f_m\rangle_{\mathbb{T}_m}$. The rational open set associated to the generators $\langle f_0, \cdots, f_m\rangle_{\mathbb{T}_m}$ is*

$$\mathcal{R}_{\operatorname{Sp}\mathbb{T}_m}(f_0|f_1, \cdots, f_m) = \{x \in \operatorname{Sp}\mathbb{T}_m \mid |f_0(x)| \geq |f_i(x)|, 1 \leq i \leq m\}.$$

Remark 1.3. *The Grothendieck topology enforces the quasi-compactness of these subsets.*

Additional points were :? introduced by van der Put in the late 1970s using a filter machinary equivalent to topos? points. He also ? equivalence with ? valuation ? ? not

Berkovich in the 1980s constructed a multiplicative seminorm $\mathbb{T}_m \to [0, \infty)$ which are similar to $f \mapsto |f(x)|$ for $x \in \mathbb{T}_m$ or $f \mapsto \|f|_{\mathbb{T}_m}\|$. One then gets a compact Hausdorff space.

Huber constructed more general valuations and one gets a spectral space with a sheaf category equivalent to sheaves on $\operatorname{Sp}\mathbb{T}_m$ (unlike Berkovich).

In the Berkovich machinery, one also has a point corresponding to

$$\|f\|_r = \max\{r^{|\alpha|}|f_\alpha|_K \mid \alpha \in \mathbb{N}^m\},$$

where $|\alpha| = \sum \alpha_i$ and $0 \leq r \leq 1$.

Remark 1.4.
$$\|f\|_1 = \|f|_{\mathbb{T}_m}\|, \|f\|_0 = 0.$$

Definition 1.10. *Let $0 \leq r \leq 1$. We define, for $f \in \mathbb{T}_m(K)\backslash\{0\}$,*

$$N_r(f) := (\|f\|_r, n)$$

where $n = \max\{i \in \mathbb{N} \mid \|f\|_r = r^i|f_i|_K\}$.

Example 1.2. *Consider the case $\mathbb{T}_1(K)$ and $f \in \mathbb{T}_1(K)\backslash\{0\}$ where*

$$f = \sum_{n=0}^{\infty} f_n T^n.$$

Then N_r defines a valuation, $\mathbb{T}_1 \to \Gamma \cup \{0\}$ where $\Gamma = (0, \infty)_{\mathbb{R}} \times \mathbb{Z}$ ordered lexicographically with the first factor being the most significant and $0 < \gamma$ for all $\gamma \in \Gamma$.

One has

$$N(a, b) = N(a)N(b), N(1) = 1, N(0) = 0, N(a + b) \leq \max\{N(a), N(b)\}. \tag{1.1}$$

When $0 < r < 1$ and $f \in \mathbb{T}_1\backslash\{0\}$, there is $R \in (r, 1]_{\mathbb{R}}$ such that

$$f(z) = \|f_r\| \left(\frac{|z|}{r}\right)^{n'}, r < |z| \leq R. \tag{1.2}$$

where N is as in Equation (1.1), and $\|f\|_r = \sup\{|f(z)| \mid |z| \leq r\}$ where $r = 1$, Equation (1.2) has no analog as no ? R extends as $f(z)$ is undefined where $|z| > 1$. This means that N_1 is extended

Definition 1.11.

$$\operatorname{Spa}(\mathbb{T}_m, \mathbb{T}_m^0) = \{\textit{continuous valuations } \gamma \textit{ of } \mathbb{T}_m \text{ s.t. } \forall f \in \mathbb{T}_m^\circ, \gamma(f) \leq 1\}.$$

Remark 1.5. $N_1(T) = (1, 1), N_1(T^{2026}) = (1, 2026)$.

1.3 Spectral Spaces

Remark 1.6. *$\emptyset$ is not irreducible.*

Proposition 1.12. *Let X be a topological space. For a point $x \in X$,*

$$\overline{\{x\}} = \{y \in X \mid \textit{every neighbourhood of } y \textit{ in } X \textit{ cotains } x\}.$$

Definition 1.12. *A point $x \in X$ is a generic point if $\overline{\{x\}} = X$, (i.e. $\{x\}$ is dense in X).*

Notation 1.3. *For a topological space X, we denote the collection of all open sets to be $\mathscr{O}_X$.*

Definition 1.13. *Let X be a topological space and $x, y \in X$. We say x, y are topologically distinguishable if there is $U \in \mathscr{O}_X$ such that either $x \in U$ and $y \notin U$ or $x \notin U$ and $y \in U$. They are otherwise indistinguishable.*

Definition 1.14. *A topological space X is T_0 or Komogolov if for any pair of points are distinguishable.*

Definition 1.15. *A topological space X is sober if it satisfies one of the following equivalent conditions.*

1). Every irreducible neighborhood of X has precisely one generic point.

2). X is T_0 and every irreducible subset has at least one generic point.

Remark 1.7. *It is easy to see that these are local conditions.*

Definition 1.16. *A topological space is T_1 if it satisfies one of the following equivalent conditions.*

1). For any $x, y \in X$, $x \neq y \Rightarrow$ there is $U \in \mathscr{O}_X$ such that $x \in U$ and $y \not\int U$.

2). Every point of X is closed.

It is T_2 or Hausdorff if it satisfies one of the following equivalent conditions.

1). Any pair of points $x, y \in X$ has disjoint neighborhoods U, V such that $x \in U$ and $y \in V$.

2). The diagonal set $\{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Definition 1.17. *X is quasi-compact if every open covering admits a fintie subcovering. A subset M of X is quasi-compact if it is a quasi-compact as a topological space with the subset topology from X.*

Proposition 1.13. *If X can be covered by finitely many quasi-compact open neighbourhoods then it is quasi-compact.*

Proposition 1.14. *Every closed subset of a quasi-compact space is again quasi-compact.*

Proposition 1.15. *Let $X \xrightarrow{f} Y$ be a continuous with X being quasi-compact and Y being Hausdorff, then f is closed (i.e. every closed set A of X has a closed image $f(A)$ in Y).*

Definition 1.18. *A subset $\mathcal{B} \subseteq \mathscr{O}_X$ is a topology basis for X if for any $U \in \mathscr{O}_X$, we have $U = \bigcup_{V \in \mathcal{B}_U} V$ where $\mathcal{B}_U = \{V \in \mathcal{B} \mid V \subseteq U\}$.*

Remark 1.8. *The subsets of the form*

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}$$

form a topology basis for $\operatorname{Spec} A$.

Proposition 1.16. *For a topological space X, the following conditions are equivalent.*

i). $\mathfrak{Q}_C(X) := \{\Omega \in \mathscr{O}_X \mid \Omega$ is quasi-compact$\}$ is a topology basis closed under finite intersection (including the empty intersection $\bigcap_\emptyset = X$).

ii). There is a topology base $\mathcal{B} \subseteq \mathfrak{Q}_C(X)$ such that $\mathcal{B}$ consists of quasi-compact and it is closed under the finite intersection in X.

Definition 1.19. *A topological space X is spectral if it is sober and satisfies one of the conditions in Proposition 1.16.*

Definition 1.20. *A spectral map is a map $X \xrightarrow{f} Y$ between spectral spaces such that for any quasi-compact subset Ω in Y, $f^{-1}(\Omega)$ is quasi-compact subset of X.*

Remark 1.9. *A spectral map is automatically continuous.*

Proposition 1.17. *If $\mathcal{B} \subseteq \mathscr{O}_Y$ is a topology base as in the second condition of Proposition 1.16 for Y, then f is spectral if and only if $\forall \Omega \in \mathcal{B}, f^{-1}(\Omega) \in \mathfrak{Q}_C(X)$.*

Example 1.3. $\operatorname{Spec} A$ *is spectral and every ring homomorphism defines a spectral map and $\{D(f) \mid f \in A\}$ are such base as in Proposition 1.16 for* $\operatorname{Spec} A$.

Example 1.4. *The underlying space of qcqs (i.e. quasi-compact and quasi-separated) scheme is spectral.*

Remark 1.10. *Every closed and every quasi-compact open sets of a spectral space is spectral.*

Remark 1.11. *A Noetherian space X (i.e. $\mathfrak{Q}_C(X) = \mathscr{O}_X$) is spectral if and only if it is sober.*

Proposition 1.18. *Every non-empty closed subset of a spectral space X contains a closed point.*

Proof. By Remark 1.10, it suffices to show that every spectral space $X \neq \emptyset$ contains a closed point as non-empty closed subsets are again spectral. Set,

$$\mathcal{M} := \{Z \subseteq X \mid Z \text{ is closed}\}.$$

Equiq $\mathcal{M}$ with a partial order by $\subseteq$. Since X is quasi-compact, $\mathcal{M}$ has a minimal element with such order. Let Z be a such minimal element in $\mathcal{M}$. If Z has two different element x, y, then by T_0 condition, we have $U \in \mathscr{O}_X$ such that $x \notin U$ and $y \in U$. Then $Z' = Z \backslash U$ is in $\mathcal{M}$. As $x \in Z'$, this contradicts to the minimality of Z as $y \in Z \backslash Z'$. $\square$

Definition 1.21. *A filter base of a set M is a non-empty subset $\mathcal{B}$ of $\mathfrak{P}(M)$ such that*

i). $\emptyset \notin \mathcal{B}$,

ii). *for $N, O \in \mathcal{B}$ there is $P \in \mathcal{B}$ with $P \subseteq N \cap O$.*

A filter is a filter base $\mathcal{F}$ such that for $N \in \mathcal{F}$, and $N \subseteq O \subseteq M$ implies $O \in \mathcal{F}$.

A filter generated by a filter base $\mathcal{B}$ is such that

$$\mathcal{F} := \{N \mid N \subseteq M \text{ and there is } O \in \mathcal{B} \text{ such that } O \subseteq N\}.$$

A direct image under a map $M \xrightarrow{f} X$ of a filter $\mathcal{F}$ is such that

$$f_*\mathcal{F} := \{Y \mid Y \subseteq X \text{ and } f^{-1}(Y) \in \mathcal{F}\}.$$

Example 1.5. *A neighbourhood filter $\mathcal{N}_x$ of $x \in X$ is such that*

$$\mathcal{N}_x := \{N \subseteq X \mid \exists U \in \mathscr{O}_X \text{ s.t. } x \in U \subseteq N\}.$$

Example 1.6. *A principal filter $\mathcal{P}_x$ of $x \in X$ is such that*

$$\mathcal{P}_x = \{N \subseteq X \mid x \in N\}.$$

Remark 1.12. *A neighbourhood base for $x \in X$ is a filter base $\mathcal{B}$ such that it generates the neighbourhood filter of x.*

Remark 1.13. *The notion of filters is introduced by Cartan.*

Proposition 1.19. *For a filter $\mathcal{F}$ of X, the following statements are equivalent.*

1). *$\mathcal{F}$ is a $\subseteq$-maximal element of the set of filters on X.*

2). *If $M, N \subseteq X$ and $M \cup N \in \mathcal{F}$ then $M \in \mathcal{F}$ or $N \in \mathcal{F}$.*

3). For $M \subseteq X$, precisely one of M and $X \backslash M$ is in $\mathcal{F}$.

Definition 1.22. *Such filter $\mathcal{F}$ satisfying the conditions in Proposition 1.19 is called an ultrafilter.*

Theorem 1.1 (Tarski)**.** *Every filter on X is contained in some ultrafilter.*

Proof. Follows from Zorn's lemma. ☐

Proposition 1.20. *Let $\mathcal{U}$ be an ultrafilter of X then all direct images of $\mathcal{U}$ are ultrafilters. If $\iota : A \to X$ is the inclusion of a subset then this induces,*

$$\{\textit{ultrafilters on } A\} \overset{\iota_*}{\to} \{\textit{ultrafilters on } X \textit{ containing } A\},$$

which is a bijectin.

Definition 1.23.

$$\operatorname{Lim} \mathcal{F} = \{x \in X \mid \mathcal{N}_x \subseteq \mathcal{F}\}.$$

is the set of limits of $\mathcal{F}$. If it contains precisely one $x \in X$, we denote it by

$$\operatorname{Lim} \mathcal{F} = x.$$

Definition 1.24. *We say that $x \in X$ is a point of accumulation (denoteds by point of accumulation, sometimes called point of condensation) if for every neighbourhood $V \in \mathcal{N}_x$ and every $M \in \mathcal{F}$, we have $V \cap M \neq \emptyset$.*

Proposition 1.21. *Let $x \in X$ then*

$$\overline{\{x\}} = \operatorname{Lim} \mathcal{P}_x = \textit{point of accumulation of the principal filter of } x = \operatorname{Lim} \mathcal{N}_x.$$

Proposition 1.22. *Let $\mathcal{F}, \mathcal{G}$ be filters such that $\mathcal{F} \subseteq \mathcal{G}$ and $\mathcal{U}$ be a ultrafilter.*

1. *Every limit of $\mathcal{F}$ is a point of accumulation of $\mathcal{F}$.*
2. *Every limit of $\mathcal{F}$ is a limit of $\mathcal{G}$.*
3. *Every point of accumulation of $\mathcal{G}$ is a point of accumulation of $\mathcal{F}$.*
4. *A point $x \in X$ is a limit of $\mathcal{U}$ if and only if it is a point of accumulation of $\mathcal{U}$.*
5. *The sets of its limits and its point of accumulations are closed for $\mathcal{F}$.*
6. *Let $\iota : A \to X$ be a canonical inclusion where A has an induced topology from X. Then $\operatorname{Lim} \mathcal{F} = A \cap \operatorname{Lim} \iota_* \mathcal{F}$.*

Proof. For the third assertion, if a is a point of accumulation of $\mathcal{U}$ and $V \in \mathcal{N}_x$, then $X \backslash V \notin \mathcal{U}$ as x is a point of accumulation. Hence $V \in \mathcal{U}$ by Proposition 1.19. For the forth assertion, let $x \notin \operatorname{Lim} \mathcal{F}$, (respectively x not a point of accumulation of $\mathcal{F}$) then there is $U \in \mathcal{N}_x$ such that $U \notin \mathcal{F}$ (respectively, such that there is $M \in \mathcal{F}$ such that $U \cap M = \emptyset$). Then no elememt of U is a limit (respectively, point of accumulation) of $\mathcal{F}$. ☐

Proposition 1.23. *For a subset M of a topological space X, the following subsets all coincide.*

a). $\{$limits of ultrafilters $\mathcal{U}$ of X such that $M \in \mathcal{U}\}$,

b). $\bigcup_{\substack{\mathcal{F} \textit{ filter} \\ M \in \mathcal{F}}} \operatorname{Lim} \mathcal{F}$.

c). $\{x \in X \mid \exists \mathcal{F} \textit{ filter s.t. } M \in \mathcal{F} \textit{ and } x \textit{ is a point of accumulation of } \mathcal{F}\}$.

d). $\bigcap_{\substack{M \subseteq Z \subseteq X \\ Z \textit{ closed}}} Z$.

Proof. Denote those sets by M_a, M_b, M_c, M_d. Then,

$$M_a \underbrace{\subseteq}_{\text{trivial}} M_b \underbrace{\subseteq}_{\text{Proposition 1.22}} M_c \underbrace{\subseteq}_{\text{definition}} M_d.$$

For the reverse direction, note that $M_d = \bigcup_{\substack{U\in\mathscr{O}_X,\\ U\cap M=\emptyset}} (X\backslash U)$. If $x \in M_d$, then

$$\{U \cap M \mid U \in \mathcal{N}_x\}$$

is a filter base containing M and in particular containing $\mathcal{N}_x$. By Theorem 1.1, it is contained in some ultrafilter $\mathcal{U}$ and by definition $x \in \operatorname{Lim}\mathcal{U}$. ☐

Definition 1.25. *The set of above is called the closure of M in X.*

Remark 1.14. *The closure of a point s a special case of this.*

Corollary 1.1. *For a subset A of a topological space X, the following conditions are equivalent.*

i). If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a limit of $\mathcal{F}$ then $x \in A$.

ii). If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a point of accumulation of $\mathcal{F}$ then $x \in A$.

iii). If $\mathcal{U}$ is a ultrafilter on X such that $A \in \mathcal{U}$, every limit of $\mathcal{U}$ in X is in A.

iv). A is closed.

Proof. Clear from Proposition 1.23 ☐

Proposition 1.24. *For a map $f : X \to Y$ of topological spaces. The following conditions are equivalent.*

i). f is continuous.

ii). If $\mathcal{F}$ is a filter on X and $x \in \operatorname{Lim}\mathcal{F}$ then $f(x) \in \operatorname{Lim} f_*\mathcal{F}$.

iii). If $\mathcal{F}$ is a filter on X and $x \in X$ is a point of accumulation of $\mathcal{F}$, then $f(x)$ is a point of accumulation of $f_*\mathcal{F}$.

iv). If $\mathcal{U}$ is a ultrafilter on X and $x \in \operatorname{Lim}\mathcal{U}$, then $f(x) \in \operatorname{Lim} f_*\mathcal{U}$.

Proof.

i) ⇒ii). This follows from that $f_*\mathcal{N}_x = \mathcal{N}_{f(x)}$ and f_* preserves inclusion.

ii) ⇒ iii). By Proposition 1.22, every limit is a point of accumulation.

iii)⇒iv). By Proposition 1.22, every point of accumulation of a ultrafilter is a limit of the filter and f_* maps ultrafilters to ultrafilters.

i)⇒iv). If $A \subseteq Y$ is closed, we want to show that $f^{-1}(A)$ is closed. Let $\mathcal{U}$ be an ultrafilter in X such that $f^{-1}(A) \in \mathcal{U}$. Then $f_*\mathcal{U}$ is a ultrafilter on Y containing A and by Corollary 1.1, any limit of $f_*\mathcal{U}$ is in A. Since limits of $\mathcal{U}$ are mapped to limits of $f_*\mathcal{U}$, $f^{-1}(A)$ contains limits of $\mathcal{U}$. Again by Corollary 1.1 as $f^{-1}(A)$ is closed. ☐

Remark 1.15. *By Definition 1.24, we have the following. The set of points of accumulation of a filter $\mathcal{F}$ is the intersection of the closure of the elements of $\mathcal{F}$ or the intersection of all $A \in \mathcal{F}$ which are closed in X.*

Proposition 1.25. *Let X be a topological space. The following conditions are equivalent.*

i). X is Hausdorff.

ii). Every filter has at most one limit.

iii). Every ultrafilter on X has at most one limit.

Proof.

i)⇒ii). If $x \neq y$, they have disjoint neighborhoods $x \in U$ and $y \in V$. If $U \in \mathcal{F}$ and $V \in \mathcal{F}$, then $\emptyset = U \cap V \in \mathcal{F}$, which is not possible. Thus precisely one of x and y is in $\operatorname{Lim} \mathcal{F}$.

ii)⇒iii) is trivial.

iii)⇒i). Assume $x \neq y$ and consider the set

$$\{U \cap V \mid U \in \mathcal{N}_x, V \in \mathcal{N}_y\}.$$

This is a filter unless it contains $\emptyset$. In that case, it is contained in some ultrafilter $\mathcal{U}$. By definition of limits, x, y are both limits of $\mathcal{U}$ thus $x = y$ which is a contradiction. □

Proposition 1.26. *Let X be a topological space. The following conditions are equivalent.*

i'). X is quasi-compact.

ii'). Every filter has at least one point of accumulation.

iii'). Every ultrafilter has at least one limit.

Proof.

i')⇒ii') By Remark 1.15,

$$\{x \mid x \text{ is a point of accumulation of } \mathcal{F}\} = \bigcap_{M \in \mathcal{F}} \overline{M}.$$

By quasi-compactness X and applying it to the complement, this is a finite intersection of elements in $\mathcal{F}$. By definition, any filter is closed under finite intersection and does not contain the emptyset, this is not empty.

ii')⇒iii'). Every filter is contained in some ultrafilter and limits and points of accumulation are the same in ultrafilter by Proposition 1.22.

iii')⇒i'). Let $(A_\lambda)_\Lambda$ be a faimly of closed subsets such that for any finite subset $F \subseteq \Lambda$, we have,

$$\bigcap_{\lambda \in F} A_\lambda \neq \emptyset.$$

Then the collection,

$$\left\{ \bigcap_{\lambda \in F} A_\lambda \;\middle|\; F \subseteq \Lambda, |F| < \infty \right\}$$

is a filter base, thus it is contained in some ultrafilter $\mathcal{U}$. Let $x \in \operatorname{Lim} \mathcal{U}$. For $\lambda \in \Lambda$, $A_\lambda \in \mathcal{U}$ hence $x \in A_\lambda$ by Corollary 1.1. Hence $x \in \bigcap_{\lambda \in \Lambda} A_\lambda$ and this intersection is not empty. This is the definition of quasi-compactness stated in terms of closed sets. □

Proposition 1.27. *If X satisfies these equivalent conditions of Proposition 1.16, then for every ultrafilter $\mathcal{U}$ on X, the closed subset $\operatorname{Lim} \mathcal{U} \subseteq X$ is irreducible.*

Proof. Let $\Omega, \Theta \in \mathfrak{Q}_C(X)$ such that $\Omega \cap \operatorname{Lim} \mathcal{U} \neq \emptyset$ and $\Theta \cap \operatorname{Lim} \mathcal{U} \neq \emptyset$. By the definition of $\operatorname{Lim} \mathcal{U}$, $\Omega, \Theta \in \mathcal{U}$. Hence $\Omega \cap \Theta \in \mathcal{U}$. As $\Omega \cap \Theta$ is quasi-compact $\mathcal{U} \cap \mathscr{O}_{\Omega \cap \Theta}$ has a limit $x \in \Omega \cap \Theta$ by Proposition 1.26. By Proposition 1.22, $x \in \operatorname{Lim} \mathcal{U}$.

If U and V are open subsets such that $U \cap \operatorname{Lim} \mathcal{U}$ and $V \cap \operatorname{Lim} \mathcal{U}$ are not emptysets, there are $\Omega \subseteq U$ and $\Theta \subseteq V$ as above (as $\mathfrak{Q}_C(X)$ is a topology base), hence $U \cap V \cap \operatorname{Lim} \mathcal{U}$ is not empty. Finally by Proposition 1.26, $\operatorname{Lim} \mathcal{U} \neq \emptyset$ as X is quasi-compact. □

Definition 1.26. *We say that $x \in X$ is a generic limit of a ultrafilter $\mathcal{U}$ if $\operatorname{Lim} \mathcal{U} = \overline{\{x\}}$.*

Definition 1.27. *Let X be a topological space in which every ultrafilter $\mathcal{U}$ has a unique generic limit. Then we write,*

$$\operatorname{glim} \mathcal{U} = \operatorname{glim}_X \mathcal{U}$$

We say that a subset $M \subseteq X$ is closed under generic limit in X if $M \in \mathcal{U}$ for every ultrafilter $\mathcal{U}$ implies $\operatorname{glim} \mathcal{U} \in M$.

Proposition 1.28.

1). X is T_0 if and only if every ultrafilter on X has at most one generic limit.

2). If every ultrafilter on X has some generic points then every closed subset $Z \subseteq X$ has some generic points.

Proof. For 1). the sufficiency of being T_0 is by Definition 1.15. If x and y are topologically indistinguishable, then $\overline{\{x\}} = \overline{\{y\}} = Z$ for some Z. Furthermore, the principal filter $\mathcal{U}$ of x which has $\mathrm{Lim}\,\mathcal{U} = Z$ from Example 1.6 has both x and y as generic limits.

2). Irreducibility of Z implies that $\mathscr{O}_Z \backslash \{\emptyset\}$ is a filter base in X. Hence it is contained in some ultrafilter $\mathcal{U}$. By Definition 1.25 and the closedness of Z implies $Z = \lim \mathcal{U}$. Hence every $\mathrm{glim}\,\mathcal{U}$ is a generic point of Z. □

Proposition 1.29. *Assume every ultrafilter $\mathcal{U}$ on X has a unique generic limit* $\mathrm{glim}\,\mathcal{U}$*. Then the following statements hold.*

1). The set of subsets of X which are closed under glim *is closed under arbitrary intersection and finite unions.*

2). An open subset $V \subseteq X$ is closed under glim *if and only if V is quasi-comapct.*

Proof. For 1), the closedness under the intersections is obvious. If $M \cup N \in \mathcal{U}$ then $M \in \mathcal{U}$ or $N \in \mathcal{U}$ by Proposition 1.19. Hence $\mathrm{glim}\,\mathcal{U} \in M \subseteq M \cup N$ or $\mathrm{glim}\,\mathcal{U} \in N \subseteq M \cup N$.

Let $\mathcal{V}$ be an ultrafilter on V and $\mathcal{U}$ be the image of $\mathcal{V}$ under the embedding $V \hookrightarrow X$. If V is closed under glim then $\mathrm{glim}\,\mathcal{V} \in V$ is a limit of $\mathcal{U}$ by Proposition 1.22. By Proposition 1.25, V is quasi-compact. If V is quasi-compact, $\mathcal{V}$ has a limit $x \in V$ and $x \in \mathrm{Lim}\,\mathcal{V} = V \cap \mathrm{Lim}\,\mathcal{U}$ by Proposition 1.22. Hence $x \in \overline{\{y\}}$ hence $y \in V$ and V is closed under glim. □

Definition 1.28. *A spectral space X is quasi-separated if for any ultrafilter $\mathcal{U}$ on X, if its limit* $\mathrm{Lim}\,\mathcal{U}$ *is non-empty then it is an irreducible closed subset.*

Proposition 1.30.

1). A topological space X is spectral if and only if every ultrafilter on X has a unique generic limit and the set of those $\Omega \in \mathscr{O}_X$ which are closed under glim *is a topology base.*

2). An open subset of a spectral space X is quasi-compact if and only if it is closed under glim*. (This is proved in Proposition 1.29).*

3). A map $f : X \to Y$ is spectral if and only if it is continuous and $\mathrm{glim}_Y\, f_*\mathcal{U} = f(\mathrm{glim}_X\, \mathcal{U})$ for every ultrafilter $\mathcal{U}$ on X. (This is proved by the above assertion).

Proof. If X is spectral, then by Proposition 1.27, $\mathrm{Lim}\,\mathcal{U}$ is always closed and irreducible subset, hence $\mathrm{glim}\,\mathcal{U}$ exists uniquely by the sobriety of X.

By Proposition 1.29, one has $\{U \in \mathscr{O}_X | U \text{ is closed under glim}\} = \mathfrak{Q}_C(X)$ is a topology base. If any ultrafilter has a unique generic limit, then X is sober by Proposition 1.29 and $\mathscr{O}_{\mathcal{C}}(X) = \{U \in \mathscr{O}_X \mid U \text{ is closed under glim}\}$ which is closed under finite intersection in X and by the assumption this is a topology base for X. □

Lemma 1.1. *Let $\mathcal{B}$ be a topology base of the topological space X.*

a) Let $\mathcal{U}$ be an ultrafilter on X and $g \in X$ such that for all $\Omega \in \mathcal{B}$, $g \in \Omega \Leftrightarrow \Omega \in \mathcal{U}$. Then g is a generic limit of $\mathcal{U}$.

b) If X is T_0 and for every ultrafilter $\mathcal{U}$ on X, there is $g \in X$ as in a), then X is spectral and the elements of $\mathcal{B}$ are quasi-compact.

Proof. Let $U \in \mathcal{N}_g$. Then there is $\Omega \in \mathcal{B}$ such that $g \in \Omega \subseteq U$. By our assumption, $\Omega \in \mathcal{U}$ hence $U \in \mathcal{U}$ thus g is a limit of $\mathcal{U}$. Let $x \notin \overline{\{g\}}$ then there is $U \in \mathcal{N}_x$ such that $g \notin U$. As $\mathcal{B}$ is a topology base there is $\Omega \in \mathcal{B}$ such that $x \in \Omega \subseteq U$. Then $g \notin \Omega$, hence $\Omega \notin \mathcal{U}$. Since Ω is a neighbourhood of x, this shows $x \notin \mathrm{Lim}\,\mathcal{U}$. Thus g is a generic limit. This proves a).

For b), it follows from our assumption and a) that X has unique generic limits of ultrafilters and the elements of $\mathcal{B}$ are closed under such generic limits. The claim now follows from Proposition 1.30. □

Theorem 1.2. *A topological space X is spectral if and only if there is a ring A such that X is homeomorphic to* $\mathrm{Spec}\,A$.

Proof. We omit the hard part which is $\Rightarrow$ direction (the existence of such ring A for every spectral space X). That $\operatorname{Spec} A$ is spectral can be seen as follows. Let $\mathcal{U}$ be an ultrafilter on $\operatorname{Spec} A$ and set

$$\mathfrak{g}_{\mathcal{U}} := \{a \in A \mid V(a) \in \mathcal{U}\} = \{a \in A \mid \{\mathfrak{p} \in \operatorname{Spec} A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

We show $\mathfrak{g}_{\mathcal{U}} \in \operatorname{Spec} A$. Let $a_1, a_2 \in \mathfrak{g}_{\mathcal{U}}$, and $M_i = \{\mathfrak{p} \in \operatorname{Spec} A \mid a_i \in \mathfrak{p}\} \in \mathcal{U}$ hence $M_1 \cap M_2 \in \mathcal{U}$, hence $\{\mathfrak{p} \mid a_1 + a_2 \in \mathfrak{p}\} \in \mathcal{U}$ as $\mathcal{U}$ is closed under finite intersection. Hence $a_1 + a_2 \in \mathfrak{g}_{\mathcal{U}}$ thus it is an ideal. Let $a_1 a_2 \in \mathfrak{g}_{\mathcal{U}}$, but $a_2 \notin \mathfrak{g}_{\mathcal{U}}$. Then $M_2 \notin \mathcal{U}$ but $M_1 \cup M_2 \in \mathcal{U}$ since $a_1 a_2 \in \mathfrak{p}$, one of them must belong to it. Hence

$$M_1 \cup M_2 = \{\mathfrak{p} \mid a_1 a_2 \in \mathfrak{p}\} \in \mathcal{U}.$$

$M_2 \notin \mathcal{U}$ implies $M_1 \in \mathcal{U}$ hence $a_1 \in \mathfrak{g}_{\mathcal{U}}$.

Since $X = V(f) \cap D(f)$, and that $f \notin \mathfrak{g}_{\mathcal{U}}$ if and only if $V(f) \notin \mathcal{U}$ if and only if $D(f) \in \mathcal{U}$. By Lemma 1.1, $\mathfrak{g}_{\mathcal{U}}$ is $\operatorname{glim} \mathcal{U}$ and all $D(f)$ are glim-closed. □

Remark 1.16. *If $\mathcal{U}$ is an ultrafilter on* $\operatorname{Spec} A$ *and*

$$\mathfrak{g}_{\mathcal{U}} = \{a \in A \mid \{\mathfrak{p} \in \operatorname{Spec} A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

Then the lemma applies with $\mathcal{B} = \{D(a) \mid a \in A\}$.

Corollary 1.2 (Of Lemma 1.1 b), and the proof of Theorem 1.2)**.** *The subset $D(a) \subseteq$ $\operatorname{Spec} A, a \in A$ are quasi-compact.*

Proof. From Remark 1.16 and Lemma 1.1. □

Definition 1.29. *Let X be a topological space. A subset U which is open and closed is called clopen.*

Corollary 1.3 (Of Proposition 1.30)**.** *For a compact (thus Hausdorff) X, the following conditions are equivalent.*

a). X is spectral.

b). The clopen subsets of X form a topology base.

In this case, the open subsets U is quasi-compact if and only if it is clopen and for a spectral space Y, a map $Y \to X$ is spectral if it is continuous.

Proof. Every ultrafilter $\mathcal{U}$ of X admits a unique limit by Proposition 1.25 and Proposition 1.26, such limits are trivially generic limits. By Corollary 1.1, the open subsets which are invariant under glim are closed.

To show these two are equivalent. If U is closed, it is quasi-compact as it is a closed subset of a quasi-compact space. If U is quasi-compact then it is closed by Proposition 1.15. If $Y \xrightarrow{f} X$ is continuous and $\Omega \in \mathfrak{Q}_C(X)$, then Ω is clopen hence $f^{-1}(\Omega)$ is clopen in Y, hence $f^{-1}(\Omega) \in \mathfrak{Q}_C(Y)$. □

Definition 1.30. *A set $\mathcal{A}$ of subsets of X is a boolean algebra if the following conditions hold.*

1. *$\emptyset \in \mathcal{A}$.*
2. *$M \in \mathcal{A} \Rightarrow X \backslash M \in \mathcal{A}$.*
3. *$\mathcal{A}$ is closed under taking finite unions.*

Remark 1.17. *From the basic axioms of boolean algebras. Clearly we have $X \in \mathcal{A}$ and $M_1, M_2 \in \mathcal{A} \Rightarrow M_1 \cap M_2 \in \mathcal{A}$.*

Proposition 1.31. *Let X be a spectral space. The following boolena algebras of subsets of X coincide.*

1. *The boolean algebra generated by $\mathfrak{Q}_C(X)$.*
2. *$\{M \subseteq X \mid M$ is closed under generic limits of ultrafilters$\}$.*

Let $\mathcal{K}_X$ be this boolean algebra of subsets of X. The topology on X obtained by taking $\mathcal{K}_X$ as a topology base coincides with the one for which taking the subsets $M \subseteq X$ which are closed under glim *of ultrafilters as closed subsets.*

Finally let X_{con} be X equipped with this topology. Then X_{con} is a spectral hausdorff space (i.e. X_{con} as in Corollary 1.2), and $X_{\mathrm{con}} \overset{\mathrm{id}_X}{\to} X$ is a spectral map.

For every ultrafilter $\mathcal{U}$ of X, $\lim_{X_{\mathrm{con}}} \mathcal{U} = \mathrm{glim}_X \mathcal{U}$.

Proof:(Caution : This is not required for the exam). Let $\mathcal{K}_1$ be th Boolean algebra generated by $\mathfrak{Q}_C(X)$ and

$$\mathcal{K}_2 := \{M \mid M \text{ and } X\backslash M \text{ closed under glim}\}.$$

By Proposition 1.28, $\mathcal{K}_2$ is a Boolean algebra. By Proposition 1.30 b), and Corollary 1.1, $\mathfrak{Q}_C(X) \subseteq \mathcal{K}_2$, hence we have an inclusion $\mathcal{K}_1 \subseteq \mathcal{K}_2$. For the time being, $\mathcal{K}_x := \mathcal{K}_1$. Let X_1 be the topological space with underlying space X and the topology generated by $\mathcal{K}_x$. Let X_2 be the topological space again with the same underlying set such that for a subset $A \subseteq X$ is closed if and only if it is closed under glim of ultrafilters. By Proposition 1.28, X_2 is indeed a topological space. Since $\mathcal{K}_x \subseteq \mathcal{K}_2$, X_2 is finer or equal to X_1.

The space X_1 is Hausdorff. If $x \neq y$ then without loss of generality exchanging x and y if necessary, there is $U \in \mathscr{O}_X$ such that $x \in U$ and $y \notin U$. As $\mathfrak{Q}_C(X)$ is a topology base, hence without loss of $U \in \mathfrak{Q}_C(X)$. Then $U \in \mathcal{K}_X$ and $X\backslash U \in \mathcal{K}_X$ and they are distinguished neighbourhoods of x and y in X_1. Moreover, X_2 is quasi-compact by Proposition 1.25. Thus it suffices to consider an ultrafilter $\mathcal{U}$ on X and to show that it has a limit in X_2. Let $g = \mathrm{glim}_X \mathcal{U}$. We claim that $g \in \mathrm{Lim}_{X_2} \mathcal{U}$. For this, let $U \in \mathscr{O}_{X_2}$ such that $g \in U$, we must show $U \in \mathcal{U}$. Otherwise $X\backslash U \in \mathcal{U}$ But by the definition of X_2, $X\backslash U$ is closed undrer glim_X of ultrafilters. Thus $g \in X\backslash U$ which is a contradiction.

By Proposition 1.15, the continuous map $X_2 \overset{\mathrm{id}_X}{\to} X$ is a homeomorphism. Thus first two parts are shown.

Let $X_{\mathrm{con}} = X_1 = X_2$. Let $\Omega \in \mathcal{K}_2$. By Corollary 1.1, Ω is clopen in X_{con} Since $\mathcal{K}_1$ is a topology base. Ω is a union of elements of $\mathcal{K}_1$ and since $\Omega \in \mathfrak{Q}_C(X_{\mathrm{con}})$, the union can be chosen to be finite. Then $\Omega \in \mathcal{K}_1$ as $\mathcal{K}_1$ is a Boolean algebra of subsets.

From now on, we will regard $\mathcal{K}_x = \mathcal{K}_1 = \mathcal{K}_2$. Since the elements of $\mathcal{K}_x$ are clopen and $\mathcal{K}_x$ is a topology base for X_{con}, and X_{con} compact. By Corollary 1.3, X_{con} is spectral. By Proposition 1.30, $X_{\mathrm{con}} \overset{\mathrm{id}_X}{\to} X$ is spectral. □

Definition 1.31. *The elements of $\mathcal{K}_X$ are called constructible subsets and the topology of X_{con} the constructible topology.*

Corollary 1.4. *A subset of X is an element of $\mathcal{K}_X$ if and only if it is clopen in X_{con}.*

Corollary 1.5 (Of Proposition 1.30)**.** *A map $X \overset{f}{\to} Y$ between spectral spaces is spectral if and only if it is continuous and $X_{\mathrm{con}} \overset{f}{\to} Y_{\mathrm{con}}$ is also continuous.*

Corollary 1.6. *For a spectral space X, $\mathfrak{Q}_C(X) = \mathscr{O}_X \cap \mathcal{K}_X$.*

Definition 1.32. *Let X be a spectral. We say that $x \in X$ is a specialization of $y \in X$ and conversely y is a generalization of x if $x \in \overline{\{y\}}$. We denote such relation by $x \preceq y$.*

Proposition 1.32.

1. *A subset $Z \subseteq X$ is closed in X if and only if it is closed in X_{con} and closed under specialization. That is $x \preceq y, y \in Z \Rightarrow x \in Z$.*
2. *A subset $U \subseteq X$ is open in X if and only if it is open in X_{con} and closed under generalization.*
3. *Let Z be closed in X_{con} and $\overline{Z}$ be its closure in X. Then $\overline{Z}$ is the set of specialization of elements of Z.*

Proof. If $z \in Z$ and $x \preceq z$, then $x \in \overline{\{z\}} \subseteq \overline{Z}$. If $x \in \overline{Z}$ by Proposition 1.23, there is an ultrafilter $\mathcal{U}$ on X such that $Z \in \mathcal{U}$ and $x \in \mathrm{Lim}\mathcal{U}$. Since Z is closed in X_{con}, $g = \mathrm{glim}\mathcal{U} \in \overline{Z}$. But $x \preceq g$, proving the third statement. The first two are equivalent and the first follows from the third. □

Corollary 1.7. *For a spectral space X, the following statements are equivalent.*

1. *X is T_0.*
2. *$X = X_{\text{con}}$ as topological space.*
3. *Every quasi-compact subset of X is closed.*
4. *X is Hausdorff.*
5. *X is T_4.*

Corollary 1.8. *For a map $X \xrightarrow{f} Y$ between spectral spaces, the following statements are equivalent.*

1. $\operatorname{glim}_Y f_*\mathcal{U} = f(\operatorname{glim}_X \mathcal{U})$ *for all ultrafilters $\mathcal{U}$ on X, and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.*
2. *$X_{\text{con}} \xrightarrow{f} Y_{\text{con}}$ and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.*
3. *f is spectral.*

Corollary 1.9. *For a subset $Z \subseteq X$, the following statements are equivalent.*

a). Z is spectral with induced topology and $Z \to X$ is spectral.

b). X is closed in X_{con}.

In this case, we dentoe Z_{con} to be the topology on Z induced from topology X_{con}

1.4 The Valuation Spectrum of a Rings

Definition 1.33. *An ordered abelian group is an abelian group Γ (under mulitplication) with a linear ordering $\leq$ such that*

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

Example 1.7. *$\{1\}$ and $(0,\infty)_{\mathbb{R}}^{\times}$ are ordered abelian groups.*

Remark 1.18. *The inequality,*

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

is strict if one of the inequalities in the left hand side is strict. In particular, $g > 1$ implies $g^k > 1$ and $g < 1$ implies $g^k < 1$ for all $k \in \mathbb{N}$. Thus Γ has no torsion.

Definition 1.34. *A (general) valuation of a ring R is a map $R \xrightarrow{\gamma} \Gamma \cup \{0\}$ where Γ is an ordered abelian group.*

Definition 1.35. *A valuation of a ring R is a map $R \xrightarrow{\gamma} \Gamma \cup \{0\}$ where Γ is an ordered abelian group with multiplication such that the following hold.*

i). $\forall x \in \Gamma \cup \{0\}, 0 \cdot x = x \cdot 0 = 0, 0 \leq x$.

ii). $\gamma(xy) = \gamma(x)\gamma(y)$

iii). $\gamma(1) = 1$

iv). $\gamma(0) = 0$

v). $\gamma(x + y) \leq \max\{\gamma(x), \gamma(y)\}$.

Remark 1.19.

$$\gamma(-1) = 1, \quad \gamma(y) < \gamma(x) \Rightarrow \gamma(x + y) = \gamma(x).$$

Definition 1.36. *Let Γ_γ be the subgroup of Γ generated by $\{\gamma(r) \mid r \in R\}\backslash\{0\}$. We say that valuations γ and ω are equivalent if there is an order preserving isomorphism $\Gamma_\gamma \overset{i,\sim}{\to} \Gamma_\omega$ such that*

$$\omega(r) = i(\gamma(r)),$$

where we put $i(0) = 0$.

Proposition 1.33. *The valuation of r and w are equivalent if and only if*

$$\{(r, s) \in R^2 \mid \gamma(r) \le \gamma(s)\} = \{(r, s) \in R^2 \mid w(r) \le w(s)\}.$$

Definition 1.37.

$$\operatorname{supp}\gamma = \{r \in R \mid \gamma(r) = 0\}.$$

Remark 1.20. $\operatorname{supp}\gamma$ *is a prime ideal. It stable under multipliation by any element since,*

$$\forall s \in R, \gamma(r) = 0 \Rightarrow \gamma(rs) = \gamma(r)\gamma(s) = 0.$$

It is closed under addition since,

$$r, s \in \operatorname{supp}\gamma \to \gamma(r + s) \le \max\{\gamma(r), \gamma(s)\} = 0.$$

Therefore it is an ideal. Clearly $1 \notin \operatorname{supp}\gamma$. If $\gamma(rs) = 0$, then $\gamma(r)\gamma(s) = 0$. Thus $\gamma(r) = 0$ or $\gamma(s) = 0$.

Remark 1.21. *The support only depends on the equivalence classes.*

Example 1.8. *If $\mathfrak{p}$ is a prime ideal and*

$$\gamma(r) = \begin{cases} 1, & r \notin \mathfrak{p}, \\ 0, quadr \in \mathfrak{p}. \end{cases}$$

the trivial valuation beloning to $\mathfrak{p}$.

Definition 1.38. *A valuation is trivial if it is equivalent to this kind of valuation. That is $\Gamma_\gamma = \{1\}$.*

Example 1.9. *If $R = K$ is a field, then $\gamma(s) = 1$ where $\gamma^k = 1$ for some $k \in \mathbb{N}$ as Γ has no torsion. Then for a finite field only admits trivial valuations.*

Example 1.10. *The p-adic valuation of $\mathbb{Q}$ or $\mathbb{Q}_p$ is a valuation.*

Definition 1.39. *Let $\operatorname{Spv}(R)$ be the set of equivalent classes of valuation of R. For $x \in \operatorname{Spv}(R)$, we choose a representative $|\cdot|_x$. Let Γ_x be the group generated by those $|r|_x$ which are non-zero. For $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in R^{m+1}$, let*

$$\begin{aligned} \mathcal{R}(f_0|f_1, \cdots, f_m) &= \mathcal{R}_{\operatorname{Spv}(R)}(f_0|f_1, \cdots, f_m), \\ &:= \{x \in \operatorname{Spv}(R) \mid 0 \ne |f_0|_x \ge |f_i|_x, 1 \le i \le m\}. \end{aligned}$$

These are called rational open subsets. We equip $\operatorname{Spv}(R)$ with the topology for which the set of rational open subsets is a topology base.

If $A \overset{\alpha}{\to} B$ is a ring homomorphism then this induces

$$\operatorname{Spv}(B) \overset{\operatorname{Spv}(\alpha)}{\to} \operatorname{Spv}(A)$$

sends $x \in \operatorname{Spv}(B)$ to the equivalence class of the valuation,

$$a \mapsto |\alpha(a)|_x$$

of A. We set,

$$V(a) = \operatorname{Spv}(A)\backslash\mathcal{R}(a|).$$

Remark 1.22. *The topology is therefore the coarsest for which*

$$\operatorname{Spv}(A) \overset{x \mapsto \operatorname{supp}(x)}{\to} \operatorname{Spec}(A)$$

is continuous and all

$$\{x \in \operatorname{Spv}(A)\backslash V(a) \mid |\alpha|_x \le |a|_x\}$$

for $(a, \alpha) \in A^2$ are open.

Remark 1.23. *The set of rational open subset is closed under finite intersections in* $\mathrm{Spv}(R)$. *Indeed,*

$$\mathcal{R}(1|) = \mathrm{Spv}(R),$$
$$\mathcal{R}(r_0|r_1, \cdots, r_m) \cap \mathcal{R}(s_0|s_1, \cdots, s_n) = \mathcal{R}(r_0 s_0 | r_i s_j, 0 \le i \le n, 0 \le j \le n).$$

If you assume $m \le 1$ *or even* $m = 1$, *you only get a topology subbase.*

One way of constructing generic limits of ultrafilters in $\mathrm{Spv}(R)$ would be to use Proposition 1.33 and

$$\{(r,s) \in R^2 \mid |r|_g \le |s|_g\} := \{(r,s) \in R^2 \mid \{x \in \mathrm{Spv}(R) \mid |r|_x \le |s|_x\} \in \mathcal{U}\}.$$

This would work and was shown by Huber. This approache will not be taken in this course.

Definition 1.40. *Let* $\mathcal{U}$ *be an ultrafilter on* Λ *and* $(X_\lambda)_{\lambda \in \Lambda}$ *be a family of sets indexed by* Λ. *We call the elements* $(x_\lambda)_{\lambda \in \Lambda}$ *and* $(\xi_\lambda)_\Lambda$ *of* $\prod_{\lambda \in \Lambda} X_\lambda$, $\mathcal{U}$*-equivalent if*

$$\{\lambda \mid x_\lambda = \xi_\lambda\} \in \mathcal{U}.$$

The ultraproduct $\prod_{\mathcal{U}} X_\lambda$ *is the set of* $\mathcal{U}$*-equivalence classes of elments of* $\prod_{\lambda \in \Lambda} X_\lambda$. *If* X_λ*-s have group structures,* $(x\xi) \in \prod_{\mathcal{U}} X_\lambda$ *is given as the* $\mathcal{U}$*-equivalece classes of* $(x_\lambda \xi_\lambda)_\Lambda$. *If* X_λ*-s are equipped with linear ordering, put*

$$(x_\lambda)_{\lambda \in \Lambda}/\sim_{\mathcal{U}} \le (\xi_\lambda)_{\lambda \in \Lambda}/\sim_{\mathcal{U}}$$

if $\{\lambda \in \Lambda \mid x_\lambda \le \xi_\lambda\} \in \mathcal{U}$.

Proposition 1.34. *Suppose* $X_\lambda \ne \emptyset$ *for all* $\lambda \in \Lambda$.

1. *If* $\mathcal{U} = \{\Theta \subseteq \Lambda \mid \lambda_0 \in \Theta\}$ *is the principal ultrafilter belonging to* λ_0, *then* $\prod_{\mathcal{U}} X_\lambda = X_{\lambda_0}$.

2. *If* F *is a finite set and all* $\lambda \in \Lambda, X_\lambda = F$. *Then* $\prod_{\mathcal{U}} F = F$ *as for an arbitrary* $x \in \prod_{\mathcal{U}} F$, $\Lambda = \bigcup_{f \in F}\{\lambda \mid x_\lambda = f\}$ *(i.e. disjoint finite union) hence there is precisely one* $f \in F$ *such that* $\{\lambda \mid x_\lambda = f\} \in \mathcal{U}$.

3. *If* $X_\lambda = X'_\lambda \cup X''_\lambda$ *(disjoint unions,* $\forall \lambda \in \Lambda, X'_\lambda, X''_\lambda \ne \emptyset$*) Then* $\prod_{\mathcal{U}}(X'_\lambda \cup X''_\lambda) = (\prod_{\mathcal{U}} X'_\lambda) \cup (\prod_{\mathcal{U}} X''_\lambda)$ *which is also a disjoint union as* $\Lambda = \{\lambda \in \Lambda | x_\lambda \in X'_\lambda\} \cup \{\lambda \in \Lambda \mid x_\lambda \in X''_\lambda\}$.

4. *Then* $\prod_{\mathcal{U}}(\Gamma_\lambda \cup \{0\}) = (\prod_{\mathcal{U}} \Gamma_\lambda) \cup \{0\}$ *and* $\prod_{\mathcal{U}} \Gamma_\lambda$ *is an ordered abelian group where all* Γ_λ *are ordered abealian groups.*

Remark 1.24. *When some of the* X_λ *may be empty, it is better to define* $\prod_{\mathcal{U}} X_\lambda$ *as the set of equivalent classes of pairs* (M, ξ) *where* $M \in \mathcal{U}$ *is a subset and* ξ *a function defined on* M *such that* $\xi(\lambda) \in X_\lambda$ *where* $\lambda \in M$. *Two pairs* $(M, \xi), (N, \nu)$ *are equivalent if*

$$\{\lambda \in M \cap N \mid \xi(\lambda) = \nu(\lambda)\} \in \mathcal{U}.$$

With this, we define,

$$\prod_{\mathcal{U}} \left(\bigcup_{i=1}^m X_\lambda^{(i)} \right) \cong \bigcup_{i=1}^m \prod_{\mathcal{U}} X_\lambda^{(i)}.$$

Remark 1.25. $\prod_{\mathcal{U}}$*(ordered abelian group) is again an ordered abelian group and*

$$\prod_{\mathcal{U}} (\Gamma_\lambda \cup \{0\}) \cong \left(\prod_{\mathcal{U}} \Gamma_\lambda \right) \cdot \{0\}.$$

Proposition 1.35. *For every ring R, $\mathrm{Spv}(R)$ is a spectral space and a subset of the form $\mathcal{R}(f_0|f_1, \cdots, f_m)$ with $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in R^{m+1}$ is quasi-compact.*

The generic limit of the ultrafiler $\mathcal{U}$ on $\mathrm{Spv}\, R$ is given by the equivalence class of the valuation,

$$R \overset{r \mapsto \left(\substack{\text{the } \mathcal{U}\text{-equivalence class of} \\ (|r|_\gamma)_{\gamma \in \mathrm{Spv}\, R}}\right)}{\longrightarrow} \left(\left(\prod_{\gamma \in \mathrm{Spv}\, R}\right)_{\mathcal{U}} \Gamma_\gamma\right) \cup \{0\} = \prod_{\mathcal{U}} (\Gamma_\gamma \cup \{0\}). \tag{1.3}$$

Moreover, the map,

$$\mathrm{Spv}(R) \overset{\gamma \mapsto \mathrm{supp}\, |\cdot|_\gamma}{\longrightarrow} \mathrm{Spec}\, R$$

is spectral and for every ring homomorphism $R \overset{\rho}{\to} S$, the map

$$\mathrm{Spv}(S) \overset{\mathrm{Spv}(\rho)}{\to} \mathrm{Spv}(R)$$

is also spectral.

Proof. It is obvious from the definition and the properties of ultrafilters that (1.3) is indeed a valuation. To show that it defines a generic limit of $\mathcal{U}$ and that the rational open subsets are quasi-compact, we use Lemma 1.1. Set

$$\mathcal{B} = \{\mathcal{R}(f_0|f_1, \cdots, f_m) \mid m \in \mathbb{N}, (f_i)_{i=0}^m \in R^{m+1}\}.$$

if g denotes the equivalence class of (1.3),

$$\begin{aligned}
[g \in \mathcal{R}(f_0|f_1, \cdots, f_m)] &\Leftrightarrow |f_0|_g \neq 0 \text{ and all } |f_1|_g \leq |f_0|_g, \\
&\overset{\text{By (1.3)}}{\Leftrightarrow} (|f_0|_x)_{x \in X} / \sim_{\mathcal{U}} \neq 0 \text{ and } (|f_i|_x)_{x \in X} / \sim_{\mathcal{U}} \leq (|f_0|_x)_{x \in X} / \sim_{\mathcal{U}}, \\
&\overset{\substack{\text{Definition of } \prod_{\mathcal{U}} \\ \text{and the valuation on it}}}{\Leftrightarrow} \{x \in X \mid |f_0|_x \neq 0\} \in \mathcal{U} \text{ and all } \{x \in X \mid |f_i(x)| \leq f_0(x)\} \in \mathcal{U}, \\
&\overset{\bigcap_{i=0}^m M_i \in \mathcal{U} \Leftrightarrow \forall M_i \in \mathcal{U}}{\Leftrightarrow} \{x \in X \mid |f_0|_x \neq 0\} \cap \bigcap_{i=1}^m \{x \in X \mid |f_i(x)| \leq |f_0(x)|\} \in \mathcal{U}, \\
&\overset{\text{Definition of rational open}}{\Leftrightarrow} \mathcal{R}(f_0 \mid f_1, \cdots, f_m) \in \mathcal{U}.
\end{aligned}$$

Lemma 1.1 can indeed be applied. That the induced map $\mathrm{Spv}(R) \overset{\mathrm{supp}}{\to} \mathrm{Spec}(R)$ is spectral follows from

$$\mathrm{supp}(D(f)) = \mathcal{R}(f|).$$

By Proposition 1.17, that $\mathrm{Spv}(\rho)$ is spectral follows immediately from

$$\mathrm{Spv}(\rho)^{-1}\mathcal{R}(f_0|f_1, \cdots, f_m) = \mathcal{R}(\rho(f_0)|\rho(f_1), \cdots, \rho(f_m)).$$

□

Proposition 1.36. *For every ideal $I \subseteq R$, if $R \overset{\pi}{\to} R/I$ denotes the canonical projection, then the induced map $\mathrm{Spv}(R/I) \overset{\pi}{\to} \mathrm{Spv}(R)$ is a homeomorphism. Explicitly, we have,*

$$\mathrm{Spv}(R/I) = V_{\mathrm{Spv}(R)}(I) := \{x \in \mathrm{Spv}(R) \mid \mathrm{supp}\, x \supseteq I\}, \tag{1.4}$$

$$= \mathrm{Spv}(R) \backslash \bigcup_{f \in I} \mathcal{R}(f). \tag{1.5}$$

If I is finitely generated (i.e. $I = \langle g_1, \cdots, g_m \rangle_R$), then the right hand side of Equation (1.5) *can be written as,*

$$\mathrm{Spv}(R/I) = \bigcup_{i=1}^m \mathcal{R}(g_i)$$

which is the embedded subset. Alternatively, we have,

$$V_{\mathrm{Spv}(R)} = \{x \in \mathrm{Spv}(R) \mid \forall f \in I, |f|_x = 0\}.$$

Furthermore, we can identify this with a homeomorphism,

$$\mathrm{Spv}(R/I) \underset{\simeq}{\overset{\mathrm{Spv}(\pi)}{\to}} V(I).$$

Let S be a multiplicative subset of S and $\iota : R \to R_S$ be a canonical inclusion then,

$$\mathrm{Spv}(R_S) \underset{\mathrm{Spv}(\iota)}{\overset{\simeq}{\to}} \bigcap_{s\in S} (\mathrm{Spv}(R)\backslash V(s)) = \bigcap_{s\in S} \mathcal{R}(s|).$$

Proposition 1.37. *Let s be a specialization of $\eta \in \mathrm{Spv}(R)$, then*

$$\operatorname{supp} \eta \subseteq \operatorname{supp} s.$$

Proof.

$$r \notin \operatorname{supp} s \iff s \in \mathcal{R}(r|) \Rightarrow \eta \in \mathcal{R}(r|) \iff r \notin \operatorname{supp} \eta.$$

□

Definition 1.41. *Let $\gamma \in \Gamma \cup \{0\}$ and $R \overset{v}{\to} \Gamma \cup \{0\}$ be a valuation. We then define,*

$$K_{v\leq\gamma} := \{r \in R \mid v(r) \leq \gamma\}.$$

For an equivalence class $x \in \mathrm{Spv}(R)$ where $v \in x$, we may denote,

$$K_{x\leq\gamma} = K_{v\leq\gamma}.$$

For $\rho \in R$, we denote,

$$K_{v\leq\rho} := K_{v\leq v(\rho)}, K_{x\leq\rho} := K_{x\leq|\rho|_x}.$$

Similarly,

$$\begin{aligned} K_{v<\gamma} = K_{x<\gamma} &:= \{r \in R \mid v(r) < \gamma\} \\ K_{v<\rho} = K_{x<\rho} &:= K_{v<v(\rho)}. \end{aligned}$$

Proposition 1.38. *$K_{v\leq1}$ is an integrally closed subring of R, and $K_{v<1}$ is a prime ideal in $K_{v\leq1}$.*

Proof. These being a subring and an ideal can be checked easily. Let $r \notin K_{r\leq1}$, then if m is a positive integer and $(p_i)_{i=1,\cdots,m} \in K^m_{v\leq1}$, we have,

$$\begin{aligned} v(r^m) = v(r)^m&, \\ &> v(r)^{m-1} \geq \max_{1\leq i\leq m-1} v(p_i)v(r)^i, \\ &= \max_{1\leq i\leq m-1} v(p_i r^i), \\ &\geq v\left(\sum_{i=1}^{m-1} p_i r^i\right). \end{aligned}$$

Hence,

$$r^m \neq \sum_{i=1}^{m-1} p_i r^i.$$

Thus r is not integral over $K_{v\leq1}$, in particular, $K_{v\leq1}$ is integrally closed.

For $K_{v<1}$ being prime, obviously $1 \notin K_{v<1}$. Let $a, b \in K_{v\leq1}\backslash K_{v<1}$. Then $v(a) = v(b) = 1$. Therefore $v(ab) = 1$. Showing $K_{v<1}$ is a prime ideal. □

Proposition 1.39. *Let s be a specialization of $\eta \in \mathrm{Spv}(R)$. We have,*

$$a \in R \backslash \operatorname{supp} s \Rightarrow K_{\eta<a} \subseteq K_{s<a}.$$

Applying this to $a = 1$, (recall that $\operatorname{supp} s$ is a prime ideal), we get,

$$K_{\eta<1} \subseteq K_{s<1}.$$

Proof. By the assumption and Proposition 1.37, we know that $|a|_\eta \neq 0$. Therefore,

$$b \notin K_{\eta<a} \iff \eta \notin \mathcal{R}(b|a) \Rightarrow s \notin \mathcal{R}(b|a) \iff b \notin K_{s<a}.$$

$\square$

Proposition 1.40. *Let s be a specilization of $\eta \in \mathrm{Spv}(R)$.*

$$a \in R \backslash \operatorname{supp}(s) \Rightarrow K_{s\leq a} \subseteq K_{\eta\leq a}.$$

In particular for $a = 1$, we get,

$$K_{s\leq 1} \subseteq K_{\eta\leq 1}.$$

Proof. By Proposition 1.37, $|a|_\eta \neq 0$. Hence,

$$b \in K_{s\leq a} \iff s \in \mathcal{R}(a|b) \Rightarrow \eta \in \mathcal{R}(a|b) \iff b \in K_{\eta\leq a}.$$

$\square$

Definition 1.42. *Let X be a linearly ordered set. For $a, b \in X$ with $a \leq b$, we define the line segment between a and b to be*

$$[a, b]_X := \{x \in X \mid a \leq x \leq b\}.$$

A subset $C \subseteq X$ is convex if for any two elements $c, d \in C, c \leq d$, we have $[c, d]_X \subseteq C$.

Definition 1.43. *A convex subgroup Θ is a subgroup of a linearly ordered group Γ which is convex.*

Proposition 1.41. *A subgroup Θ of a linearly ordered group Γ is convex if and only if for any $\theta \in \Theta$ with $\theta < 1$, $[\theta, \theta^{-1}]_\Gamma \subseteq \Theta$.*

Recall that for a valuation $v : R \to \Gamma \cup \{0\}$, Γ_v is the subgroup of Γ generated by $v(R)\backslash\{0\}$.

Definition 1.44. *$c\Gamma_v$ is the convex subgroup of Γ_v generated by $\{v(r) \mid r \in R, v(r) \geq 1\}$.*

Remark 1.26. *Equivalently, we can define $c\Gamma_v$ to be a subgroup generated by $\{v(r) \mid r \in R, v(r) > 1\}$.*

Proposition 1.42.

$$c\Gamma_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \min\{\gamma, \gamma^{-1}\}v(r) \geq 1\}$$

Proof. Let $c\Gamma'_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \min\{\gamma, \gamma^{-1}\}v(r) \geq 1\}$. We will show that $c\Gamma_v = c\Gamma'_v$. If $r \in R$ is as in the construction of $c\Gamma'_v$, then $v(r) \geq 1$ thus $v(r) \in c\Gamma_v$. If $\gamma \in c\Gamma'_v$, then

$$v(r)^{-1} \leq \gamma \leq v(r).$$

Hence $\gamma \in c\Gamma_v$. In particular $c\Gamma'_v \subseteq c\Gamma_v$. On the other hand $c\Gamma'_v$ is a convex subgroup of Γ_v containing all $v(r)$ such that $v(r) \geq 1$. Thus by the minimality of $c\Gamma_v$, we have shown the statement. $\square$

Remark 1.27. *Even if H is a convex subgroup of Γ_v, it does not imply that $c\Gamma_v \subseteq H$.*

Remark 1.28. *If R is a field then $c\Gamma_v = \Gamma_v$.*

Definition 1.45. *Let $A \xrightarrow{v} \Gamma$ be a valuation of A and $H \subseteq \Gamma_v$ a convex subgroup containing $c\Gamma_v$. Let*

$$v|_H(r) := \begin{cases} v(r) & (v(r) \in H) \\ 0 & (\textit{otherwise}). \end{cases}$$

Proposition 1.43. *Under the assumption of Definition 1.42, $v|_H$ is indeed a valuation of A and $v|_H$ is a specialization of v in* $\mathrm{Spv}(A)$.

Proof. Let M be the convex hull of $\{0\} \cup H$ in $\Gamma_v \cup \{0\}$. For $m \in M$, set,

$$\varphi(m) = \begin{cases} m, & (m \in H), \\ 0, & (\text{otherwise}). \end{cases}$$

Since $H \supseteq c\Gamma_v$, we have $v(a) \in M$ for all $a \in A$. (note that $[0,1]_{\Gamma_v \cup \{0\}} \supseteq M$ and $v(a) \in c\Gamma_v \subseteq H \subseteq M$ where $v(a) \geq 1$) Set $w := v|_H = \varphi v$. Since v is monotonic, $\varphi(0) = 0$, and $\varphi(1) = 1$, the axioms

$$w(0) = 0, w(1) = 1, w(a+b) \leq \max\{w(a), w(b)\}$$

follows from their counterparts for v. Also M is multiplicativly closed and $\varphi(m\mu) = \varphi(m)\varphi(\mu)$. (This is obvious when $\varphi(m)\varphi(\mu) \neq 0$. Otherwise, without loss of generality $\varphi(m) = 0$, hence $m \notin H$. In particular, $m < 1$. Also $hm < 1$ for all $h \in H$ by the convexity of H. Thus, $nm < 1$ for all $n \in M$, in particular $m\mu < 1$. Now $m\mu \in H$ would imply $\mu \in H$ as $m\mu \leq \mu$ and $\mu \in M$ (hence μ is bounded from above by some $h \in H$). But then $m \in H$ a contradiction).

To see w is a specialization of r, observe that

$$\begin{aligned} w \in \mathcal{R}(f_0|f_1, \cdots, f_m) &\iff [w(f_0) \neq 0 \text{ and } w(f_0) \geq w(f_i), i = 1, \cdots, m], \\ &\iff [v(f_0) \in H \text{ and } v(f_0) \geq w(f_i), i = 1, \cdots, m], \\ &\iff [v(f_0) \in H \text{ and } v(f_0) \geq v(f_i), i = 1, \cdots, m], \\ &\iff [v(f_0) \neq 0 \text{ and } v(f_0) \geq v(f_i), i = 1, \cdots, m] \\ &\iff (v/\sim) \in \mathcal{R}(f_0|f_1, \cdots, f_m). \end{aligned}$$

Note that $v(f_i) > v(f_0) \in H$ would imply $v(f_i) \in H$ by the convexity of H, as $v(f_i) \in H$, hence $v(f_i)$ bounded from above by some elements of H. Hnce $w(f_i) = v(f_i) > v(f_0) = w(f_0)$ if $v(f_i) > v(f_0)$. ☐

Remark 1.29. *Motivated by the above notion, we say $s \in \mathrm{Spv}(R)$ is a horizontal specialization of $\eta \in \mathrm{Spv}(R)$ if there is $v \in \eta$ and a convex subgroup $c\Gamma_v \subseteq H \subseteq \Gamma_v$ such that s is equivalent to $v|_H$.*

Corollary 1.10. *If x is a closed point of* $\mathrm{Spv}(A)$*, then $c\Gamma_x = \Gamma_x$.*

Proof. If this is not the case, $x|_{c\Gamma_x}$ is a proper specialization of x. ☐

Proposition 1.44. *If $s \in \mathrm{Spv}(A)$ is a horizontal specialization of $\eta \in \mathrm{Spv}(A)$ and $\Omega \in \mathcal{R}(f_0|f_1, \cdots, f_m)$ then $s \in \Omega \iff (\eta \in \Omega$ and $|f_0|_s \neq 0)$.*

Proof. Indeed, the construction of the end of the proof of Proposition 1.43 imply

$$(w/\sim) \in \Omega \iff [(v/\sim) \in \Omega \text{ and } w(f_0) \neq 0].$$

☐

Definition 1.46. *Let $A \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation and a convex subgroup $H \subseteq \Gamma$ or $H \subseteq \Gamma_v$. The vertical specialization of v with respect to H is*

$$(v/H)(a) = \begin{cases} v(a)H, & (v(a) \neq 0), \\ 0, & (\textit{ otherwise }), \end{cases}$$

where $v(a)H$ denotes the coset represented by $v(a)$.

Remark 1.30. *s is a vertical specilization of $\eta \in \mathrm{Spv}(A)$ if and only if s is a specialization of η and* $\mathrm{supp}\, s = \mathrm{supp}\, \eta$.

Remark 1.31. *In Huber's literature, he refers horizontal specialization as primary specialization and the vertical specializations as the secondary specializations.*

Remark 1.32. *Note that $v|_H$ is a specialization of v while v is a specialization of v/H.*

1.5 Valuation Rings

Definition 1.47. *Let K be a field. A subring $R \subseteq K$ is a valuation ring of K if for all $x \in K^\times, x \in R$ or $x^{-1} \in R$.*

Example 1.11. *Every discrete valuation rings is a valuation ring of its field of fraction. More concretely, $\mathbb{Z}_p$ is a valuation ring of $\mathbb{Q}_p$.*

Example 1.12. *If S is a valuation ring of L and $K \subseteq L$ is a subfield then $S \cap K$ is a valuation ring of K.*

Example 1.13. *If $\mathcal{L}$ is a set of valuation rings which is linearly ordered by $\subseteq$. Then $\bigcap_{R\in\mathcal{L}} R$ is a valuation ring of K (if $\mathcal{L} = \emptyset$ then the intersection would be K itself which is trivially a valuation ring) and if $\mathcal{L} \neq \emptyset$ then $\bigcup_{R\in\mathcal{L}} R$ is also a valuation ring of K.*

Example 1.14. *If R is a valuation ring of K and $R \subseteq S \subseteq K$ then S is a valuation ring of K.*

Proposition 1.45. *Let R be a valuation ring of K. We have the following.*

a). The field of fractions of R is K.

b). For $a, b \in R$, either $a \mid b$ or $b \mid a$.

c). R is a Bézout domain. That is every finitely generated ideals are principal.

d). R is integrally closed in K.

e). The following conditions are equivalent for R

- *i). R is Noetherian.*
- *ii). R is a discrete valuation ring.*
- *iii). The unique maximal ideal $\mathfrak{m}_R$ is principal.*
- *iv). R is a principal ideal domain.*

f). The set of ideals of R is linearly ordered by the inclusion.

g). $\mathrm{Spv}(R)$ is linearly orded by specizalization.

h). An ideal $I \subseteq R$ is prime if and only if $I = \sqrt{I}$ and $1 \notin I$.

i). We have an order reversing bijection such that

$$\mathrm{Spec}(R) \underset{R\cap\mathfrak{m}_S \leftarrow\!\shortmid S}{\overset{\mathfrak{p}\mapsto R_\mathfrak{p}}{\rightleftarrows}} \{S \mid R \subseteq S \subseteq K \text{ and } S \text{ is a subring of } K\}$$

j). The set of S such that $R \subseteq S \subseteq K$ and S is a subring of K is linearly ordered by inclusion.

Proposition 1.46. *If R and S are valuation ring of K then $R \subseteq S$ is equivalent to $\mathfrak{m}_S \subseteq \mathfrak{m}_R$.*

Proof.

$$\mathfrak{m}_R = \{0\} \cup \{x \in K^\times \mid x \notin R\}.$$

$\square$

Proposition 1.47. *We have following bijections,*

$$\mathrm{Spv}(K) \underset{v_R/\sim \leftarrow\!\shortmid R}{\overset{x\mapsto\{k\in K \mid |k|_x\leq 1\}}{\rightleftarrows}} \{\text{valuation rings } R \text{ of } K\}$$

where

$$v_R(x) = \begin{cases} x \cdot R^\times, & (x \neq 0), \\ 0, & (x = 0). \end{cases} \quad , \Gamma_R = K^\times / R^\times.$$

The ordering on Γ_R is given as follows,

$$xR^\times \leq yR^\times \iff x \in yR.$$

Remark 1.33. *Under the bijection in Proposition 1.47, if the embedding of K into a field extension L is denoted i, $\mathrm{Spv}(i)(x_R) = x_{K\cap R}$ where R is a valuation ring of L, $x_R \in \mathrm{Spv}(L)$. Moreover, $x_R \in \mathcal{R}(f_0|f_1, \cdots, f_m)$ if and only if $f_0 = 0$ and $f_i \in f_0\mathcal{R}$.*

Proposition 1.48. *x_R be a specialization of x_S if and only if $R \subseteq S$.*

Proof.
If $R \subseteq S$, then by Remark 1.33 and

$$\begin{aligned} x_R \in \mathcal{R}(f_0|f_1, \cdots, f_m) &\iff [f_0 \neq 0 \text{ and } \forall 1 \leq i \leq m, f_i \in f_0 R] \\ &\Rightarrow [f_0 \neq 0 \text{ and } \forall f_i \in f_0 S], \\ &\iff [x_S \in \mathcal{R}(f_0|f_1, \cdots, f_m)]. \end{aligned}$$

If x_R is a specialization of x_S, then $R = K_{x_R \leq 1} \subseteq K_{x_S \leq 1} = S$ by Proposition 1.40. ☐

Lemma 1.2. *Let B be an A-algebra and $M \subseteq B$ be a finitely generated A-submodule of B such that $bM = 0$ for any $b \in B$. Then it implies $aM = 0$. If $b \in B$ is such that $bM \subseteq M$ then b is integral over A.*

Proof. Let $(m_i)_{i=1}^l$ be generators of M as an A-module.

$$bm_i = \sum_{i=1}^{l} \alpha_{ij} m_j.$$

Set $p \in A[T]$ be the characteristic polynomial of (α_{ij}). Then $p(b) = 0$ by Cauchy-Hamilton. ☐

Proposition 1.49. *Let A be a subring of the filed K. Then the integral closure A in K is equal to the intersection of the valuation rings of K containing A.*

Proof. Let A' be the intersection of all valuation rings containing A. $A' \supseteq A$ is clear as all valuation rings are integrally closed. Thus it is sufficient to consider an element $x \in K$ which is not integral over A and show there is a valuation ring R of K which doesn't contain x. To this end, set

$$\mathcal{M} := \{A \subseteq B \subseteq K \mid B \text{ is an intermediate ring with } x \text{ is not integral over } B\}.$$

Since $A \in \mathcal{M}$, $\mathcal{M} \neq \emptyset$. Let $\mathcal{L} \subseteq \mathcal{M}$ be a non-empty set which is linearly ordered with respect to inclusion. Then $C = \bigcup_{B\in\mathcal{L}} B$ is in $\mathcal{M}$. That is $\mathcal{L}$ is bounded in $\mathcal{M}$. By Zorn's lemma, there is a maximal element R. It remains to show that R is a valuation ring.

Let $k \in K \backslash R$ such that $k^{-1} \notin R$. Then x is integral over $R[x^\pm]$, hence there are polynomials $(p_j^\pm)_{j=0}^{d_\pm - 1}$ such that

$$x^{d_\pm} = \sum_{j=0}^{d_\pm - 1} x^j p_j^{(\pm)}(k^{\pm 1}).$$

Let $d = \max\{d_+, d_-\}$ and e the maximum of all $\deg p_j^{(\pm)}$ with $0 \leq j < d^{(\pm)}$. Let $N \subseteq K$ be the R-submodule generated by the $x^i k^j$ where $0 \leq i < d, 0 \leq j < 2e$. Then N is a finitely generated R-module and $xN \subseteq N$. Indeed, consider a generator $y = x^i k^j$ of N. We can assume that $j < e$. If $i < d_+ - 1$ then xy is in the set generators of N. Otherwise,

$$xy = x^{i+1-d_+} \sum_{l=0}^{d_+ - 1} x^l p_l^{(+)}(k) k^j \in N.$$

Now assume $j \geq e$. If $i < d-1$, then xy is in the set of generators of N. Otherwise,

$$xy = x^{i+1-d_-} \sum_{l=0}^{d_- -1} x^l k^j p_l^{(-)}(k^{-1}) \in N.$$

By Lemma 1.2, x is integral over R. This contradicts that $R \in \mathcal{M}$. $\square$

Remark 1.34. *The book of Kniebuch and Schrecker also has a proof.*

2 G_+-topological span and their sheaf cohomology

2.1 Grothendieck topologies and G_+-topological span

Let $\mathscr{C}$ be a category. In the application we are interested in, it will be given by a poset $(P, \preceq)$. Objects of $\mathscr{C}$ are the all of P and a morphism $x \xrightarrow{\mu} x$ exists if and only if $x \preceq y$, in which case μ is unique. In fact $P = \mathscr{O}_X$ (all open subsets of X) or P is some topology base of X, and $\preceq$ is containment $\subseteq$. However, the case where therer may be more than one morphism between objects of $\mathscr{C}$ are very important in other applications like étale or crystalline cohomology.

Definition 2.1. *A sieve on (or over) an object X of $\mathscr{C}$ or an X-sieve is a class $\mathscr{S}$ of morphisms $U \to X$ such that $[U \xrightarrow{\nu} X] \in \mathscr{S}$ then $\nu\omega \in \mathscr{S}$ where $V \xrightarrow{\omega} U$ is any morphism in $\mathscr{C}$.*

Definition 2.2. *If $\mathscr{S}$ is a X-sieve and $Y \xrightarrow{\xi} X$ then $\xi^*\mathscr{S}$ is the Y-sieve of all morphism $U \xrightarrow{\nu} Y$ such that $\xi\nu \in \mathscr{S}$.*

Definition 2.3. *A Grothendieck topology on $\mathscr{C}$ is a collection $(\mathcal{T}_X)_{X \in \mathrm{Ob}(\mathscr{C})}$ of X-sieves (where $\mathcal{T}_X$ being called the class of covering sieves of X) satisfying the following conditions.*

I). (GT-Triv) The all-sieve $\mathscr{S}$ on X (i.e. the class of all morphisms with its target to X) is in $\mathcal{T}_X$.

II). (GT-Trans) If $\mathscr{S} \in \mathcal{T}_X$, then for any morphism $Y \xrightarrow{\nu} X$ in $\mathscr{C}$, we have $\nu^*\mathscr{S} \in \mathcal{T}_Y$.

III). (GT-Loc) Let $\mathscr{T}$ be a X-sieve. If there is $\mathscr{S} \in \mathcal{T}_X$ such that $\nu^*\mathscr{T} \in \mathcal{T}_U$ for any morphism $(U \xrightarrow{\nu} X) \in \mathscr{S}$, then $\mathscr{T} \in \mathcal{T}_X$.

Remark 2.1. *Recall that a presheaf on $\mathscr{C}$ is a contravariant functor $\mathscr{C} \xrightarrow{\mathcal{G}} (Sets)$. The idea is that $\mathcal{G}$ is a sheaf if and only if $\mathcal{G}(X) \to \lim_{\mathscr{S}} \mathcal{G}$ where morphisms in $\mathscr{S}$ are morphisms of X-objects, is an isomorphism for all $\mathscr{S} \in \mathcal{T}_X$.*

Remark 2.2. *If $\mathscr{C}$ is given by a poset, then $\mathscr{S}$ is also a poset (of $\preceq$-predecessor of X) and*

$$\varprojlim_{\mathscr{S}} \mathcal{G} = \left\{ (g_Y)_{Y \in \mathscr{S}} \in \prod_{Y \in \mathscr{S}} \mathcal{G}(Y) \,\middle|\, g_Y|_Z = g_Z \text{ where } Z \preceq Y \right\}.$$

Here $\mathcal{G}$(the unique morphisms $Z \to Y$) are denoted by $\cdot|_Z$.

Proposition 2.1.

1. *If $\mathscr{S} \subseteq \mathscr{T}$ and $\mathscr{S} \in \mathcal{T}_X$, then $\mathscr{T} \in \mathcal{T}_X$.*
2. *If $\mathscr{S}_1, \mathscr{S}_2 \in \mathcal{T}_X$, then $\mathscr{S}_1 \cap \mathscr{S}_2 \in \mathcal{T}_X$.*

Proof. If $[U \xrightarrow{\nu} X] \in \mathscr{S}$,

$$\nu^*\mathscr{S} = \{\sigma : V \to U \mid \nu\sigma : V \to X \in \mathscr{S}\}.$$

Thus for any $\sigma : V \to U \in \mathscr{C}$, $\sigma \in \nu^*\mathscr{S}$ by using the definition of sieves. Thus $\nu^*\mathscr{S}$ is the all-sieve on U. Observing that $\nu^*\mathscr{T} \supseteq \nu^*\mathscr{S}$, hence $\nu^*\mathscr{T} =$ (all sieve on U) $\in \mathcal{T}_U$ by GT-Triv. By GT-Loc, $\mathscr{T} \in \mathcal{T}_X$.

Let $\mathscr{T} = \mathscr{S}_1 \cap \mathscr{S}_2$. If $[U \xrightarrow{\nu} X] \in \mathscr{S}_1$, as in the previous argument $\nu^*\mathscr{S}_1$ is the all-sieve, thus $\nu^*\mathscr{T} = \nu^*\mathscr{S}_2 \in \mathcal{T}_U$. By GT-Trans and $\nu^*\mathscr{S}_2 = \mathscr{T} \in \mathcal{T}_X$. By GT-Loc $\mathscr{T} \in \mathcal{T}_X$. $\square$

From now on, $\mathscr{C}$ will be given by some topology base $\mathcal{B}$ in a topology space X turned into a poset by the inclusion. We then give a more specific definition for sieves in $\mathscr{C}$.

Definition 2.4. *Let $\mathscr{C}$ be a category whose objects are open subsets of X and morphisms are inclusions.*

1). A sieve $\mathscr{S}$ on $U \in \mathscr{C}$ is a set of open subsets of U such that for any $V \in \mathscr{S}$, every open subsets W of V is also in $\mathscr{S}$.

2). Let $V \subseteq U$ and $\mathscr{S}$ be a sieve on U, then $\mathscr{S}|_V = \mathscr{S} \cap \{$open subsets of $V\}$.

Definition 2.5. *Let X be a topological space and $\mathcal{B}_X$ be the topology base of X. Let $(U_i)_{i\in I}$ be a collection of elements of $\mathcal{B}_X$. Then,*

$$\begin{aligned}[U_i|i \in I] &:= \bigcap_{\substack{X\text{-}sieve\ \mathscr{S} \\ \text{s.t. } \forall i \in I, U_i \in \mathscr{S}}} \mathscr{S}, \\ &= \{U \in \mathcal{B}_X \mid \exists i \in I \text{ s.t. } U \subseteq U_i\}.\end{aligned}$$

Remark 2.3. *In order to see the two definitions given above note that the latter is a sieve and contains all U_i and indeed a sieve on X. Note that this is a sieve in the case I is finite by Proposition 2.1.*

Definition 2.6. *A W-sieve is finitely generated if it has the form $[\Omega_1, \cdots, \Omega_n]$ where all $\Omega \in \mathcal{B}_W$. If several topology bases are under consideration, we will use $[\Omega_1, \cdots, \Omega_n]_{\mathcal{B}}$ for disambiguation.*

Let $U \in \mathscr{O}_X$. U-sieve $\mathscr{S}$ is admissible (denoted by $\mathscr{S} \vDash U$) by the Grothendieck group $(\mathcal{T}_U)_{U\in\mathscr{O}_X}$ if $\mathscr{S} \in \mathcal{T}_U$. Also for an arbitrary sieve $\mathscr{S}$, we write $\mathscr{S}//U$ if $\mathscr{S}|_U \vDash U$.

Proposition 2.2. *Let X be a topological space and $\mathcal{B}$ be a topology base for it. Then we have a bijection between the following two sets.*

1). Grothendieck topologies $\mathcal{T}$ on $\mathcal{B}$.

2). Grothendieck topologies $\mathcal{T}$ on $\mathscr{O}_X$ with the additional property that $[\mathcal{B}_U] \in \mathcal{T}_U$ holds for all $U \in \mathscr{O}_X$.

The bijection is explicitly given by the following way.

a). If a Grothendieck topology $\mathcal{T}^{(\mathcal{B})}$ on $\mathcal{B}$ is given, $\mathscr{S} \in \mathcal{T}_U$ if and only if $\mathscr{S} \cap \mathcal{B}_\Omega \in \mathcal{T}_\Omega^{(\mathcal{B})}$ holds for all $\Omega \in \mathcal{B}_U$.

b). If a Grothendieck topology $\mathcal{T}$ on $\mathscr{O}_X$ is given, $\mathscr{S} \in \mathcal{T}_\Omega^{(\mathcal{B})}$ if and only if $[\mathscr{S}] \in \mathcal{T}_\Omega$, (where $[\mathscr{S}] = [\Theta \mid \Theta \in \mathscr{S}]$ is the sieve on $\mathscr{O}_\Omega$ generated by $\mathscr{S}$).

Proof. Ommited. Hint : it becomes easier if it is assumed that $\Omega, \Theta \in \mathcal{B} \Rightarrow \Omega \cap \Theta \in \mathcal{B}$. $\square$

Definition 2.7. *A G_+-topological space is a T_0-space X with a Grothendieck topology on $\mathscr{O}_X$ such that*

1. *$\emptyset \vDash \emptyset$ (implies $\mathcal{G}(\emptyset) = \{0\}$ for sheaves $\mathcal{G}$ of abelian groups).*
2. *If $\mathscr{S} \vDash U$ then $U = \bigcup_{V\in\mathscr{S}} V$.*

Definition 2.8. *If X is a G_+-space then a G_+-base for X is a subset $\mathcal{B} \subseteq \mathscr{O}_X$ such that $\mathcal{B}$ is a topology base in the ordinary sense and $[\mathcal{B}_U] \vDash U$ for all $U \in \mathscr{O}_U$. We say that $\mathcal{B}$ is G_{++} if in addition,*

a). $\Omega, \Theta \in \mathcal{B} \Rightarrow \Omega \cap \Theta \in \mathcal{B}$.

b). Membership in $\mathcal{B}$ is local in the sense that $\Omega \in \mathcal{B}, U \in \mathscr{O}_\Omega$ such that $U \subseteq \Omega$ and $[\Theta \in \mathcal{B}_\Omega \mid U \cap \Theta \in \mathcal{B}] \vDash \Omega$ then $U \in \mathcal{B}$.

Definition 2.9. *If X is a G_+-space then $V \in \mathscr{O}_X$ is called G_+-quasicompact (abbreviated as (G_{+qc})) if for all $\mathscr{S} \in \mathcal{T}_U$, there are finitely many $(V_i)_{i=1,\cdots,n}$ such that*

$$[V_1, \cdots, V_n] \vDash V.$$

(i.e. every covering sieve of U has a finitely generated covering subsieve).

Example 2.1. *If X is a T_0-space it gets the structure of a G_+-space by*

$$\mathcal{T}_U = \left\{ \text{sieves } \mathscr{S} \text{ in } U \text{ such that } U = \bigcup_{V \in \mathscr{S}} V \right\}.$$

Proposition 2.3. *Let X be a T_0-space, $\mathcal{B}$ a topology base for it such that $\Omega, \Theta \in \mathcal{B}$ implies $\Omega \cap \Theta \in \mathcal{B}$. Put*

$$\mathcal{T}_U = \left\{ \text{sieves } \mathscr{S} \text{ on } U \,\middle|\, \forall \Omega \in \mathcal{B}_U, \exists (\Theta_i)_{i=1}^n \in \mathscr{S}^n \text{ such that } \Omega = \bigcup_{i=1}^n \Theta_i \right\}$$

Then X becomes a G_+-topological space and $\mathcal{B}$ is a G_+-base and the elements of $\mathcal{B}$ are G_{+qc}.

Proof. GT-Triv follows since for any open set U, U itself is a subset of U. Let $\mathscr{S}$ be an admissible U-sieve and $\nu : V \hookrightarrow U$. Then as $\mathcal{B}_V \subseteq \mathcal{B}_U$, we have GT-Trans. Let $\mathscr{S}, \mathscr{T}$ be U-sieves such that $\mathscr{S}$ is admissible. Then for $\nu : V \hookrightarrow U \in \mathscr{S}$, $\nu^* \mathscr{T} \in \mathcal{T}_V$ means that $\{\Theta_1, \cdots, \Theta_n\}$ as in the construction exists for each element of $\mathcal{B}_V$ in $\mathscr{T}$. Thus $\mathscr{T} \in \mathcal{T}_U$.

To show that it is a G_+-topological space. First, $\emptyset \vDash \emptyset$ is clear from the definition. The other parts of G_+ for X is also clear. Let $U \in \mathscr{O}_X$ and $\mathscr{S} = [\mathcal{B}_U]$. If $\Omega \in \mathcal{B}_U$ then $\Omega \in \mathscr{S}$ hence $n = 1, \Theta_1 = \Omega$ can be taken.

To show that the membership of $\mathcal{B}$ is local, note that for $\Omega \in \mathcal{B}$ and $U \in \mathscr{O}_X$,

$$[\Theta \in \mathcal{B}_\Omega | U \cap \Theta \in \mathcal{B}] \vDash \Omega$$

is generated in $\mathcal{B}$, thus it implies that

$$\Omega = \bigcup_{i=1}^n \Theta_i \cap U = \Omega \cap U = U.$$

Thus $U \in \mathcal{B}$.

Finally let $\mathscr{S} \vDash \Omega$ for $\Omega \in \mathcal{B}$. If $\Theta_1, \cdots, \Theta_n$ are as in the formulation of the Proposition. Then $\mathscr{T} = [\Theta_1, \cdots, \Theta_n]$ then $\mathscr{T} \vDash \Omega$ as when $\Xi \in \mathcal{B}_\Omega$ and $\Xi_i = \Xi \cap \Theta_i \in \mathcal{B} \cap \mathscr{T}$ by the definition of topology bases and $\Xi = \bigcup_{i=1}^n \Xi_i$. □

Remark 2.4. *The assertion of the Proposition 2.3 will still hold when we relax the condition as follows.*

$$\Omega, \Theta \in \mathcal{B} \Rightarrow \exists n \in \mathbb{N}, \exists (\Xi_i)_{i=1,\cdots,n} \in \mathcal{B}^n \text{ s.t. } \Omega \cap \Theta = \bigcup_{i=1}^n \Xi_i.$$

Definition 2.10. *This is called the G_+-topology on X enforcing the quasicompactness of the elements of $\mathcal{B}$.*

Definition 2.11. *A sieve in $\mathscr{O}_X$ is called a prime sieve if $n \in \mathbb{N}$ and $\bigcap_{i=1}^n U_i \in \mathscr{S}$, (intersection in X and all $U_i \in \mathscr{O}_U$) implies that there is $1 \leq i \leq n$ such that $U_i \in \mathscr{S}$.*

Remark 2.5. *It is sufficient to require this when $n = 0, (X \notin \mathscr{S})$ and $n = 2, (U \cap V \in \mathscr{S} \Rightarrow U \in \mathscr{S}$ or $V \in \mathscr{S})$.*

Proposition 2.4. *Let X be a G_+-space. For a subset $\xi \subseteq \mathscr{O}_X$ the following conditions are equivalent.*

a). ξ satisfies

- *i). $U \in \xi, V \in \mathscr{O}_X, U \subseteq V \Rightarrow V \in \xi$.*
- *ii). $U, V \in \xi \Rightarrow U \cap V \in \xi$ and moreover $X \in \xi$.*
- *iii). If $U \in \xi$ and $\mathscr{S} \in \mathcal{T}_U$ then $\mathscr{S} \cap \xi \neq \emptyset$.*

b). $\mathscr{T} = \mathscr{O}_X \backslash \xi$ is a prime sieve containing every open subset covered by it ($U \in \mathscr{O}_X$ and $\mathscr{T} // U$ implies $U \in \mathscr{T}$).

Proof. The primeness follows from i) and ii). $\square$

Definition 2.12. *We call ξ a van der Put point.*

Definition 2.13. *The set of vander Put points of X is denoted by X^*.*

Proposition 2.5. *Let X be a G_+-space and $U \in \mathscr{O}_U$ be an open subset. Equip U with the G_+-structure $\mathcal{T}^{(U)} = \mathcal{T}|_{\mathscr{O}_U}$, we then have a bijection*

$$\{\text{van der Put poitns } \xi_U \text{ of } U\} \underset{\xi_U = \xi \cap \mathscr{O}_X \leftarrow\!\shortmid \xi}{\overset{\xi_U \mapsto \xi = \{V \in \mathscr{O}_X \mid V \cap U \in \xi_U\}}{\rightleftarrows}} U^* = \{\xi \in X^* \mid U \in \xi\}$$

We have $(U \cap V)^* = U^* \cap V^*$.

Proof. $\xi_U \mapsto \xi$ is justified since $\cdot \cap U$ preserves inclusion and $\cap$ is associative. $\square$

Definition 2.14. *We equip X^* with the topology for which $\{U^* \mid U \in \mathscr{O}_X\}$ is a topology base.*

Proposition 2.6.

a). If $\mathscr{S} \vDash U$ then $U^* = \bigcup_{V \in \mathscr{S}} V^*$. (This follows from Proposition 2.4 a) iii)).

b). If $\mathcal{B}$ is a G_+ base then $\mathcal{B}^* = \{\Omega^* \mid \Omega \in \mathcal{B}\}$ is a topology base for X^*. (Follows from Definition 2.8).

c). $\xi \in X^*$ is a specialization (i.e. $\xi \in \overline{\{u\}}$ of $u \in X^*$) if and only if $\xi \subseteq u$.

d). If $\mathscr{S}$ is a maximal element of the set of sieves on X which are not in $\mathcal{T}_X$. Then $\xi = \mathscr{O}_U \backslash \mathscr{S}$ is a closed point of X^*.

Proof. We show the proof of d). Obviously, $X \notin \mathscr{S}$ by GT-Triv. Let $U_1, U_2 \notin \mathscr{S}$ and set $U = U_1 \cap U_2 \in \mathscr{S}$. By maximality of $\mathscr{S}$, $\mathscr{S}_i = \mathscr{S} \cup \mathscr{O}_{U_i} \in \mathcal{T}_X$. But $\mathscr{S}|_V = \mathscr{S}_2|_V$ for all $V \in \mathscr{S}_1$. Then $\mathscr{S} \in \mathcal{T}_X$ by GT-Loc, a contradiction. Thus $\mathscr{S}$ is a prime sieve. if $\mathscr{S}//V$ then $\mathscr{S} \cup \mathscr{O}_X \notin \mathcal{T}_X$ (otherwise $\mathscr{S} \vDash X$ by GT-Loc). Then $V \in \mathscr{S}$ by maximality of $\mathscr{S}$. By Proposition 2.4, $\xi = \mathscr{O}_X \backslash \mathscr{S}$ is a van der Put point. If ξ has a proper specialization u then $\mathscr{T} = \mathscr{O}_X \backslash u \notin \mathcal{T}_X$ and $\mathscr{T} \supsetneq \mathscr{S}$ a contradition.. $\square$

Definition 2.15. *Let X be a G_+-space. We say X has sufficiently many van der Put points if for all $U \in \mathscr{O}_X$, and every U-sieve $\mathscr{S}$, $U^* = \bigcup_{V \in \mathscr{S}} V^*$ implies $\mathscr{S} \vDash U$.*

Remark 2.6. *Definition 2.15 can be restated by saying that the converse of a) in Proposition 2.6 also holds.*

Lemma 2.1. *If X has sufficently many van der Put points, then it is G_+-quasicompact and if $\mathscr{S}$ is an $\mathscr{O}_X$-sieve such that $X^* = \bigcup_{U \in \mathscr{S}} U^*$ then $\mathscr{S} = X$.*

Proof. Assume that $\mathscr{S}$ is not covering X. Set,

$$\mathcal{M} := \{\mathscr{T} \mid \mathscr{S} \subseteq \mathscr{T}, \mathscr{T} \notin \mathcal{T}_X\}.$$

We apply Zorn's lemma to $(\mathcal{M}, \subseteq)$. This is possible as $\mathcal{M} \neq \emptyset$ because $\mathscr{S} \in \mathcal{M}$ and if $\mathcal{L} \subseteq \mathcal{M}$ is non-empty subset linearly ordered by $\subseteq$, then

$$\widetilde{\mathscr{S}} = \bigcup_{\mathscr{T} \in \mathcal{L}} \mathscr{T}$$

is a sieve containinig $\mathscr{S}$. If $\widetilde{\mathscr{S}} \in \mathcal{T}_X$, then by the quasi-compactness of X, there are $U_1, \cdots, U_n \in \widetilde{\mathscr{S}}$ such that $[U_1, \cdots, U_n] \vDash X$. Then there is $\mathscr{T} \in \mathcal{L}$ such that all $i = 1, \cdots, n, U_i \in \mathscr{T}$ therefore

$$[\Theta_1, \cdots, \Theta_n] \vDash x, [\Theta_1, \cdots, \Theta_n] \subseteq \mathscr{T} \overset{\text{Proposition 2.1}}{\Rightarrow} \mathscr{T} \vDash X.$$

This is a contradiction to $\mathscr{T} \in \mathcal{M}$.

It follows that $\mathcal{M}$ has a $\subseteq$-maximal element. By Proposition 2.6, $\xi = (\mathscr{O}_X \backslash \mathscr{S}_{\max}) \in X^*$. If $U \in \mathscr{S} \subseteq \mathscr{S}_{\max}$, then $\xi \notin U^*$ (otherwise $U \in \xi$, but by Proposition 2.5, $\xi \cap \mathscr{S}_{\max} \neq \emptyset$). A contradiction to our assumption that $X^* = \bigcup_{U \in \mathscr{S}} U^*$. $\square$

Proposition 2.7. *If X is a G_+-space which has a G_+-base $\mathcal{B}$ where elemnts are G_+-quasicompact. Then X has sufficiently many van der Put points. If $\mathcal{B}$ is closed under finite intersection in X (including intersection over an empty set thus $X \in \mathcal{B}$), then X^* is a spectral space.*

Proof. Let $U \in \mathscr{O}_X$ and $\mathscr{S}$ any U-sieve such that $U^* = \bigcup_{V \in \mathscr{S}} V^*$. Let $\Omega \in \mathcal{B}_U$ and $\mathscr{S}_\Omega = \mathscr{S}|_\Omega$. Then $\Omega^* \subseteq U^*$ hence $\Omega^* = \bigcup_{V \in \mathscr{S}} \Omega^* \cap V^* = \bigcup_{V \in \mathscr{S}} (\Omega \cap V)^*$. (Note that $(V \cap W)^* = V^* \cap W^*$ by definition). Since $\Omega \cap V \in \mathscr{S}_\Omega$, it follows that $\Omega^* = \bigcup_{V \in \mathscr{S}_\Omega} V^*$. By Lemma 1.1, $\mathscr{S} \vDash \Omega$. Let $\mathcal{T}$ be the sieve generated by $\mathcal{B}_U$. From GT-Trans and what we have seen $\mathscr{S}//W$ where $W \in \mathcal{T}$. By GT-Loc, $\mathscr{S} \vDash U$ hence X has sufficiently many van der Put points.

For the spectrality of X^*, it is easy to see that Ω^* is quasicompact if Ω is Ω is G_+-quasicompact. Then $\{\Omega^* \,|\, \Omega \in \mathcal{B}\}$ satisfy the requirement for the quasicompactness part of the definition of spectrality. If $Z \subseteq X^*$ is a van der Put point, then $\xi = \{U \in \mathscr{O}_X \,|\, U^* \cap Z \neq \emptyset\}$ is a generic point of Z. T_0 is also easy to show. $\square$

Remark 2.7.

$$\operatorname{glim}_{X^*} \mathcal{U} = \{U \in \mathscr{O}_X \mid U \textit{ contains some } \Omega \in \mathcal{B} \textit{ such that } \Omega^* \in \mathcal{F}\}$$

All Ω^* with $\Omega \in \mathcal{B}$ are closed under this.

2.2 Sheaves

Definition 2.16. *A presheaf on X with values in (sets, abelian groups, rings) is a contravariant functor $\mathcal{G}$ from $\mathscr{O}_X$ to one of these categories. Then all $\mathcal{G}(U)$ are objects in these categories and we have*

$$V \subseteq U \Rightarrow \mathcal{G}(U) \overset{g \mapsto g|_V}{\to} \mathcal{G}(V),$$

which is a morphism in the target category such that $g \in \mathcal{G}$ then $g|_U = g, (g|_V)_W = g|_W$ if $W \subseteq V \subseteq U$. If this is only given on elements of a certain topology base $\mathcal{B}$, we speak of a presheaf on $\mathcal{B}$. In all cases, if a Grothendieck topology in $\mathcal{B}$ is given, then a sheaf is a presheaf $\mathcal{G}$ satisfying the additional assumption that

$$\mathcal{G}(\Omega) \overset{\sim}{\to} \varprojlim_{\Theta \in \mathscr{S}} \mathcal{G} = \left\{ (g_\Theta)_{\Theta \in \mathscr{S}} \in \prod_{\Theta \in \mathscr{S}} \mathcal{G}(\Theta) \mid g_\Theta|_\Xi = g_\Xi \textit{ where } \Theta \in \mathscr{S}, \Xi \in \mathcal{B}_\Theta \right\}.$$

If the map is only injective, then

Definition 2.17.

1. *A stalk of $\mathcal{G}$ at $\xi \in X^*$ is such that*

$$\mathcal{G}_\xi = \varinjlim_{\Omega \in \xi} \mathcal{G}(\Omega) = \{(\Omega, g) \mid \Omega \in \xi \cap \mathcal{B}, g \in \mathcal{G}(\Omega)\}/\sim$$

where the equivalent relation is given by $(g, \Omega) \sim (r, \Theta)$ if there is $\Xi \subseteq \Omega \cap \Theta$ such that $\Xi \in \mathcal{B}$ and $g|_\Xi = g|_\Xi$.

2. *A G-skyscraper sheaf of $\xi \in X^*$ is*

$$([G]_\xi)(U) = \begin{cases} G, & U \in \xi, \\ \{0\} \end{cases}$$

Definition 2.18. *Let X be a G_+-space with sufficiently many van der Put points and $\mathcal{G}$ be a sheaf on X. For an open subset $U \subseteq X$ and*

$$r \in \prod_{\xi \in U^*} \mathcal{G}_\xi.$$

We denote the sieve $\mathscr{S}_r$ on U generated in $\mathscr{O}_X$ to be such that

$$\mathscr{S}_r := \left[\Omega \in \mathcal{B}_U \,\middle|\, \exists g \in \mathcal{G}(\Omega) \text{ s. t. } \forall \xi \in \Omega^*, r_\xi = g_\xi \right].$$

Then we define,

$$(\mathrm{Gerb}(\mathcal{G}))(U) = \left\{ r \in \prod_{\xi \in U^*} \mathcal{G}_\xi \,\middle|\, \mathscr{S}_r \vDash U \right\}.$$

For the well-definedness, see Proposition 2.2.

Proposition 2.8. *Let X be a G_+-space with sufficiently many van der Put points.*

a) The stalk is exact on presheaves.

b) $[G]_\xi$ is a sheaf and $\mathcal{G} \mapsto \mathcal{G}_\xi$ is left adjoint to $G \mapsto [G]_\xi$ on (pre)sheaves.

c) If $\mathcal{G}$ is a presheaf, the morphism $\mathcal{G} \to \mathrm{Sheaf}(\mathcal{G})$ sending $g \in \mathcal{G}(U)$ to (image of g in $\mathcal{G}_\xi)_{\xi \in \mathcal{U}^*} \in \prod_{\xi \in U^*} \mathcal{G}_\xi$ defines an isomorphisms on stalks.

d) $(\mathcal{G}_\mathcal{B})_\xi \overset{\simeq}{\to} \mathcal{G}_\xi$ for all $x \in\in X$.

e) A morphism of presheaves defining isomorphism on stalks defines an isomorphism on sheafification.

f) If $\mathcal{G}$ is a sheaf, $\mathcal{G} \to \mathrm{Sheaf}(\mathcal{G})$ is an isomorphism. If $\mathcal{G}$ is only defined on $\mathcal{B}$, $\mathcal{G} \overset{\simeq}{\to} \mathrm{Sheaf}(\mathcal{G})|_\mathcal{B}$.

g) A morphism of sheaves defining isomorphism on all stalks is an isomorphism.

h) Sheafification (Presheaves on $\mathcal{B}$) $\to$ (sheaves on $\mathscr{O}_X$) is left adjoint to the functor of restriction to $\mathcal{B}$.

i) One has an equivalence of category between sheaves on X (for the Grothendieck topology on X) and (ordinary) sheaves on X^*, sending a sheaf $\mathcal{G}$ on X to the sheafification of the (pre)sheaf $U^* \to \mathcal{G}(U)$ on $\{U^* \mid U \in \mathscr{O}_X\}$ (a topology base for X^*) and a sheaf $\mathcal{H}$ on X^* to $\mathcal{G}(U) \to \mathcal{H}(U^*)$.

Proof. a)-h) are showwn in some ways as in derived sheaf theory. i) follows from these ☐

3 General non-archimedian topological ring

3.1 General topological rings and modules

Definition 3.1. *A topological group is a group G with a topology on the underlying set such that multiplication and inversion are continuous. That is*

$$\mu(g, h) = gh, \iota(g) = g^{-1}$$

are continuous.

Definition 3.2. *A topological ring is a ring R with a topology on the underlying set such that it is a topological group under addition and*

$$R \times R \ni (a, b) \mapsto ab \in R$$

is continuous. (We do not assume the continuity of $a \mapsto a^{-1}$ for $a \in R^\times$).

Definition 3.3. *Let R be a topological ring. A topological R-module M is an R-module with topology on its underlying set such that $(M, +)$ is a topological group and*

$$R \times M \ni (a, m) \mapsto am \in M$$

is continuous.

Proposition 3.1. *For a topological group G, the following conditions are equivalent.*

a). G is T_0 (Komogolov).

b). G is T_1 (Fréchet).

c). G is T_2 (Hausdorff).

d). $\{0\}$ is closed in G.

Proof.
If $x \in \overline{\{0\}}$ then $x^{-1} \in \overline{\{0\}}$ by the continuity of inversion. Hence $1 \in \overline{\{x\}}$ by continuity of $\mu(x, \cdot)$ hence T_0 is satisfied. For a)⇒d), let

$$\Delta = (\text{preimage of } \{u\} \text{ under } G \times G \overset{(g,h)\mapsto g^{-1}h}{\to} G)$$

is closed thus T_2. ☐

Proposition 3.2. *Every open subgroup H of a topological group G is also closed, and every group $H \subseteq H' \subseteq G$ is also clopen.*

Proof. We have,

$$G \backslash H = \bigcup_{g \in G/H \backslash \{0\}} gH$$

is open therefore H is closed. Every H' subset is a union of right or left H-cosets in G which is also open. ☐

Definition 3.4. *Let R be a topological ring and M be a topological R-module. A subset $X \subseteq M$ is bounded or R-bounded if for every neighborhood U of 0 in M, there is a neighborhood V of 0 in R such that*

$$V \cdot X = \{vx \mid v \in V, x \in X\} \subseteq U.$$

Example 3.1. *When R has the discrete topology, every subset is bounded (take $V = \{0\}$).*

Example 3.2. *This defines a boundary on X. If X_1, X_2 are bounded then so is $X_1 \cup X_2$. Every $\{m\}$ is bounded, $M \subseteq N$ and N bounded then M is also bounded.*

Example 3.3. *If $R = K$, equipped with an valuation $|\cdot|$ and M is an (valuative) K-vector space and K is not discrete, then a subset X of M is bounded if and only if there is $B \in [0, \infty)_{\mathbb{R}}$ such that*

$$\forall x \in X, \|x\| \leq B.$$

For example, take $U = \{x \in X \mid \|x\| \leq 1\}, B = |x|^{-1}$ where $x \in V \backslash \{0\}$ and V is as in the definition.

Definition 3.5. *For subsets $X_1, X_2 \subseteq R$, we define,*

$$X_1 X_2 = \{x_1 x_2 | x_1 \in X_1, x_2 \in X_2\}.$$

Let $X^0 = \{1\}$. We define recursively X^n such that

$$X^{n+1} = X \cdot X^n.$$

We say that. a subset $X \subseteq R$ is power bounded if

$$\bigcup_{n \in \mathbb{N}} X^n$$

is bounded. It is topologically nilpotent if for every neighborhood U of 0 in R, there is $n \in \mathbb{N}$ such that

$$\bigcup_{m=n}^{\infty} X^m \subseteq U.$$

We call an element $r \in R$ topologically nilpotent / power bounded if $\{r\}$ satisfies these properties.

Proposition 3.3. *Let R be a topological ring and M be a topological R-module. We have the following statements.*

a). A finite union of bounded subsets of M is bounded.

b). Trivially, every finite subset of M is bounded.

c). If $X_1, X_2 \subseteq M$ are bounded then so is $X_1 + X_2 = \{x_1 + x_2 \mid x_1 \in X_1, x_2 \in X_2\}$.

d). If $X \subseteq R$ and $Y \subseteq M$ are bounded then so is $XY \subseteq M$.

Proof. For c), let U be a neighborhood of 0 in M, there are a neighborhood $\tilde{U}$ of 0 in M such that $\tilde{U} + \tilde{U} \subseteq U$ and another neighborhoods V_1, V_2 of 0 in R such that $V_1 X_1, V_2 X_2 \subseteq \tilde{U}$. Take $V = V_1 \cap V_2$ in Definition 3.4. Proofs for the other statements are also easy. □

Proposition 3.4. *If $X \subseteq M$ is R-bounded and $M \xrightarrow{f} N$ a continuous morphism of R-modules then $f(X)$ is bounded in N.*

Proof. If $U \subseteq N$ is a neighborhood of 0, let $\tilde{U} = f^{-1}(U)$ and V a neighborhood of 0 in R such that $VX \subseteq \tilde{U}$. Then $Vf(X) \subseteq U$. □

It is not always true that the image of a bounded subset of a R under. a morphism $R \to S$ of topological ring is bounded. (The image is R-bounded but not necessarily S-bounded) for example take $R = \mathbb{Q}_p, S = \mathbb{Q}_q$. However, the following conditions on R asserts that this does not happen.

$$R \text{ has a topologically nilpotent unit} \tag{T}$$

$$R \text{ has a sequence of units converging to } 0 \tag{H}$$

$$\text{Every neighborhood of } 0 \text{ intersects } R^\times \tag{E}$$

Proposition 3.5. *Let $R \xrightarrow{f} S$ be a continuous morphism of topological ring. If R has* (T),(H),(E)*, then so does S and the image $f(X)$ of a bounded subset of R is a bounded subset of S.*

Proof. If U is a neighbourhood o 0 in S and R has (E), put $\tilde{U} = f^{-1}(U)$ and $r \in R^\times \cap \tilde{U}$. $f(r) \in S^\times \cap U$. If

$$\lim_{j \to \infty} r_j = 0$$

in R and all $r_j \in R^\times$,

$$\lim_{j \to \infty} f(r_j) = 0$$

in S and all $f(r_j) \in S^\times$. showing that (H). For (T), take a topologically nilpotent unit $r \in R$, then by setting $r_j = r^j$ in the previous argument, we see (T) is inheritted.

If R has (E) and $X \subseteq R$ bounded and U is a neighborhood of 0 in S, there is $\tilde{U}$ a neighborhood 0 in S such that $\tilde{U}\tilde{U} \subseteq U$. Let $U' = f^{-1}(\tilde{U}) \subseteq R$ a neighborhood of 0. There is a neighbourhood V' of 0 in R such that $V'X \subseteq U'$. As R has (E), we can take $\rho \in R^\times \cap f^{-1}(\tilde{U})$. Take $V = \rho V'$ then $Vf(X) \subseteq U$. □

Proposition 3.6. *Let R be a topological ring.*

a). Every power bounded subset is bounded.

b). A finite union of power bounded subsets is power bounded.

c). A subset $X \subseteq R$ which is both bounded and topologically nilpotent is power bounded.

d). A finite union of subsets which are both bounded and topologically nilpotent is topologically nilpotent.

e). If X is power bounded or topologically nilpotent and Y is topologically nilpotent, XY is then topologically nilpotent.

f). The image of a topologically nilpotent $X \subseteq R$ under a morphism $R \to S$ of topological rings is topologically nilpotent.

g). If R has (E), *$R \xrightarrow{f} S$ is a morphism of topological rings and $X \subseteq R$ is bounded in R. Then $f(X)$ is bounded in S.*

Proof. Mostly straightforward. $\square$

Definition 3.6. *Let R be a topological ring, we define the following subsets.*

$$R^\circ = \{r \in R \mid r \text{ is powewr bounded}\}, R^{\circ\circ} = \{r \in R \mid r \text{ is topologically nilpotent}\}.$$

Remark 3.1. *Without further conditions, these may fail to be additionally closed. By Proposition 3.3 b), and Proposition 3.6 a), $R^{\circ\circ} \subseteq R^\circ$ always holds.*

Definition 3.7. *Let $(m_n)_{n\in\mathbb{N}}$ be a sequence in M. It is Cauchy if for every neighbourhood U of 0 in M, there is k such that $m_i - m_j \in U$ where $\min\{i,j\} \geq k$. M is sequentially complete if every Cauchy sequence has a limit in M.*

Definition 3.8. *A filter $\mathcal{F}$ of M is Cauchy if every neighborhood U of 0 in M, there is $X \in \mathcal{F}$ such that $x - y \in U$ where $x, y \in X$. M is complete if every Cauchy filter has a limit.*

Definition 3.9. *The sequential completion of M is a set of equivalence classes of Cauchy sequences in M where two Cauchy sequences $(m_i), (n_i)$ are equivalent if for every neighborhood U of M, there is $k \geq 0$ such that $i \geq k$ then $m_i - n_i \in U$.*

Remark 3.2. *If G is a non-commutative topological group, then it typically requires that both $g_i g_j^{-1} \in U$ and $g_i^{-1} g_j \in U$ where $\min\{i,j\} \geq k$. Respectively, $x^{-1}y \in U$ and $xy^{-1} \in U$ for $x, y \in X$ for filters.*

Remark 3.3. *One can complete by taking equivalence classes of Cauchy filters. But how to continus multiplication in that completion of R^2. (Sequential completion does not have this problem).*

Remark 3.4. *It is known that topological vector space over $\mathbb{C}$ or $\mathbb{R}$ is the quotient of a complete topological vector space V over that field such that every bounded subset of V is contained in a finite dimensional subspace.*

Proposition 3.7. *Every quotient of a complete topological group with a countable neighborhood base of 1 by a closed normal divsor is complete.*

3.2 Continuous Valuations

For an ordered group Γ equip $\Gamma \cup \{0\}$ with the topology where U is open if and only if $0 \notin U$ or there is $\gamma \in \Gamma$ such that $(0, \gamma)_\Gamma \subseteq U$.

Consider a valuation $A \xrightarrow{v} \Gamma$ where A is a topological ring. Recall that Γ_v is the convex subgroup of Γ such that

$$\Gamma_v = \langle \{v(a) \mid a \in A \cdot \operatorname{supp} v\} \rangle_\Gamma.$$

Also recall that for $a \in A$ and $\gamma \in \Gamma$,

$$K_{v<a} = \{\alpha \in A \mid v(\alpha) < v(a)\}, K_{v<\gamma} = \{\alpha \in A \mid v(\alpha) < \gamma\}.$$

Similarly,

$$K_{v\leq a} = \{\alpha \in A \mid v(\alpha) \leq v(a)\}, K_{v\leq\gamma} = \{\alpha \in A \mid v(\alpha) \leq \gamma\}.$$

They come with an exception such that

$$K_{v<a} = \emptyset,$$

when $a \in \operatorname{supp} v$. They are subgroups of $(A, +)$. Note that a subgroup $H \subseteq G$ of a topological group G is open if and only if it is a neighbourhood of 0. Also recall that subgroups containing open subgroups are also open.

Proposition 3.8. *The following conditions are equivalent.*

a). $A \xrightarrow{v} \Gamma_v \cup \{0\}$ is continuous.

b). For all element γ of the convex hull of Γ_v in Γ, $K_{v<\gamma}$ is open.

c). For all $\gamma \in \Gamma_v$, $K_{r<\gamma}$ is open.

d). For all $a \in A\backslash \operatorname{supp}(v)$, $K_{r<a}$ is open.

e). For all $a \in A\backslash \operatorname{supp}(v)$, $K_{r<a}$ is a neighbourhood of 0.

Proof. By the previous argument d) and e) are equivalent. d)⇒c) is trivial. For c)⇒a), consider $\gamma \in \Gamma_v \cup \{0\}$ and a neighbourhood U of γ in $\Gamma_v \cup \{0\}$. We must show that $v^{-1}(U)$ is a neighbourhood of a in A. If $v(a) \neq 0$, it is sufficient to consider

$$U = \{v(a)\}.$$

Then,

$$v(b) = v(a)$$

where $b \in K_{r<a} + a$, and the set of such b is a neighborhood of a in A. If $v(a) = 0$, we may assume $U = [0, \gamma)_{\Gamma_v \cup \{0\}}$ with $\gamma \in \Gamma_v$. Then $v^{-1}(U) = K_{v<\gamma}$ is open by the assumption c).

a)⇒d) is trivial by the definition of the topology on $\Gamma_v \cup \{0\}$, $K_{v<a}$ being $v^{-1}([0, v(a))_{\Gamma_v \cup \{0\}})$ where

$$[0, v(a))_{\Gamma_v \cup \{0\}}$$

is a neighbourhood of 0 in $\Gamma_v \cup \{0\}$.

d)⇒b), every $\gamma \in \Gamma_v$ has the form,

$$\gamma = \frac{v(a)}{v(b)},$$

with $a, b \in A\backslash \operatorname{supp} v$. Then $K_{v<\gamma}$ is the inverse image under the endomorphism $A \xrightarrow{\cdot b} A$ of the topological group $(A, +)$ of the open subgroup $K_{r<a}$. Hence it is also an open subgroup. □

Definition 3.10. *A valuation of A is called continuous if the equivalent conditions in Proposition 3.8 hold.*

Remark 3.5.

1. *This only depends on the equivalent class of v.*

2. *$\Gamma_v \cup \{0\}$ carries the topology induced from $\Gamma \cup \{0\}$ only if Γ is the convex hull of Γ_v in Γ. Hence Γ can be replaced by Γ_v in c) only if Γ is equal to that convex hull (in which case the c) is b)).*

Example 3.4.

1. *$\mathbb{Q}_p \xrightarrow{|\cdot|_p} [0, \infty)_{\mathbb{R}}$ is a continuous valuation.*

2. *The trvial valuation for $\mathfrak{p} \in \operatorname{Spec} A$ is continuous if and only if $\mathfrak{p}$ is open.*

3. *If $\mathfrak{p}$ is an open prime ideal, then any valuation with support $\mathfrak{p}$ is open.*

4. *For $r \in (0, 1]_{\mathbb{R}}$, the valuation N_r from 1.1(1) (need to look for the reference) is continuous.*

Proposition 3.9. *If v is a continuous valuation. Then $\operatorname{supp} v$ is closed.*

Proof. Indeed $\operatorname{supp} v = v^{-1}(\{0\})$ and $\{0\}$ is closed in $\Gamma_v \cup \{0\}$. □

Definition 3.11. *We denote the subset equivalence classes of continuous valuations of A to be $\operatorname{Cont}(A) \subseteq \operatorname{Spv}(A)$. Let $A^+ \subseteq A$ be a subset we then denote $\operatorname{Spa}(A, A^+)$ to be the set of equivalence classes of continuous valuations v of A such that $a \in A^+ \Rightarrow v(a) \leq 1$.*

Remark 3.6.

1. *At this point, we do not yet introduce a topology on these sets.*

2. *If $A = \mathbb{T}_1$ as in 1.1.(1)(look at the first part and take the reference) and $A^+ = A^\circ$. $N_r \in \mathrm{Spa}(A, A^+)$ if and only if $r \in (0,1)_{\mathbb{R}}$.*

Definition 3.12. *A Oka system is a ring A and a set $\mathfrak{M}$ of ideals in A such that for any ideal I in A, $a \in A$ and the ideal $I + aA$ and*

$$(I : a) := \{\alpha \in A \mid a\alpha \in I\}$$

both belong to $\mathfrak{M}$ then $I \in \mathfrak{M}$.

Example 3.5. *The set of finitely generated ideals in A form a Oka system. This observation gives Eakin-Nagata or Cohen's theorem which state that a ring is Noetherian if and only if all of its prime ideals are finitely generated.*

Proposition 3.10. *Let $\mathfrak{M}$ be an Oka system and $\mathfrak{p} \neq A$ be a $\subseteq$-maximal element of the set of ideals of A not belonging to $\mathfrak{M}$. Then $\mathfrak{p} \in \mathrm{Spec}\, A$.*

Proof. Assume $ab \in \mathfrak{p}$. Let $a \notin \mathfrak{p}, b \notin \mathfrak{p}$. Then $\mathfrak{p} + aA$ and $(\mathfrak{p} : a)$ contain $\mathfrak{p}$ and not equal to $\mathfrak{p}$ as $b \in (\mathfrak{p} : a)$. By the maximality, both belong to $\mathfrak{M}$ thus $\mathfrak{p} \in \mathfrak{M}$ which is a contradiction. □

Proposition 3.11. *Let $\mathfrak{M}$ be a set of ideals such that $I \subseteq J$ and $I \in \mathfrak{M}$ implies $J \in \mathfrak{M}$. Suppose further that $\mathfrak{M}$ is closed under products, that is $I, J \in \mathfrak{M}$ then $I \cdot J \in \mathfrak{M}$. Then $\mathfrak{M}$ is an Oka system.*

Proof. Indeed $I \supseteq (I + aA)(I : a)$ holds. Therefore by definition, it is Oka. □

Remark 3.7. *The Oka system from Example 3.5 satisfies the assumptions if and only if A is Noetherian.*

Example 3.6. *The set $\mathfrak{M}_S$ of ideals not disjoint form a multiplicative subset $S \subseteq A$ is an Oka system. Then every $\subseteq$-maximal non-element of $\mathfrak{M}_S$ is a prime ideal. (This could also be seen by assuming that all maximal ideals of A_S are prime). By Zorn's lemma, every ideal not belonging to $\mathfrak{M}_S$ is contained in an ideal which is maximal with respect to this non-containment. By Proposition 3.10, such ideal is prime.*

Proposition 3.12. *Let v be a valuation of a topological ring A satisfying the following conditions.*

1. *$K_{v\leq 1}$ is a neighbourhood of 0 in A.*
2. *For each $a \in A \setminus \mathrm{supp}\, v$, there is $b \in A$ with $v(ab) \geq 1$.*

Then v is continuous.

Proof. If $a \in A \setminus \mathrm{supp}\, v$, and b is as above, then $K_{v<a}$ cotains the inverse image under the continuous endomorphism, $A \xrightarrow{\cdot b} A$ of $K_{r\leq 1}$ which is a neighbourhood of 0. Then Proposition 3.8 e) is satisfied. □

Proposition 3.13. *Let A be a topological ring, A^+ be an open subring of A which is integrally closed in A and $\alpha \in A \setminus A^+$. Then there is $v \in \mathrm{Spa}(A, A^+)$ such that $v(\alpha) > 1$.*

Proof. By assumption, α is not integral over A^+ ,hence $S = \{p(\alpha) \,|\, p \in A^+[T] \text{ monic}\}$ is a multiplicative subset of A not containg 0. By Example 3.6, there is $\mathfrak{p} \in \mathrm{Spec}\, A$ maximal among all ideals of A disjoint from S. It follows from the maximality of $\mathfrak{p}$, that for every $a \in A \setminus \mathfrak{p}$, there is $b \in A$ and a monic polynomial $P \in A^+[T]$ such that $P(\alpha) - ab \in \mathfrak{p}$. Since $\mathfrak{p}$ is disjoint from S, the image $\alpha \mod \mathfrak{p}$ of α in $k(\mathfrak{p})$ is not integral over the image $A^\#$ of A^+ in $k(\mathfrak{p})$. By Proposition 1.4.1, this is a valuation ring R of $k(\mathfrak{p})$ such that $A^\# \subseteq R$ and $\alpha \notin R$. Consider v, the compsition,

$$A \to k(\mathfrak{p}) \overset{\text{valuation defined by R Prop 1.4.1}}{\to} \Gamma.$$

Since $\alpha \mod \mathfrak{p} \notin R, v(\alpha) > 1$. By the definition, $K_{v\leq 1}$ contains A^+ as A^+ and $\mathfrak{p} \subseteq A^\# \subseteq R$, hence $K_{v\leq 1}$ is open. If $a \in A$ and $v(a) \neq 0$¡ then $a \notin \mathfrak{p}$ and we have $b \in A$ and a monic $P \in A^+[T]$ such that $P(\alpha) - ab \in \mathfrak{p}$. Since P is monic and $v(A^+) \subseteq [0,1]_{\Gamma \cup \{0\}}$, $P(\alpha) > 1$. Since $\mathrm{supp}\, v = \mathfrak{p}$, $v(ab) = v(P(\alpha)) > 1$. By Proposition 3.12, v is continuous. □

Corollary 3.1. *For a subset A^+ of A, the following conditions are equivalent.*

a). $A^+ = \bigcap_{x \in \mathrm{Spa}(A,A^+)} K_{x\leq 1} = \{a \in A \mid |a|_x \leq 1 \textit{ for all } x \in Spa(A,A^+)\}$

b). *A^+ is the intersection of a set of open and integrally closed subrings of A.*

Proof. For b)⇒a), note that $\subseteq$ in a) is trivial and $\supseteq$ is by Proposition 3.13. For a)⇒b), note that $K_{x\leq 1}$ for $x \in \mathrm{Spa}(A,A^+)$ are integrally closed (by Fact 1.3.5) and open (by Proposition 3.8 a)) subrings of A. □

Proposition 3.14. *Let $\eta \in \mathrm{Cont}(A)$ and $s \in \mathrm{Spv}(A)$ a specialization of η. Then $s \in \mathrm{Cont}(A)$.*

Proof. Let $a \notin \operatorname{supp} s$. Then by Proposition 1.37, $\operatorname{supp} \eta \subseteq \operatorname{supp} s$. Hence $|a|_\eta \neq 0$. Thus $K_{\eta<a}$ is a neighbourhood of 0 in A by Proposition 3.8 e). By Fact 1.3.6, $K_{\eta<a} \subseteq K_{s<a}$ hence $K_{s<a}$ is a neighbourhood of 0 in A and again by Proposition 3.8 e), hence for s as well. □

WIthout further assumptions, Spa and Cont (to be equipped with a topology below) fail to be spectral spaces and there are ultrafilters $\mathcal{U}$ on Spa such that $\mathrm{Cont}(A) \in \mathcal{U}$ but $\mathrm{glim}_{\mathrm{Spa}\,A}\,\mathcal{U}$ fails to have any continuous specializations.

The following condition on A are often useful.

If $I \subseteq A$ is an open ideal, I^2 is also open (oisq)

Every open ideal $I \subseteq A$ contains a finitely generated open ideal of A. (oifin)

Proposition 3.15. *If I and J are open ideals of a topological ring A satisfying (oisq), then IJ is open. In particular, by induction on n, all I^n are open.*

Proof. As $I \cap J$ is open $(I \cap J)^2 \subseteq I \cdot J$ is also open by (oisq). □

Proposition 3.16. *If $a \in A^{\circ\circ}$ and $x \in \mathrm{Cont}(A)$, then $|a|_x < 1$.*

Proof. By assumption, we have,

$$\lim_{n\to\infty} |a|_x^n = 0$$

in $\Gamma_x \cup \{0\}$. By the continuity of x, the above equality does not hold if $|a|_x \geq 1$. □

We further introduce following conditions.

The ideal generated by $A^{\circ\circ}$ of A is open (tn;0)

$A^{\circ\circ}$ is a neighbourhood of 0 in A. (tn;1)

Obviously (tn;1)⇒(tn;0).

Proposition 3.17. *If A contains an open subring B satisfying one of (oifin),(oisq), (tn;0),(tn;1), then A also has that property.*

Definition 3.13. *We say that $x \in \mathrm{Spa}(A)$ can be made continuous if $K_{x<a}$ is a neighbourhood of 0 in A.*

Remark 3.8. *The condition in Definition 3.13 is already satisfied if x is continuous by Propositoin 3.8 e).*

Proposition 3.18.

a). *If $x \in \mathrm{Spv}(A)$ can be made continuous then $K_{x<\gamma}$ is a neighbourhood of 0 where $\gamma \in c\Gamma$.*

b). *If $K_{x\leq 1}$ is a neighbourhood of 0 then $K_{x<\gamma}$ is a neighbourhood of) where $\gamma \in c\Gamma \backslash \{0\}$.*

c). *If $K_{x\leq 1}$ is a neighbourhood of 0, then $c\Gamma_x = \{1\}$ or x can be made continuous.*

Proof. For b), If $\gamma > 1$ then $K_{x\leq 1} \subseteq K_{x<\gamma}$ and the claim follows. Otherwise $\gamma < 1$ and there is $a \in A$ such that $\gamma|a|_x \geq 1$. In that case, $\gamma|a|_x^2 > 1$ and $K_{x<\gamma}$ contains the preimage under the continuous map $A \xrightarrow{a^2} A$ of $K_{x\leq 1}$, a neighbourhood of 0.

a). For $\gamma = 1$, this follows from Definition 3.13 and for $\gamma \neq 1$, we can apply b).

c). Assume $c\Gamma_x \neq \{1\}$. There is $\gamma \in c\Gamma_x \backslash \{1\}$. Without loss of generality, assume $\gamma < 1$. Then $K_{x<1}$ is a neighbourhood of 0 as it contains $K_{x<\gamma}$, a neighbourhood of 0 by b). ☐

Proposition 3.19. *Let A be a topological ring with (oisq) and (tn;0), and let $\eta \in \mathrm{Spv}(A)$. Assume $|x|_\eta < 1$ for all $x \in A^{\circ\circ}$. In the following the word "specialization" always means "specialization in* Spv*".*

a). There is a continuous specialization of η if and only if η can be made continuous in the sense of Definition 3.13.

b). If this is the case, then,

I). The following subset of Γ_x is a convex subgroup containing $c\Gamma_x$.

$$s\Gamma_\eta := \{\gamma \in \Gamma_x \mid K_{\eta<r} \text{ is a neighbourhood of } 0 \text{ in } A\}.$$

II). The horizontal specialization $s = \eta|_{s\Gamma_\eta}$ (which by I) is well-defined) is continuous.

III). The sets of specializations of s and of continuous specializations of η coincide.

IV). Let $\Omega = \mathcal{R}(f_0|f_1, \cdots, f_m)$ with $(f_i)_{i=0}^{m+1} \in A^{m+1}$ be such that the ideal of A generated by $K_{\eta\leq f_0}$ is open, then $s \in \Omega$ if and only if $\eta \in \Omega$.

Proof. Assume that η has the continuous specialization s. As $K_{s\leq 1} \subseteq K_{\eta\leq 1}$ by Fact 1.3.7 and $K_{s<1}$ is a neighbourhood of 0 by Proposition 3.8 e)., $K_{\eta\leq 1}$ is a neighbourhood of 0. By Proposition 3.18, $c\Gamma_\eta = \{1\}$ or η can be made continuous. If $c\Gamma_x = \{1\}$, then $A = K_{\eta\leq 1}$ and $K_{\eta<1}$ is an ideal in A. By assumption about η, $A^{\circ\circ} \subseteq K_{\eta<1}$ and by (tn0), $K_{\eta<1}$ is an open ideal of A. Then η can be made continuous.

Assume that η can be made continuous. To finish a), and to show b), it suffices to show I)-IV) under this assumption. I) follows from Proposition 3.18 a). That is $c\Gamma_\eta \subseteq s\Gamma_\eta$. The convexity of $s\Gamma_\eta$ as a neighbourhood of Γ_η is obvious. One easily checks that the subgroup property follows (and I) is fully shown) where it is shown that $\gamma^2 \in s\Gamma_\eta$ when $\gamma \in s\Gamma_\eta$. Note that

$$\min\{\gamma_1, \gamma_2\}^2 \leq \gamma_1\gamma_2 \leq \max\{\gamma_1, \gamma_2\}^2.$$

It is sufficient to do this where $\gamma \leq 1$. When $\gamma \in c\Gamma_\eta$, then $\gamma^2 \in c\Gamma_\eta$ and the claim follows from a). Otherwise, $\gamma|a|_\eta < 1$ for all $a \in A$. Consider the (open as $\gamma \in s\Gamma$) ideal of A generated by $K_{\eta<\gamma}$. By (oisq) and Proposition 3.15, I^3 is also open. The elements of I^3 have the form $\alpha = \sum_{i=1}^n a_ib_ic_id_i$ with $a_i \in A$ and $b_i, c_i, d_i \in K_{\eta<\gamma}$. By b), $|a|_\eta, |b|_\eta < |a_i|_\eta\gamma < 1$. Hence

$$|\alpha|_\eta \leq \max_{1\leq i\leq n}\{|a_i|_\eta, |b_i|_\eta, |c_i|_\eta, |d_i|_\eta\} < \max_{1\leq i\leq n}\{|c_i|_\eta, |d_i|_\eta\} < \max_{1\leq i\leq n}\gamma\cdot\gamma = \gamma^2.$$

Thus, $K_{\eta<\gamma^2}$ contains the open ideal I^3, and the claim follows.

For II), note that $|a|_s \neq 0$ implies $|a|_\eta \in s\Gamma_\eta$. hence by the definition,

$$K_{s<a} = K_{\eta<a}.$$

This is a neighbourhoodof 0 by definition of s.

For IV), by Fact 1.3.10, it suffices to show that $|f_0|_s \neq 0$. That is $|f_0|_\eta \in s\Gamma_\eta$. if $|f_0|_\eta \in c\Gamma_\eta$, then this follows from a). Otherwise $|a|_\eta|f_0|_\eta < 1$ for all $a \in A$. Moreover, the ideal I of A generated by $K_{\eta<f_0}$ is open by our assumption about f_0. By (0;aq), I^2 is also open. Its elements have the form, $\alpha = \sum_i^n a_ib_ic_i$ with all $a_i \in A, b_i, c_i \in K_{\eta\leq f_0}$. Therefore,

$$|\alpha|_\eta \leq \max_{1\leq i\leq n}\{|a_i|_\eta, |b_i|_\eta, |c_i|_\eta\} < \max_{1\leq i\leq n}|c_i|_\eta \leq |f_0|_\eta.$$

Then $K_{\eta<f_0}$ contains the open ideal I^2, hence is a neighbourhoodof 0. Thus $|f_0|_\eta \in s\Gamma_\eta$ as claimed.

For III, every specialization of the continuous. (II) s is continuous by 3.14. Let $t \in \mathrm{Spv}(A)$, be any continuous specialization of η. For every rational open subset $\Omega = \mathcal{R}(f_0|f_1, \cdots, f_n)$, we want to show that $t \in \Omega$ implies $s \in \Omega$. But $t \in \Omega$ implies $|f_0|_t \neq 0$ by definition, hence $K_{t<f_0}$ is a neighbourhood of 0 by **??** d) (and the containing of 0), hence $K_{t\leq f_0} \supseteq K_{t<f_0}$ is a neighbourhood of 0 By Fact 1.3.7 $K_{\eta\leq f_0} \supseteq K_{t\leq f_0}$ hence this is a neighbourhood of 0 as well. Hence IV is applied and $\eta \in \Omega$ follows since $t \in \Omega$ and t is a specialization of η. Thus, $\eta \in \Omega$ by IV and II is shown. □

Definition 3.14. *The strong topology on* $\mathrm{Cont}(A)$ *or on* $\mathrm{Spa}(A, A^+)$ *is the one induced from* Spv. *That is for which the set of all* $\mathcal{R}(f_0|f_1, \cdots, f_m)$ *for arbitrary* $n \in \mathbb{N}$ *and* $(f_i)_{i=0}^n \in A^{n+1}$ *form a topology base.*

If A satisfies (oisq), the weak topology on both spaces is the one for which the set $\mathfrak{Rat}_A = \mathfrak{Rat}_A^{(t)}$ *of* $\mathcal{R}(f_0|f_1, \cdots, f_n)$ *for which* $\langle f_0, \cdots, f_n\rangle_A$ *is an open ideal of A is a topology base.*

Remark 3.9. *Note that* $\mathcal{R}(f_0|f_1, \cdots, f_m) \cap \mathcal{R}(g_0|g_1, \cdots, g_m) = \mathcal{R}(f_0g_0|f_ig_j)$ *and*

$$\langle f_ig_j\rangle_A = \langle f_0, \cdots, f_n\rangle_A\langle g_0, \cdots, g_m\rangle_A$$

is open by (oisq). Hence $\mathfrak{Rat}_A^{(t)}$ *is closed under finite intersection. In most case, A will be equipped with a topology and may the notation* (t) *be dropped. When the ring is equipped with a topology only the elements of* $\mathfrak{Rat}_A^{(t)}$ *will be called rational open subsets of* $\mathrm{Cont}(A)$ *or* $\mathrm{Spa}(A, A^+)$.

Proposition 3.20. *If A has (oisq) and (oifin), then these topologies coincide.*

Proof. Let $U = \mathcal{R}(f_0|f_1, \cdots, f_n) \subseteq X$ where $X = \mathrm{Cont}(A)$ or $X = \mathrm{Spa}(A, A^+)$. For $x \in U$, we want to show that there is $\Omega \in \mathfrak{Rat}_A^{(t)}$ such that $x \in \Omega \subseteq U$. As x is a valuation, $K_{x<f_0}$ is open and by (oifin), there are finitely many $\varepsilon_1, \cdots, \varepsilon_l \in K_{x<f_0}$ such that $\langle\varepsilon_1, \cdots, \varepsilon_l\rangle_A$ is open. Then $x \in \Omega = \mathcal{R}(f_0|f_1, \cdots, f_n, \varepsilon_1, \cdots, \varepsilon_l) \subseteq U$ and $\Omega \in \mathfrak{Rat}_A^{(t)}$. □

Proposition 3.21. $\mathrm{Cont}(A)$ *and* $\mathrm{Spa}(A, A^+)$ *are* T^0 *for the strong topology. This also holds for the weak topology when it coincides with the strong one.*

Proposition 3.22. *Let A be a topological ring with (oisq), (oifin) and (tn0). For an ultrafilter* $\mathcal{U}$ *on* $\mathrm{Cont}(A)$*, let* $\mathscr{V}$ *be its image on* $\mathrm{Spv}\,A$*. The following conditions are equivalent.*

a). $\mathcal{U}$ has a generic limit in $\mathrm{Cont}(A)$.

b). $\mathcal{U}$ has some limit in $\mathrm{Cont}(A)$.

c). The generic limit $g = \mathrm{glim}_{\mathrm{Spv}(A)}\mathscr{V}$ *of* $\mathscr{V}$ *in* $\mathrm{Spv}\,A$ *can be made continuous in the sense of Definition 3.13.*

d). The set

$$\{a \in A \mid \{v \in \mathrm{Cont}(A) \mid |a|_x < 1\} \in \mathcal{U}\} \tag{3.1}$$

is a neighbourhood of 0 *in A.*

In this case, a generic limit γ *of* $\mathcal{U}$ *in* $\mathrm{Cont}(A)$ *is given by the continuation* $g|_{s\Gamma_g}$ *of* g*. If* $\mathrm{Spa}(A, A^+) \in \mathcal{U}$*, then* $\gamma \in \mathrm{Spa}(A, A^+)$ *and* γ *is a generic limit of the intersection of* $\mathcal{U}$ *iwth the set of subsets of* $\mathrm{Spa}(A, A^+)$.

Proof. a)⇒b) is obvious. For b)⇒c), if λ is any limit of $\mathcal{U}$, it is a limit of $\mathscr{V}$ in $\mathrm{Spv}(A)$ by the continuity of (recall that we are dealig with the strong topology Propositio n3.20). Hence λ is a specialization of g in $\mathrm{Spv}(A)$. By Proposition 3.19 a), g can be made continuous.

For c)⇒ a). If g can be made continuous and γ is its continuation. This is in the closure of $g = \mathrm{glim}_{\mathrm{Spv}(A)}\mathscr{V}$, hence is a limit of $\mathscr{V}$. As $\mathrm{Cont}(A)$ carries the induced topology, it is a limit of $\mathcal{U}$ in $\mathrm{Cont}(A)$. Let λ be any limit of $\mathcal{U}$ in $\mathrm{Cont}(A)$. Then λ is a limit of $\mathscr{V}$ in $\mathrm{Spv}(A)$. Hence Proposition 3.19, a specialization of γ in $\mathrm{Spv}(A)$. Using once again the fact that $\mathrm{Cont}(A)$ carries the induced topology, λ is a specialization of γ in $\mathrm{Cont}(A)$, showing that γ is a $\mathrm{glim}_{\mathrm{Cont}(A)}\mathcal{U}$.

We have shown the equivalence of first three statement and also the

Since
$$\{x \in \mathrm{Cont}(A) \mid |a|_x < 1\} = \{x \in \mathrm{Spv}(A) \mid |a|_x\} \cap \mathrm{Cont}(A)$$
as $a \in A$ belongs to the subset considered in (3.1) if and only if
$$M = \{x \in \mathrm{Spv}(A) \mid |a|_x < 1\} \in \mathscr{V}.$$
By the description of g in Proposition 1.3.1,
$$\{a \mid M_a \in \mathscr{V}\}$$
coincides with $K_{g<a}$. By Definition 3.13, thisis a neighbourhood of 0 in A if and only if g can be made continuous, and c)⇒d) follows.

Finally let $\mathrm{Spa}(A, A^+) \in \mathcal{U} \subseteq \mathscr{V}$. We have
$$\mathrm{Spa}(A, A^+) = \left[\bigcap_{\alpha \in A^+} \mathcal{R}_{\mathrm{Spv}(A)}(1|\alpha)\right] \cap \mathrm{Cont}(A) \tag{3.2}$$
and since the open sets $\mathcal{R}_{\mathrm{Spv}(A)}(1|\alpha)$ are all closed under intersection. g is contained in all of them, hence $|\alpha|_g \leq 1$ where $\alpha \in A^+$. Since γ is a horizontal specialization of g, this implies $|\alpha|_\gamma \leq 1$ and γ belongs to the filter generated in the outher $\bigcap$ in (3.2). Since γ is also continuous, $\gamma \in \mathrm{Spa}(A, A^+)$ by (3.2). Since $\mathrm{Spa}(A, A^+)$ carries the topology induced from $\mathrm{Cont}(A)$, γ is a limit of
$$\mathcal{U}' = \mathcal{U} \cap (\text{subsets of } \mathrm{Spa}(A, A^+)).$$
If λ is any limit of $\mathcal{U}'$ in $\mathrm{Spa}(A, A^+)$, it is a limit of the image $\mathcal{U}$ of $\mathcal{U}'$ under the continuous map $\mathrm{Spa}(A, A^+) \to \mathrm{Cont}(A)$. Hence it is a specialization of γ in $\mathrm{Cont}(A)$. Hence also in $\mathrm{Spa}(A, A^+)$ which has the induced topology. □

Corollary 3.2. *If A is a topological ring satisfying (oisq),(oifin), and (tn;1). Then $X = \mathrm{Cont}(A)$ and all $X = \mathrm{Spa}(A, A^+)$ are spectral spaces and the elements of $\mathfrak{Rat}_X^{(t)}$ are quasicompact open subsets of X.*

Proof. If $a \in A^{\circ\circ}$, then
$$\{x \in \mathrm{Cont}(A) \mid |a|_x < 1\} = \mathrm{Cont}(A) \in \mathcal{U}$$
(in the situation of Proposition 3.22), hene the subset of A contained in (3.1) contains $A^{\circ\circ}$ which is a neighbourhood of 0 if (tn1) holds. It follows that all ultrafilters on X has generic limits. To show that the set of open subsets of X which are closed under generic limit of ultrafilters in X contains $\mathfrak{Rat}_X^{(t)}$ (which is a topology base). Let
$$\Omega = \mathcal{R}_X(f_0|f_1, \cdots, f_m) = \Omega' \cap X,$$
where
$$\Omega' = \mathcal{R}_{\mathrm{Spv}(A)}(f_0|f_1, \cdots, f_m),$$
with $I = \langle f_0, \cdots, f_m\rangle_A$ being open. Let $\mathcal{U}'$ be an ultrafilter on X, $\mathcal{U}$ its image in $\mathrm{Cont}(A)$ and the remaining situation as in Proposition 3.22. Then $g \in \Omega'$ as $\Omega' \in \mathscr{V}$ (note $\Omega' \cap X = \Omega \in \mathcal{U}'$) and Ω' is closed under $\mathrm{glim}_{\mathrm{Spv}(A)}$ by Proposition 1.3.1. We claim that

$K_{g\leq f_0}$ is a neighbourhood of 0 in A. Then $g \in \Omega'$ implies $\mathrm{glim}_X\, \mathcal{U}' = \gamma \in \Omega' \cap X = \Omega$. Assume ? that there is $a \in A$ such that $|af_0| \geq 1$. Then $K_{g\leq f_0}$ contains the preimage under $A \overset{f_0}{\to} A$ of $K_{g\leq 1}$, which is in a neighbourhood of 0 as it contains $K_{g\leq 1}$ and g ca nbe made continuous. Hence $Kg \leq f_0$ is a neighbourhood of 0 in this case.

If no such a exists , $K_{g\leq f_0}$ contains I^2, as elements of I^2 have the form

$$\alpha = \sum_{i,j=1}^{m} a_{ij} f_i f_j.$$

Hence,

$$|\alpha|_g \leq \max_{\substack{0\leq i\leq m\\ 0\leq j\leq m}} |a_{ij}|_g |f_i|_g |f_j|_g \leq |a_{00}|_g |f_0|_g^2 < |f_0|_g$$

as $g \in \Omega$. $\square$

Remark 3.10. *Even when* $\mathrm{Spa}(A, A^+) \in \mathcal{U}$ *is the situation of Proposition 3.22,* $\mathrm{glim}_{\mathrm{Cont}(A)} = \mathrm{glim}_{\mathrm{Spa}(A,A^+)}(\mathcal{U} \cap \mathscr{O}_{\mathrm{Spa}(A,A^+)})$ *may have specialization in* $\mathrm{Cont}(A)$ *which do not belong to* $\mathrm{Spa}(A, A^+)$*. (However, horizontal specialization of g are the elements of* $\mathrm{Spa}(A, A^+)$*). For example $A = \mathbb{T}_1, A^+ = A^\circ$ and g is given by $\|\cdot|\mathbb{T}_1\|$, $\mathcal{U}$ is the principal filter for g, then $g = \mathrm{glim}_{\mathrm{Cont}(\mathbb{T}_1)}$ and the valuation (1.1.1) (denoted by N_1) is nto in* $\mathrm{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ)$.

Proposition 3.23. *If* $X = \mathrm{Cont}(A)$ *or* $\mathrm{Spa}(A, A^+)$*, and $x \in X$ is closed then $c\Gamma_x = \Gamma_x$.*

Proof. Problem 5 from Sheet 8 $\square$

Proposition 3.24. *If* $X = \mathrm{Cont}(A)$ *or* $\mathrm{Spa}(A, A^+)$*, and $s \in X$ is a specialization of $\eta \in X$ in X. Let $x \in \mathrm{Spv}(A)$ is a specialization in $\mathrm{Spa}(A)$ of η such that s is a specialization in $\mathrm{Spv}(A)$ of x, then $x \in X$ and the two specializations in $\mathrm{Spv}(A)$ involving x also hold in X*

Proof. Sheet 8 Problem 7. There are ? Huber (1.5. of the script) for Spv also hold in X. Definition **??** and Fact 13 about this are omitted $\square$

Proposition 3.25. *Let A be a topological ring and $\mathcal{M}$ be the set of open ideals in A.*

a). If A has (oifin) then every ideal I not in $\mathcal{M}$ is contained in some ideal of A which is $\subseteq$-maximal in the set of ideals of A which are not in $\mathcal{M}$.

b). If A has (oisq), then $\mathcal{M}$ is an Oka-system. If $\mathfrak{p}$ is any ideal of A which is $\subseteq$-maximal with the property of begin $\notin \mathcal{M}$. Then $\mathfrak{p}$ is a prime ideal which is closed but not open and such that every ideal strictly contains it is open, or $\mathfrak{p}$ is a dense maximal ideal.

Proof. Let $\mathcal{N}$ be the set of ideals in A which are not open. For a), let $\mathcal{L} \subseteq \mathcal{N}$ be linearly $\subseteq$-ordered non-empty subset. Let $I = \bigcup_{J\in\mathcal{L}} J$. Then I is an ideal in A. If I is open then by (oifin), there are $f_1, \cdots, f_m \in I$ such that

$$K = \langle f_1, \cdots, f_m \rangle_A$$

is open. As $\mathcal{L}$ is linearly orderd, there is $J \in \mathcal{L}$ containing all f_i. Hence $K \subseteq J$ and J is open. This contradicts to that $J \in \mathcal{N}$. Applying Zorn's lemma, we derived the statement.

For b),that $\mathcal{M}$ is an Oka system follows from Proposition Fact 3 and Proposition Fact 6. That $\mathfrak{p}$ must be prime follows from 3.10. If $\mathfrak{p}$ is not closed let $a \in \overline{\mathfrak{p}}\backslash\mathfrak{p}$. As $\overline{\mathfrak{p}}$ is an ideal and $\mathfrak{p}$ is prime, $a^2 \in \overline{p}\backslash\mathfrak{p}$. Now, $\mathfrak{p} + a^2A$ is open by the maximality of $\mathfrak{p}$, hence closed hence $\overline{p} = \mathfrak{p} + a^2A$. Sinc $a \in \overline{p} = \mathfrak{p} + a^2A$, there is $b \in A$ such that $a = a^2b \in \mathfrak{p}$. As $a \in \mathfrak{p}, 1 - ab \in \mathfrak{p}$. Thus $1 \in \overline{\mathfrak{p}}$ as $a \in \overline{\mathfrak{p}}$ and $\overline{\mathfrak{p}}$ is an ideal. Now the previous consideration can be applied to all $a \in A$, and show that $A/\mathfrak{p}$ is a field. Thus $\mathfrak{p}$ is a dense maximal ideal. $\square$

Definition 3.15. *A continuous varuation v of A and its equivalence class is called analytic if the closed prime ideal* $\mathrm{supp}\, v$ *fails to be open. Let* $\mathrm{Cont}(A)_a$ *and* $\mathrm{Spa}(A, A^+)_a$ *be the sets of* $x \in \mathrm{Cont}(A)$ *or* $x \in \mathrm{Spa}(A, A^+)$ *which are analytic.*

Proposition 3.26. *For a topological ring A with (tn;0), the following conditions are equivalent.*

a). The ideal of A generated by $A^{\circ\circ}$ is A.

b). A has no open maximal ideal.

c). A has no open prime ideal.

d). A has no open proper ideal.

e). For every subset $A^+ \subseteq A$, all elements of $\mathrm{Spa}(A, A^+)$ are analytic.

Proof. a)⇒d), as every open ideals is contained in $A^{\circ\circ}$. d)⇒c)⇒b) and c)⇒d) are all trivial. d)⇒a) is by (tn;0). b)⇒d), every proper ideal I is contained in some maximal ideal which is open when I is.

e)⇒c) If ¶ is an open ideal, its finite valuation given as elements of $\mathrm{Spa}(A, A^+)$. □

Definition 3.16. *A ring is called analytic if the ideal of A generated by $A^{\circ\circ}$ is A.*

Example 3.7. *Eery ring with (E) from 3.1 ($A^\times$ intersects every neighbourhood of 0) is analytic. In particular, every ring with (T) is analytic.*

Proposition 3.27. *When A is a topological ring with (oifin) and (tn;0) and $X = \mathrm{Cont}(A)$ or $X = \mathrm{Spa}(A, A^+)$. Then X_0 is a finite union of elements of $\mathfrak{Rat}_X$. In particular, it is open.*

Proof. By our assumption, there are finitely man yelements $f_1, \cdots, f_m \in A^{\circ\circ}$ generating an open ideal in A. Then $\mathfrak{p} \in \mathrm{Spec}\, A$ is open if and only if $f_1, \cdots, f_m \in \mathfrak{p}$. Then

$$X_a = \bigcup_{i=1}^{m} \mathcal{R}_x(f_i | f_1, \cdots, f_m).$$

□

Remark 3.11. *If only (tn;0) holds then,*

$$X_a = \bigcup_{f \in A^{\circ\circ}} \mathcal{R}_X(f|)$$

is still open.

Proposition 3.28. *Let $A \xrightarrow{p} B$ be a morphism of topological rings which is open. If $M \subseteq A$ is bounded then $p(M)$ is bounded in B.*

Proof. Exercise. Sheet 8 problem 1. □

Proposition 3.29. *Let A be a topological ring with (oisq), (oifin), and (tn0) and without dense maximal ideals. Let A^+ be an open subring of A which is a subset of A°. For $m \in \mathbb{N}$ and $(a_i)_{i=1}^m \in A^m$, the following conditions are equivalent.*

a). The ideal $I = \langle a_1, \cdots, a_n \rangle_A$ is open.

b). For all $x \in \mathrm{Spa}(A, A^+)$, there is $1 \le i \le m$ such that $|a_i(x)| \ne 0$.

Proof. By the definition, $\max_{1 \le i \le m} |a_i|_x \ne 0$ if and only if $I \subseteq \mathrm{supp}\, x$. If I is open, no such x can be analytic. Thus a)⇒b).

Assume that I is not open. By Proposition 3.25, there is a closed prime ideal $\mathfrak{p}$ such that $I \subseteq \mathfrak{p}$ and such that $\mathfrak{p}$ is not open but every larger ideal is open. By (tn;0), there is $x \in A^{\circ\circ} \backslash \mathfrak{p}$. As A^+ is a neighbourhood of 0, without losing generality, ($s \hookleftarrow s^j$ for some positive $j \in \mathbb{N}$), $s \in A^+$. We denote the image of A^+ in $A/\mathfrak{p} \subseteq k(\mathfrak{p})$ by B. By Proposition 3.28,

$$B \subseteq (A/\mathfrak{p})^\circ. \tag{1}$$

Now $(s \mod \mathfrak{p})^{-1} \in k(\mathfrak{p})$ be not integral over B. Otherwise, there will be $(p_i)_{i=1}^d \in B$ such that

$$1 = \sum_{i=1}^{d} p_i (s \mod \mathfrak{p})^i = (s \mod \mathfrak{p}) \sum_{i=0}^{d-1} p_{i+1} (s \mod \mathfrak{p})^i \in (s \mod \mathfrak{p})B.$$

By a) and $(s \mod \mathfrak{p}) \in (A/\mathfrak{p})^{\circ\circ}$, this implies that 1 is topologically nilpotent in $A/\mathfrak{p}$. This contradicts the fact that $\mathfrak{p}$ is closed in A. By Proposition 1.4.2, there is a valuation ring R of $k(\mathfrak{p})$ containing B but not $(s \mod \mathfrak{p})^{-1}$. Let γ denote the composition,

$$A \to A/\mathfrak{p} \to k(\mathfrak{p}) \overset{\text{valuation defined by } R}{\to} (k(\mathfrak{p})^\times / R^\times) \cup \{0\}.$$

There is a valuation and its support $\mathfrak{p}$ is not open and $\gamma(a) \leq 1$ for $a \in A^+$, as a mod $\mathfrak{p} \in B \subseteq R$, for such a. Thus the equivalence class of γ is in $\mathrm{Spa}(A, A^+)_a$ and all $\gamma(f_i) = 0$, contradicting b), provided that we can show that γ is continuous.

We first show that $K_{\gamma \leq s^m}$ is a neighbourhood of 0, for all $m \in \mathbb{N}$¿ If $m = 0$, this follows from $A^+ \subseteq K_{\gamma \leq 1}$ and the openness of A^+. If there is $a \in A$ such that $\gamma(as^m) \geq 1$, then $K_{\gamma \leq s^m}$ contains the preimage under $A \overset{a}{\to} A$ of $K_{\gamma \leq 1}$, hence it is a neighbourhood of 0. Let $\alpha \in A \backslash \mathfrak{p}$. Then $\mathfrak{p} + \alpha A$ is an open ideal. Since $s \in A^{\circ\circ}$, $s^n \in \mathfrak{p} + \alpha A$ for some positive integer. Thus there is $b \in A$ such that $s^n - b\alpha \in \mathfrak{p}$ or

$$\gamma(b)\gamma(\alpha) = \gamma(s^r)$$

hence $\gamma(b)\gamma(\alpha) > \gamma(s^{n+1})$. (Note that $(s \mod \mathfrak{p}) \in \mathfrak{m}_R$ as $(s \mod \mathfrak{p})^{-1} \in R$, hence $\gamma(s) < 1$). Thus $b^{-1} K_{\gamma \leq s^{n+1}}$ which is a neighbourhood of 0 which is contained in $K_{\gamma < \alpha}$. By Fact 1 d), γ is continuous. □

Corollary 3.3. *Under the assumption of Proposition 3.29 on A and A^+, $\langle a_1, \cdots, a_m \rangle_A = A$ if and only if*

$$\max_{1 \leq i \leq m} |a_i|_x > 1$$

for all $x \in \mathrm{Spa}(A, A^+)$.

Proof. The necessity is obvious. If the condition holds, $I = \langle a_1, \cdots, a_m \rangle_A$ is open by Proposition 3.29. If it is different from A, then there is an open maximal ideal $\mathfrak{m}$ such that $I \subseteq \mathfrak{m}$. If $x \in \mathrm{Spa}(A, A^+)$ is given by the trivial valuation for $\mathfrak{m}$, then $|a_i|_x = 0$, contradicting to our assumption. □

Remark 3.12. *In particular, $a \in A^\times$ if and only if $|a|_x \neq 0$ for all $x \in \mathrm{Spa}(A, A^+)$ under the same assumption as in Proposition 3.29.*

Lemma 3.1. *If the map gf is open then g is open provided that f and g are continuous homomorphism of topological groups.*

Proof. Such g are open if and only if $g(U)$ is a neighbourhood of 1 if U is a neighbourhoodof 1. (Then as $g(U) = g(Ux^{-1})g(x)$ is a neighbourhood of $g(x)$ under U is a neighbourhood of x). But $g(U)$ contains $(gf(f^{-1}(U)))$ and this is a neighbourhood of 1 as $f^{-1}(U)$ is a neighbourhood of 1 and gf is open.. □

Proposition 3.30. *Let A be a topological ring and M be a topological A-module. Let $m, n \in \mathbb{N}$ and $\vec{a} = (a_i)_{i=1}^m \in A^m$, $I = \langle a_1, \cdots, a_m \rangle$, $\vec{b} = (b_i)_{i=1}^n \in I^n$. If*

$$M^n \overset{\vec{b}}{\to} M, (m_i)_{i=1}^n \mapsto \sum_{i=1}^{n} b_i m_i$$

is open, then $M^m \overset{\vec{a}}{\to} M$ is also open.

Proof. Set

$$b_i = \sum_{j=1}^{m} c_{ij} a_j.$$

Then we have the commutative diagram,

$$\begin{tikzcd} M^n \arrow[d, "(c_{ij})_{ij}"'] \arrow[dr, "\vec{b}"] & \\ M^m \arrow[r, "\vec{a}"'] & M \end{tikzcd}$$

Now apply Lemma 3.1. $\square$

Remark 3.13. *In particular, the openess of the map $\vec{a} : M^m \to M$ depends only on $I = \langle a_1, \cdots, a_m \rangle$.*

Proposition 3.31. *Let A be a topological ring and M be a topological A-module.Then the following conditions are equivalent.*

a). For every neighbourhood U of 0, there are $m \in \mathbb{N}$ and $\vec{a} \in U^m$ such that $M^m \xrightarrow{\vec{a}} M$ is open.

b). When $I \subseteq A$ is an open ideal, then a) holds with $U = I$. (A direct corollary of Proposition 3.30).

If A has (oifin), this is equivalent to

c). If $m \in \mathbb{N}$ and $\vec{a} \in A^m$ generates an open ideal, then $M^m \xrightarrow{\vec{a}} M$ is open. (Also trivial as I in b) can be assumed to be finite generated under (oifin)).

Definition 3.17. *An A-module M is called adic if and only if the equivalent conditions a) and b) hold in Proposition 3.31. An A-algebra is called adic if and only if it is adic as an A-module.*
A is called id*-adic if and only if* id_A *is an adic morphism.*

Proposition 3.32. *Every* id*-adic ring satisfies (oifin) and (oisq).*

Proof. If I is an open ideal. Then there are $(a_j)_{j=1}^m \in I^m$ such that $A^m \xrightarrow{\vec{a}} A$ is open. Then $\langle a_1, \cdots, a_m \rangle_A$ is open and it is the image of $\vec{a}$. Thus (oifin). Moreover, the image of I^m under $\vec{a}$ is also open and contained in the ideal square gI, which is therefore open. $\square$

Proposition 3.33. *If A is analytic or satisfies (3.1.E), then very A-module (hence every A-algebra) is adic. in particular, A is* id*-adic.*

Proof. If A is analytic, there are $(a_i)_{i=1}^m \in (A^{\circ\circ})^m$ such that $\langle a_1, \cdots, a_m \rangle = A$. Then also $\langle a_1^j, \cdots, a_m^j \rangle = A$ for all $j \in \mathbb{N}$. Thus $I = A$ for all open ideal I, and one can take $n = 1, a_1 = 1$ in Proposition 3.31 b). Condition (3.7 E0)likewise implies that A is the only open ideal. $\square$

Proposition 3.34. *Let $A \xrightarrow{f} B$ be a morphism of topological rings and M be a topological B-module which is adic as an A-module .The nM is adic as a B-module. In particular, if B is an adic A-algebra, then B is* id*-adic.*

Proof. Let $I \subseteq B$ be an open ideal. There are $\vec{a} \in (f^{-1}I)^m$ such that $M^m \xrightarrow{\vec{a}} M$ is open (induced by multiplicatio n$A \times M \to M$). Then $M^m \xrightarrow{f(\vec{a})} M$ (induced from the multiplication $B \times M \to M$) is also open. $\square$

Proposition 3.35. *Let $B \subseteq A$ be an open subring, M be a topoological A-module, N an open sub-B-moudle of M. Then M is adic as an A-module if and only if N is adic over B.*

Proof. $\square$

3.3 Nat-rings, Nat-modules and Huber Ring

Definition 3.18. *A commutative (additively written) topological group is called nat (for non-Archimedian topological) if and only if it has a neighbourhood base of* 0 *where each element is an open subgroup. A nat-ring is a topological ring A such that $(A,+)$ is a nat. For a nat-ring A, a nat-A-module is a topological A-module M such that $(M,+)$ is a nat. A morphism of nat-rings (of nat-A-modules) is a morphism of topological rings (A-modules) where source and target are nat.*

Example 3.8. *$\mathbb{Q}_p$ with topology given by $|\cdot|_p$ or any other field with ultrametric absolute value. Any ring with its discrete topology.*

Remark 3.14. *For any topological ring A, A^{nat} is a topological ring A equipped with the topology where the open subgroups of $(A,+)$ are a neighbourhood base of* 0*. Then A^{nat} is a nat-ring. Every morphism $A \to B$ of topological ring with B being nat factors over A^{nat}, and $\mathrm{Cont}(A^{\mathrm{nat}}) \to \mathrm{Cont}(A)$ is bijective.*

Proposition 3.36. *Let R be a topological ring M be a R-module which is nat, $X \subseteq M$ a subset of M, $X' \subseteq M$ be the subgroup generated by X. Then X is bounded if and only if X' is.*

Proof. $X'X = X$, thus $\Leftarrow$ is clear. For the other, let $U \subseteq M$ be a neighbourhood of 0. Without loss of generality, U is a subgroup of $(M,+)$. If X is bounded, there is a neighbourhood V of 0 in R such that $VX \subseteq U$. Since U is a subgroup, $VX' \subseteq U$. Thus X' is bounded. □

Proposition 3.37. *Let A be a nat, $X \subseteq A$ be a subset, X' be a subgroup generated by X. Then X is power-bounded if and only if X' is. (Respectively, it is topologically nilpotent and power bounded if X' is).*

Proof. Let $Y = \bigcup_{n=0}^{\infty} X^n$ and $Y' \subseteq A$ be the subgroup generated by Y. Then Y'' also be the subgropu generated by $\bigcup_{n=0}^{\infty} X'^m$. $Y' \subseteq Y''$ is clear. Moreover, $X'Y' \subseteq Y'$, $(XY' \subseteq Y')$ is clear and

$$\{\xi \in A \mid \xi Y \subseteq Y\}$$

is a subgroup. Thus $X'^m \subseteq Y'$ by induction on m. Thus , $Y'' \subseteq Y'$. By Proposition 3.36, X is powerbounded if and only if Y is bounded if andonly if $Y' = Y''$ bounded if and only if X' is power bounded.The latter case is similar. □

Proposition 3.38. *Let A be a nat-ring, $X \subseteq A$ be a subset and $X' \subseteq A$ be the subring generated by X. Then the following condition are equivalent.*

a). X is powerbounded.

b). X' is bounded.

c). X' is powerbounded.

Proposition 3.39. *Let A be a nat ring, $(X_i)_{i=1}^m$ are power bounded subsets. Then the subring generated by X_i is powerbounded. (Apply Proposition 3.38 to $\bigcup_{i=1}^m X_i$ which is bounded, see 3.1).*

Proposition 3.40. *For bounded subgroups B_1, B_2 of a nat ring A are contained in a bounded subring $B \subseteq A$.*

Proposition 3.41. *For a nat ring A, A° is a subring of A which is integrally closed in A, $A^{\circ\circ}$ is an ideal in A° and $\sqrt{A^{\circ\circ}} = A^{\circ\circ}$. If A is (tn;1), then A° is clopen in A.*

Proof. That $A^{\circ\circ}$ is a subgroup of $(A^\circ,+)$ follows from Proposition 3.36 and $A^\circ A^{\circ\circ} \subseteq A^{\circ\circ}$ follows from Fact 3.1.5. That $\sqrt{A^{\circ\circ}} = A^{\circ\circ}$ follows from the definition. If $r^n \in A^{\circ\circ}$ with $n > 0$, then

$$\lim_{j\to\infty} r^{i+jn} = 0$$

for all $0 \leq i < n$. Hence $\lim_{j\to\infty} r^j = 0$. If $x \in A$ is integral over A°, then there are $n \in \mathbb{N}$ and $(p_j)_{j=0}^{n-1} \in A^\circ$ such that

$$x^n = \sum_{j=0}^{n-1} p_j x^j.$$

By Proposition 3.38, there is a bounded subring $S \subseteq A$ containning all p_j. Then X^N is contained in the subgroup generated by S and $\{(x_i)_{i=0}^{n-1}\}$ which is bounded by Proposition 3.36 and Fact 3.1.3. If (tn;1) holds then $A^{\circ\circ}$ is a neighbourhood of 0 hence as in A° and A° is open, hence closed subgroup of $(A, +)$. ☐

Remark 3.15. *Note that A° can be unbounded (e.g. $A = \mathbb{Q}_p[T]/T^2\mathbb{Q}_p[T]$, $A^\circ = \mathbb{Z}_p + T \cdot \mathbb{Q}_p$) and $A^{\circ\circ}$ will often happen to be topologically nilpotent. For example, $A = \mathbb{C}_p$, then*

$$A^{\circ\circ} = \{x \in \mathbb{C}_p \mid |x| < 1\}$$

such that $(A^{\circ\circ})^2 = A^{\circ\circ}$.

Proposition 3.42. *Let A be a complete nat with (tn;1) (e.g. $A^{\circ\circ}$ is open subgroup). If A is sequentially compact, then $A^\times$ is an open subset of A, $A^\times \overset{a\mapsto a^{-1}}{\to} A$ is continuous, and A has no dense proper ideals (hence no dense maximal ideals).*

Proof. $A^\times$ contains $A^{\circ\circ} + 1$ (a neighbourhood of 1), hence by multiplicative invariance, it is a neighbourhood of each of its elements, by $(1+n)^{-1} = \sum_{j=0}^\infty n^j$. If $I \subseteq A$ is a dense ideal, $1 \in \overline{I}$, hence $\overline{I} \cap A^\times \neq \emptyset$, hence $I \cap A^\times \neq \emptyset$, hence $I = A$. ☐

Proposition 3.43. *Let A be a nat ring, B be a nat A-algebra which is adic over A, $X \subseteq AA$ bounded. Then the image of X in B is bounded.*

Proof. Let U be a neighbourhood of 0 in B, we need to find a neighbourhood V of 0 in B such that $Vf(X) \subseteq U$, where $A \xrightarrow{f} B$ is the homomorphism making B into an A-algebra. There is a neighbourhood U of 0 in B such that $U'U' \subseteq U$ and without loss of generality, U and U' can be assumed to be subgroups of $(A, +)$. Let $W = f^{-1}(U') \subseteq A$, there is $\Omega \subseteq A$ such that $\Omega X \subseteq W$ and by the assumption, there are $(\omega_i)_{i=1}^n := \vec{\omega}$ such that $B^n \xrightarrow{\vec{\omega}} B$ is an open map. Then V which is the image of U'^n under this open map will do. For $r = f(\omega_i)u$ with $u \in U'$, then $rf(X) \subseteq f(\Omega X)U' \subseteq f(W)U' \subseteq U'U' \subseteq U$, and write? U as a subgroup and the general elements of V is a sum of such r, we indeed have $Vf(X) \subseteq U$. ☐

Remark 3.16. *In general, A° may well be unbounded in A for example,*

$$A = \mathbb{Q}_p[T]/T^2, A^\circ = \mathbb{Z}_p + \mathbb{Q}_p T.$$

But every finite subseet of it is contained in some bounded subring.

Definition 3.19. *Let A be a ring and $I \subseteq A$ be an ideal. The I-adic topology on A is the topology for which $(A, +)$ is a topological group and $(I^n)_{n=0}^\infty$ is a neighbourhood base of* 0.

Proposition 3.44. *For a nat ring A, the following conditions are equivalent.*

a). A is id*-adic satisfies (tn;1) and has an open bounded subring B.*

b). A is id*-adic, satisfies (tn;0) and has an open bounded subring B.*

c). A has an open bounded subring B for which there is a finitely generated ideal $I \subseteq B$ such that the topology on B is the I-adic one.

Proof. For b).$\Rightarrow$ a), as A is id-adic and $A^{\circ\circ}$ generates an open ideal ((tn;0)), there are $(a_i)_{i=1}^m \in A^{\circ\circ m}$ such that $A^m \xrightarrow{\vec{a}} A$ is an open map. The image of B^m under this map is the open and topologically nilpotent since B is powerbounded and all a_i are topologically nilpotent.

a).$\Rightarrow$ b), is trivial.

b),$\Rightarrow$ c), If the (a_i) are as above, then for each $n \in \mathbb{N}$, the ideal generated by the a_i^n is also open, as A has (oisq). Then without loss of generality, all $a_i \in B$ (replace a_i by a_i^n which is in B when n is large). Then $I = \langle a_1, \cdots, a_m \rangle_B$ can be taken. Indeed, $B^m \xrightarrow{\vec{a}} B$ is open and $I^{\cdot(j+1)}$ is the image of $(I^{\cdot j})^{\oplus m}$ (using $\cdot$, $\oplus$ to distinguish ideal powers and Cartesian powers) under this open map, hence open by induction on j. If U is a neighbourhood of 0 in A, then since B is bounded, there is a neighbourhood U' of 0 such that $U'B \subseteq U$ where U is a subgroup of $(A, +)$ without loss of generality. As the a_i are topologically nilpotent, there is N such that $\vec{a}^\alpha = a_i^{\alpha_i} \cdots, a_m^{\alpha_m} \in U'$ where $|\alpha| \geq N$. Then $I^{\cdot N} \subseteq U$, showing I^n are indeed a neighbourhood base of 0.

c)$\Rightarrow$ a). Since $I \subseteq A^{\circ\circ}$ and (tn;1) holds. Then, only id-adicity need to be verified. If J is an open ideal in A there is N such thta $I^{\cdot N} \subseteq J$ (as the I^n form a neighbourhood base of 0). Then if $I^{\cdot N}\langle b_1, \cdots, b_l \rangle_B$ then $A^l \xrightarrow{\vec{b}} A$ is open. (it maps $(I^{\cdot j})^{\oplus l}$ onto $I^{\cdot j+N}$ which is open and hence $I^{\cdot j}$ form a neighbourhood base of 0 in A). Since all $b_i \in J$, A is id-adic as claimed. ☐

Definition 3.20. *A nat-ring with these properties in Proposition 3.44 is called a Huber ring (In Huber's note, it is refered as f-adic).*

Definition 3.21. *A ring of Def of a Huber ring is an open bounded subring of A. A pair of def is a pair (B, I) where B is an open bounded subring and $I \subseteq B$ a finitely generated ideal such that the topology on B is I-adic.*

Corollary 3.4. *Every ring of def of a Huber ring can be extended to a pair of def.*

Proof. Proved in b)$\Rightarrow$c) in Proposition 3.44. ☐

Proposition 3.45. *Every Huebr ring is (oisq), (oifin) (from Fact 3.1.19), and (tn;1) (from Proposition 3.44 b)). Moreover A7○ is clopen and integrally closed subring of A. If A is (sequentially) complete (note that the $I^{\cdot j}$ form a countable neighbourhood base of 0) then $A^\times$ is an open subset of A and A has no proper dense ideal (every maximal ideal is then closed).*

Proposition 3.46. *Let B be an open subring of the nat ring A. Then B is Huber if and only if A is. In this case, B has a pair of def which is a pair of def for a).*

Proposition 3.47. *Let (B, I) be a pair of def for the Huber ring A. Then a neighbourhood X of A is bounded if andonly if there is $l \in \mathbb{N}$ suchthat $I^{\cdot l}X \subseteq B$ (where $I^{\cdot l}$ is an ideal power of I in B as before).*

Proposition 3.48. *Every Huber ring has a countable neighbourhood base of 0, the completion of a Huber ring is a Huber ring (If (B, I) is a pair of def, then $\hat{B}$ is an open subring of $\hat{A}$ and the topology of $\hat{B}$ is $\hat{I}$-adic with $\hat{I} = \langle f_1, \cdots, f_n \rangle_{\hat{B}}$ where $I = \langle f_1, \cdots, f_m \rangle_B$).*

Definition 3.22. *A Huber pair (affnoid pair in Huber) is a pair $A = (A^\triangleleft, A^+)$ such that $A^\triangleleft$ is a Huber ring and $A^+ \subseteq A^{\triangleleft\circ}$ is an open and integrally closed subring of $A^\triangleleft$ (hence $A^{\triangleleft\circ\circ} \subseteq A^+$).*

A Huber pair is complete if $A^\triangleleft$ (or, equivalently A^+) is (sequential) complete. A morphism $A \to B$ of Huber pairs is a continuous ring homomorphism $A^\triangleleft \to B^\triangleleft$ mappign A^+ to B^+. it is adic if $A^\triangleleft \to B^\triangleleft$ is adic

Theorem 3.1. *Let $A = (A^\triangleleft, A^+)$ be a Huber pair and $X = \operatorname{Spa} A$.*

a). $A^+ = \{a \in A^\triangleleft \mid |a|_x \leq 1$ for all $x \in X\}$.

b). X is a spectral space.

If A is complete, we also have,

c). The ideal $\langle a_1, \cdots, a_m\rangle_{A^\triangleleft}$ is open if and only if $\max_{1\le i\le m}|a_i|_x > 0$ for all $x \in X_a$ (the analytic points of X). This is the case if and only if $A^{\triangleleft m} \xrightarrow{\vec{a}} A$ is an open map.

d). $A^\triangleleft = \langle a_1, \cdots, a_m\rangle_{A^\triangleleft}$ if and only if $\max_{1\le i\le m}|x_i|_x > 0$ for all $x \in X$.

A morphism of Huber pairs defines a continuous map $\operatorname{Spa} B \to \operatorname{Spa} A$ which is the case of an adic morphisms is a spectral map.

Proof. a) follows from Propositin 3.2.2. b) is Corollary 3.2.2. c) follows from Proposition 3.2.5 + id-adicity. d) follows from Corollary 3.2.3. That $Y = \operatorname{Spa} B \to X = \operatorname{Spa} A$ is continuous is easily observed for the strong topologies. If $B^\triangleleft/A^\triangleleft$ is adic and $\Omega \in \mathfrak{Rat}_X$, $\Omega = \mathcal{R}(a_0|a_1, \cdots, a_m)$ with $\langle a_0, \cdots, a_m\rangle_A$ open. As B/A is adic, $B^{m+1} \xrightarrow{\vec{a}} B$ is open hence $\langle a_0, \cdots, a_m\rangle_B$ is open hence $\mathcal{R}_Y(a_0|a_1, \cdots, a_m)$ the preimage of Ω in Y is rational open. It follows from Fact 1.2.6 that such maps are spectral. ☐

Remark 3.17. *Note that $D(T) = \mathcal{R}(T|)$ is quasicompact in $\operatorname{Spv} \mathbb{T}_{1,\mathbb{Q}_p}$ but not in $\operatorname{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ)$ as*

$$D(T) = \bigcup_{l=0}^{\infty} \mathcal{R}(T|p^l).$$

Thererfore $\operatorname{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ) \to \operatorname{Spv}(\mathbb{T}_1)$ is not spectral.

Proposition 3.49. *Let A be a complete Huber ring (or sequentially complete, nat,* id-*adic, (tn;1)). Let $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in A^{m+1}$ such that*

$$\langle f_0, \cdots, f_m\rangle_A$$

is open in A. Le $tX = \operatorname{Cont}(A)$. Put $f_i = 0$ when $i > m$. Then there is a neighbourhood U of 0 in A such that for $n \ge m$ and $(g_i)_{i=0}^m \in A^{m+1}$ such that $g_i - f_i \in U$ for $0 \le i \le n$, we have,

1). $I = \langle g_0, \cdots, g_n\rangle_A$,

2). $\mathcal{R}_X(f_0|f_1, \cdots, f_m) = \mathcal{R}_X(g_0|g_1, \cdots, g_n)$.

Proof. Let $\vec{f} = (f_i)_{i=0}^m$ and $\vec{g} = (g_i)_{i=0}^m$. By the assumption, $A^{m+1} \xrightarrow{\vec{f}\cdot} A$ is open, hence the image U of $(A^{\circ\circ})^{m+1}$ under $\vec{f}\cdot$ is a neighbourhood of 0. By our assumption about $\vec{g}$, all $g_i - f_i \in U \subseteq I$, hence all $g_i \in I$. We have $\vec{g} = (I_{m+1} + N)\vec{f}$ where I_{m+1} is an identity $(m+1)$-square matrix and N is an $(m+1)$-matrix with coefficients from $A^{\circ\circ}$. Then $det(I_{m+1} + N) \in 1 + A^{\circ\circ} \subseteq A^\times$ as the completeness of A is assumed. Hence $1_{m+1} + N$ is invertible and,

$$\langle f_0, \cdots, f_m\rangle = \langle g_0, \cdots, g_m\rangle.$$

For 2)., we distinguish the case $x \notin X_a$, ($\mathfrak{p} = \operatorname{supp} x$ open) and $x \in X_a$, ($\mathfrak{p} \in \operatorname{supp} x$ not open). In the first case, $A^{\circ\circ} \subseteq \mathfrak{p}$ ($a \in A^{\circ\circ} \Rightarrow a^n \in \mathfrak{p}$ when n is large $\Rightarrow a \in \mathfrak{p}$) hence $U \subseteq \mathfrak{p}$ hence $g_i - f_i \in \mathfrak{p}$ and $|f_i|_x = |g_i|_x$ and x is in the right hand side of 2).

In the second case, $M = \max_{0\le i\le m}|f_i|_x$ is positive (otherwise $I \subseteq \mathfrak{p}$ hence $\mathfrak{p}$ is open ad I is). For $a \in A^{\circ\circ}$, $|a|_x < 1$ as,

$$\lim_{n\to\infty}|a|_x^n = \lim_{n\to\infty}|a_i^n|_x = \left|\lim_{n\to\infty} a^n\right|_x = 0$$

by the continuity. Let $\rho = \max_{\substack{0\le i\le n\\ 0\le j\le m}}|n_{ij}|_x$ where

$$g_i - f_i = \sum_{j=1}^m n_{ij} f_j.$$

Then $|f_i - g_i|_x \le \rho M$ hence,

$$\max(|f_i|_x, \rho M) = \max(|g_i|_x, \rho M).$$

Thus, $|g_i|_x = M$ when $M = |f_i|_x$. Hence M ☐

Remark 3.18. *Of course from 2), the ? about* $\mathrm{Spa}(A, A^+) \subseteq X$ *follows.*

Definition 3.23. *A Tate ring is the sense of Hueber is a nat ring A such that* $A^{\circ\circ} \subseteq A^\times \neq \emptyset$ *(ie. 3.1 (T)) and such that A contains a bounded open subring. A pair of definition of a Tate ring A is a pair* (B, s) *where B is a bounded open subring of A and* $s \in B^{\circ\circ} \cap A^\times$*. A ring of definition is (as in the general case of Huber rings) as open bounded subring.*

Proposition 3.50.

a). If $s \in A^{\circ\circ} \cap A^\times$ *and* $B \subseteq A$ *open bounded then* $s^j \in B$ *for large enough* $j \in \mathbb{N}$*. Then* (B, s^j) *is a pair of definition. Hence every ring of definition can be completed to a pair of definition.*

b). If (B, s) *is a Tate pair of definition for A then* (B, sB) *is a pair of definition in the sense of Huber rings. Hence any Tate ring is Huber.*

c). If (B, s) *is a Tate pair of definition for A,* $(s^j B)_{j \in \mathbb{Z}}$ *or* $(s^j B)_{j=n}^{\infty}$ *for any* $n \in \mathbb{N}$ *are neighbourhood bases of* 0*, and moreover* $X \subseteq A$ *is bounded if and only if* $X \subseteq s^j B$ *for some* $j \in \mathbb{Z}$*.*

Remark 3.19. *Even in the Tate case,* A° *can fail to be bounded. For example,*

$$A = \mathbb{Q}_p[T]/(T^2), A^\circ = \mathbb{Z}_p + T\mathbb{Q}_p.$$

Remark 3.20. *If A is a Tate ring then the image of a bounded or power bounded subset of A under the morphism* $A \to B$ *of topological rings is bounded or power bounded. Then the universal property of* $A\langle X_1, \cdots, X_0\rangle$ *(Lemma 3.4.1 or Proposition 3.4.3) will take a*

3.4 The structure presheaf on $\mathrm{Spa}(A^\triangleleft, A^+)$

We start with the special case of a topological localization of $A^\triangleleft$ with respect to $f \in A^\triangleleft$ such that $A^\triangleleft \xrightarrow{f} A^\triangleleft$ is open, giving $\mathscr{O}_X(\mathcal{R}_X(f|))$.

Proposition 3.51. *Let A be a complete nat ring with a countable neighbourhood base of* 0 *(e.g. a complete Huber ring). Let* $f \in A$ *such that* $A \xrightarrow{f} A$ *is an open map and* $I = \bigcup_{j=0}^{\infty} \mathrm{Ker}(A \xrightarrow{f^j} A)$ *and* $\overline{I}$ *the closure of I in A. Then* $B = A/\overline{I}$ *is a complete nat ring,* $B \xrightarrow{f \bmod \overline{I}} B$ *open and* $A \to B$ *has the universal property for morphisms* $A \xrightarrow{\tau} T$ *such that* $T \xrightarrow{\tau(f)} T$ *is injective and T complete nat ring.*

Proof. It is clear that any such τ ?. $\ker(\tau)$ is an ideal and $\ker(\tau) f = \ker(\tau)$. Thus, the ? easily follows if $\overline{I} \cdot f = \overline{I}$ (or equivalently, $B \xrightarrow{f} B$ injective). For this, we have to consider $a \in A$ such that $fa \in \overline{I}$ and have to show $a \in \overline{I}$.

For this, let $(U_k)_{k=0}^{\infty}$ be a neighbourhood base of 0 in A, such that the U_k are open subgroups of $(A, +)$ and

$$A = U_0 \supseteq U_1 \supseteq \cdots$$

By the assumption $f \cdot U_k$ is open, hence ? $fa \in \overline{I}$. There are $i_k \in I$ such that $fai_k \in fU_K$, where $i_0 = 0$. Let $(d_k)_{k=1}^{\infty}$ be such that

$$d_k \in U_k, fd_k = i_{k-1} - i_k.$$

Then $fd_k \in I$ hence $d_k \in I$ as $I \cdot f = I$. Then

$$j = \sum_{k=0}^{\infty} d_k \in \overline{I} \tag{3.3}$$

and $f(a - j) = 0$. Hence $a - j \in I \subseteq \overline{I}$ hence $a \in \overline{I}$. ∎

Proposition 3.52. *If A is Huber (Tate, Analytic), then so is B (as these are identified by A/I).*

Lemma 3.2. *If $C \xrightarrow{\gamma} C$ is open then $(C/I) \overset{r \bmod I}{\to} (C/I)$ is always open.*

Proof. Let U be a neighbourhood of 0 in C/I then $V = \pi^{-1}U$ is also a neighbourhood of 0 in C where $C \xrightarrow{\pi} C/I$ then $\pi^{-1}(rU) \supseteq rV$ and rV is also a neighbourhood of 0 in C. □

Thus in Proposition 3.51, $B \xrightarrow{f} B$ is indeed open.

Corollary 3.5. *Let C be a complete nat ring with a countable neighbourhood base of 0, $J_0 \subseteq C$ a closed ideal and $f \in C$ such that $C \xrightarrow{f} C$ is an open map. Let $J = \bigcup_{j=0}^{\infty} J_0 f^j$ and $\overline{J}$ be the colosure of J in C, then $B = C/\overline{J}$ is a complete nat ring and $B \overset{f \bmod \overline{J}}{\to} B$ an injectiev open map. Moreover $G \to B$ has the universal property for morphismsms $C \xrightarrow{\tau} T$ to complete topological rings such that $\tau(J_0) = 0$ and $T \overset{\tau(f)}{\to} T$ is injective. In addition, if C is Huber (Tate, analytic), then so is B.*

Proof. We would like to apply Proposition 3.51 to $A = C/J_0$. Let

$$I = \bigcup_{j=0}^{\infty} (0(f \bmod J_0)^j)_A = \bigcup_{j=0}^{\infty} J_j/I = J/I$$

where $J_j = J \cdot f^j$. ? the ideal contained in Proposition 3.51 ? ? follow if $\pi(\overline{J}) = \overline{I}$ whereLet $B \xrightarrow{\pi} A$ be a projection. We have $\pi(J) = I$. We have $\pi(\overline{J}) \subseteq \overline{I}$. On the other hand,

$$\overline{I}/J_0 = \pi(\overline{J})$$

is also closed as $\pi^{-1}(\overline{I}/J_0) = \overline{I}$. Indeed in ? $\pi(\overline{I})$ is closed but $I \subseteq \pi(J) \subseteq \pi(\overline{J})$ hence also $\overline{I} \subseteq \pi(\overline{J})$. Then $\overline{I} = \pi(\overline{J})$ as needed. □

Proposition 3.53. *Let A be a nat ring and $f \in A$ such that $A \xrightarrow{f} A$ is an injective and open map. Consider A as a subring of the localization $A_f = A[T]/\langle 1 - T \cdot f \rangle_{A[T]} = A[f^{-1}]$. Equip $(A_f, +)$ with the topology fro which U is a neighbourhood of 0 in A_f if and only if $U \cap A$ is a neighbourhood of 0 in A. Then we have the following.*

a). A_f is a nat ring and has the universal property for morphisms $A \xrightarrow{\tau} T$ of topological rings such that $\tau(f) \in T^{\times}$.

b). A is an open subring of A_f.

c). A is Huber (respectively comp)

Proof. The only hard point is the continuity of $A_f \times A_f \to A_f$. Certainley $A \times A \to A \subseteq A_f$ is continuous. Note that $(A_f, +) \xrightarrow{f \cdot} (A_f, +)$ is an isomorphism of topological groups. If U is a neighbourhood of 0, $(f \cdot U) \cap A \supseteq f(U \cap A)$ which is open by out assumption. Thus fU is a neighbourhood of 0, showing the continuity of the inverse of $A_f \xrightarrow{f \cdot} A_f$. Now take $m \in \mathbb{Z}, (f^{-m}A, +)$ is an open subgroup of $(A, +)$ and the continuity of

$$(f^{-m}A) \times (f^{-m}A) \to A_f$$

follows from the diagram belowin which the vertical arrows are homeomorphisms and the bottom horizontal arrow is continuous by our initial observation. As $(A_f, +)$ is the ascending union of its open subgroups $f^{-m}A$, this shows the continuity of $A_f \times A_f \to A_f$.

$$\begin{tikzcd}
(f^{-m}A) \times (f^{-m}A) \arrow[r, "\cdot"] \arrow[d, "(f^m \cdot) \times (f^m \cdot)"'] & A_f \arrow[d, "f^{2m}"] \\
A \times A \arrow[r] & A_f
\end{tikzcd}$$

If $A \xrightarrow{\tau} T$ is a morphism of topological rings and $\tau(f) \in T^\times$, then we have a ring homomorphism $A_f \xrightarrow{\tilde{\tau}} T$. This is continuous. If U is a neighbourhood of 0 in T. Then $\tilde{\tau}^{-1}U \cap A = \tau^{-1}U$ is a neighbourhood of 0 in A by continuity of τ, hence $\tilde{\tau}^{-1}U$ is a neighbourhood of 0 in A_f. The rest is also easy. ☐

Corollary 3.6. *In the situation of Corollary 3.5, $B_{f \bmod \overline{J}}$ is a complete nat-ring and $C \to B_{f \bmod \overline{J}}$ has the universal property for morphisms $C \xrightarrow{\tau} T$ such that $\tau(J_0)$ and $\tau(f) \in T^\times$. If C is Huber, (Tate, analytic), then so is $B_{f \bmod \overline{J}}$.*

Definition 3.24. *Let A be a nat-ring. We equip $A[X_1, \cdots, X_n]$ with the topology such that $(A[X_1, \cdots, X_n])$ is a topological group and a neighbourhood base of 0 is given by*

$$\{U[X_1, \cdots, X_n] \mid U \text{ a neighbourhood of } 0 \text{ in } A\},$$

where

$$U[X_1, \cdots, X_n] = \left\{ f = \sum_{\alpha \in \mathbb{N}^n} f_\alpha X^\alpha \in A[X_1, \cdots, X_n] \mid \forall \alpha \in \mathbb{N}^n, f_\alpha \in U \right\}.$$

We define $A\langle X_1, \cdots, X_n\rangle$ as the completion of $A[X_1, \cdots, X_n]$. If M is an A-module, then $M[X_1, \cdots, X_n]$ a module over $A[X_1, \cdots, X_n]$, is the set of formal sum $\sum_{\alpha \in \mathbb{N}^n} m_\alpha X^\alpha$ where $m_\alpha \in M$, all but finitely many $m_\alpha = 0$. The neighbourhood base of 0 is givenby $U[X_1, \cdots, X_n]$, where U is a neighbourhood of 0 in M, as above. $M\langle X_1, \cdots, X_n\rangle$ is defined as the compltion of $M[X_1, \cdots, X_n]$.

Lemma 3.3. *Let A be a nat-ring, B be a nat-A-algebra.*

a). $A[X_1, \cdots, X_n]$ is a nat ring in which $\{X_1, \cdots, X_n\}$ are powerbounded.

b). If $C \subseteq A$ is an open subring, $C[X_1, \cdots, X_n]$ is an open subring of $A[X_1, \cdots, X_n]$. (Take $U = C$ in Definition 3.24).

c). If A is Huber(Tate, analytic), then so is $A[X_1, \cdots, X_n]$. (If (C, I) is a pair of definition for A, so is $(C[X_1, \cdots, X_n], I[X_1, \cdots, X_n])$ for $A[X_1, \cdots, X_n]$. If $(s_1, \cdots, s_n) \in A^{\circ\circ n}$ such that $\langle s_1, \cdots, s_n\rangle_A = A$ then they do the same in $A[X_1, \cdots, X_n]$).

d). If $(b_i)_{i=1}^n$ are power bounded in B, then the homomorphism,

$$A[X_1, \cdots, X_n] \ni p \mapsto p(b_1, \cdots, b_n) \in B,$$

is continuous.

e). If B is an adic A-algebra (e.g. as A is Tate or analytic), then we have a bijection,

$$\mathrm{Hom}_{(A-\mathrm{alg})^{\mathrm{top}}}(A[X_1, \cdots, X_n], B) \ni \varphi \overset{\cong}{\mapsto} (\varphi(X_i))_{i=1}^n (B^\circ)^n.$$

(use d) for the surjectivity and well-definedness follows from Fact 3.3.6).

f). If $\langle f_0, \cdots, f_n\rangle_A$ is open, $I = \langle f_0X_1 - f_1, \cdots, f_0X_n - f_n\rangle_{A[X_1, \cdots, X_n]}$ then $A[X_1, \cdots, X_n]/I \xrightarrow{f_0} A[X_1, \cdots, X_n]/I$ is an open map.

g). If M is an adic A-module, then so is $M[X_1, \cdots, X_n]$. in particular $A \to A[X_1, \cdots, X_n]$ is adic if A is id*-adic.*

Proof. That $(A, +)$ is indeed a nat-group follows from the fact that the U in Definition 3.24 can be taken to be open subgroups of $(A, +)$. Then $(U[X_1, \cdots, X_n], +)$ is a subgroup of $A[X_1, \cdots, X_n]$.

If $P, Q \in A[X_1, \cdots, X_n]$ and U an open subgroup of $(A, +)$, we have a neighbourhood of 0, V such that

· All $P_\alpha V \subseteq U$,

· All $Q_\alpha V \subseteq U$,

· $V \cdot V \subseteq U$.

(Note that only finitely many $\alpha \in \mathbb{N}^n$, $P_\alpha, Q_\alpha \neq 0$). Then

$$(P + V[X_1, \cdots, X_n])(Q + V[X_1, \cdots, X_n]) \subseteq (PQ) + U[X_1, \cdots, X_n].$$

Thus, multiplication is continuous.

For the powerboundedness assertion, we need to show that

$$\Theta = \{X^\alpha \mid \alpha \in \mathbb{N}^n\}$$

is bounded. For every neighbourhood U' of 0, we must find a neighbourhood V' of 0 such that $\Theta V' \subseteq U'$. By Definition 3.24, it suffices to consider $U' = U[X_1, \cdots, X_n]$ in which case $\Theta U' \subseteq U'$ such that $V' = U'$ will do.

d). Consider $A[X_1, \cdots, X_n] \overset{\varphi}{\to} B$ such that $\varphi(P) = P(b_1, \cdots, b_n)$. This is a ring homomorphism. To show its continuity, consider a neighbourhood Ω of 0 in B. We must find a neighbourhood V' of 0 in $A[X_1, \cdots, X_n]$ such that $\varphi(U') \subseteq \Omega$. Without loss of generality, Ω is a subgroup of $(B, +)$. By the powerboundedness of $\{b_1, \cdots, b_n\}$, there is a neighbourhood V of 0 in B such that $b^\alpha V \subseteq \Omega$. Lt $U' = U[X_1, \cdots, X_n] \subseteq A$ where U is the preimage of V in A. If $P \in U[X_1, \cdots, X_n]$, then all $P_\alpha b^\alpha \in b^\alpha V \subseteq \Omega$ hence $\varphi(U') \subseteq \Omega$ as Ω is a subgroup of $(B, +)$.

f). Let U be an open subgroup of $(A, +)$ and $V \subseteq A$ the image of U^{n+1} under $A^{n+1} \overset{\vec{f}}{\to} A$. if $B = A[X_1, \cdots, X_n]/I$ then the image of $U[X_1, \cdots, X_n]$ in B under $(f_0 \bmod I)$, contains the image of $V[X_1, \cdots, X_n]$ in B, which is an open subgroup of $(B, +)$. This will prove f) as the image of $V[X_1, \cdots, X_n]$, U running over open subgroups of A, form a neighbourhood base of 0 in B.

For the proof of our claim, let $P = \sum_{\alpha \in \mathbb{N}^n} P_\alpha X^\alpha \in V[T]$. There are $(q_{\alpha,j})_{j=0}^n \in U^{n+1}$ for each $\alpha \in \mathbb{N}^{n+1}$ such that $p_\alpha = \sum_{j=0}^n f_\alpha q_{\alpha,j}$, where we assume $q_{\alpha,j} = 0$ when $p_\alpha = 0$. Then,

$$\begin{aligned} Q_0 &= \sum_{\alpha \in \mathbb{N}^n} \left(q_{\alpha,j} + \sum_{j=1}^n q_{\alpha - d_j, j} \right) X^\alpha, \\ Q_j &= \sum_{\alpha \in \mathbb{N}^n} q_{\alpha,j} X^\alpha \end{aligned}$$

where $d_j = (\delta_{ij})_{i=0}^n$, are elements of $U[X_0, \cdots, X_m]$ and $P = f_0 Q_0 + \sum_{j=1}^n (f_j - f_0 X_j) Q_j$, hence $(P \bmod I) = f_0(Q_0 \bmod I)$ as needed.

If Ω is a neighbourhood of 0 in A, then by the assumption, there are $(\omega_i)_{i=1}^n \in \Omega$ such that $M^m \overset{\vec{\omega}}{\to} M$ is an open map. We claim that $M[X_1, \cdots, X_n]^m \overset{\vec{\omega}}{\to} M[X_1, \cdots, X_n]$ is also open which shows g). But this is obvious. If U is an open subgroup of M then so is the image V of U^m under $\vec{\omega}$, and the $\vec{\omega}$ maps $U[X_1, \cdots, X_n]^n$ onto $V[X_1, \cdots, X_n]$ which is an open subgroup of $M[X_1, \cdots, X_n]$ by Definition 3.24. ∎

Proposition 3.54. *Let $X \overset{f}{\to} Y$ be a continous homomorphism between abelian nat-groups which have countable neighbourhood bases of 0. Let $\hat{X}$ and $\hat{Y}$ be their completions. If f is an open map then so is $\hat{X} \overset{\hat{f}}{\to} \hat{Y}$.*

Proof. There are open subgroups $(X_i)_{i=1}^\infty$ of X forming a neigh base of 0, with

$$X_1 \supseteq X_2 \supseteq \cdots$$

As $f(X_i)$ is open, there is a countable neighbourhood base,

$$Y_1 \supseteq Y_2 \supseteq \cdots$$

of Y such that $Y_i \subseteq f(X_i)$ is an open subgroup of Y. It sufices to show that $\hat{f}(\hat{X}_i)$ is an open subgroup of $\hat{Y}$ and for this, it suffices to show that $\hat{Y}_i \subseteq \hat{f}(\hat{X}_i)$. An element

y of $\hat{Y}_i$ is represented as an equivalence class of Cauchy sequence $(y_j)_{j=1}^{\infty}$ with elements $y_i \in Y_i$. Without loss of generality (taking subsequences), we assume $y_{j+i} - y_j \in Y_{i+j}$ for all $j \geq 1$. By assumption, there is $x_1 \in X_1$ such that $f(x_1) = y_1$ and there are $(\xi_j)_{j=1}^{\infty}$ such that $\xi_j \in X_{i+j}$ such that

$$f(\xi_j) = y_j + iy_j.$$

Let

$$x_j = x_1 + \sum_{l=1}^{j-1} \xi_l.$$

Then

$$x_{j+1} - x_j = \xi_j \in X_{i+j}. \tag{3.4}$$

Therefore $(x_j)_{j=1}^{\infty}$ is Cauchy and $f(x_j) = y_j$. The equivalence class x of $(x_i)_{i=1}^{\infty}$ is al element $\hat{X}_i$ such that $f(x) = y$. □

Proposition 3.55. *Let A be a nat-ring with a countable neighbourhood base of 0 and B be a complete nat-A-algebra.*

a). $A\langle X_1, \cdots, X_n\rangle$ is a nat ring in which $\{X_1, \cdots, X_n\}$ is power bounded. (use $U\langle \cdots\rangle = \widehat{U[\cdots]}$ are X_i·-invariant)

b). If C is an open subring of A then $C\langle X_1, \cdots, X_n\rangle$ is an open subring of $A\langle X_1, \cdots, X_n\rangle$ in which $C[X_1, \cdots, X_n]$ is dense.

c). If A is Huber (Tate, analytic) then so is $A\langle X_1, \cdots, X_n\rangle$ (For Huber if C in b) is $\langle c_1, \cdots, c_n\rangle_C$-adic, then $C\langle\cdots\rangle$ is $\langle c_1, \cdots, c_n\rangle$-adic.

d). If $(b_i)_{i=1}6n \in B^{\circ n}$, then there is a unique continuous $A\langle X_1, \cdots, X_n\rangle \to B$ sending X_i to b_i (Lemma 3.3 + Completeness).

e). If B is adic overr A (e.g. A is Tate or analytic or ? 3.1(E)), then we have a bijection,

$$\mathrm{Hom}_{\mathrm{nat}A-\mathrm{alg}}(A\langle X_1, \cdots, X_n\rangle, B) \ni \varphi \overset{\simeq}{\mapsto} (\varphi(X_i))_{i=1}^n \in (B^\circ)^n.$$

(All such φ are adic Fact 3.2.21. use Fact 3.3.6 for powerboundedness).

f). If $\langle f_1, \cdots, f_n\rangle_A$ is open and $I = A\langle X_1, \cdots, X_n\rangle$ is an arbitrary closed ideal containing all $f_0 X_i - f_i$ and $B = A\langle X_1, \cdots, X_n\rangle / I$. We have $B \xrightarrow{f_i} B$ is open. (Proposition 3.54 applied with $I = \overline{\langle f_0 X_i - f_i\rangle}_{A\langle X_1, \cdots, X_n\rangle}$ Then any larger I can be derived from this (Proposition 3.54)).

g). if M is an adic A-module with a countable neighbourhood base of 0 (e.g. $M = A$ if A is id-adic). Then $M\langle X_1, \cdots, X_n\rangle$ is an adic A-module (Proposition 3.54).

Theorem 3.2. *Let $A = (A^{\triangleleft}, A^+)$, $X = \mathrm{Spa}\, A, \Omega = \mathcal{R}_X(f_0|f_1, \cdots, f_m)$ a rational open subset, $\langle f_0, \cdots, f_n\rangle_A$ being open in A. Let $\mathcal{P}$ be the category of complete Huber pair $(B^{\triangleleft}, B^+)$.*

a). The functor F_Ω such that

$$C \mapsto F_\Omega(C) = \{\varphi \in \mathrm{Hom}(A, C) \mid (\mathrm{Spa}(\varphi)(\mathrm{Spa}(C))) \subseteq \Omega\} \tag{3.5}$$

is represented by an object $B = (B^{\triangleleft}, B^+) = (\mathscr{O}_X(\Omega), \mathscr{O}_X^+(\Omega))$. We denote the universal element of $F_\Omega(B)$ by $b \in \mathrm{Hom}(A, B) \subseteq \mathrm{Hom}(A^{\triangleleft}, \mathscr{O}_X(\Omega))$ and $y = \mathrm{Spa}\, B$ and $Y \xrightarrow{\beta} X$ where $\beta = \mathrm{Spa}(b)$.

b). If $\langle f_0, \cdots, f_m\rangle_A = A$ (which is automatic if A is analytic or Tate), then one can take

$$\mathscr{O}_X(\Omega) = B^{\triangleleft} = A\langle X_1, \cdots, X_m\rangle / I$$

when $I \subseteq A\langle X_1, \cdots, X_n\rangle$ is the closure of the ideal generated by $f_0 X_i - f_i$ for $1 \leq i \leq m$. For $\mathscr{O}_X^+(\Omega)$, one must then take the integral closure of the image of $A^+\langle X_1, \cdots, X_m\rangle$.

c). In general, one can take,

$$\mathscr{O}_X(\Omega) = (A\langle X_1, \cdots, X_m\rangle / J)_{f_0} \tag{3.6}$$

where $J \subseteq A\langle X_1, \cdots, X_n\rangle$ is the closure of $\bigcup_{l=0}^{\infty} I \cdot f_0^l$. For $\mathscr{O}_X^+(\Omega)$, one can take the integral closure of the image of $A^+\langle X_1, \cdots, X_n\rangle$ in Equation (3.6).

d). The image of f_0 in $\mathscr{O}_X(\Omega)$ is a unit and the image of the ? morphism $A^{\triangleleft}_{f_0} \to \mathscr{O}_X(\Omega)$ is a dense subring of $\mathscr{O}_X(\Omega)$.

e). The subring $\mathscr{O}_X^+(\Omega)$ is integral over the closure in $\mathscr{O}_X(\Omega)$ of the subring generated by $b(A)$ and the $\frac{b(f_i)}{b(f_0)}$.

f). The map $Y \xrightarrow{\beta} \Omega$ is a homeomorphism sends following $\mathfrak{Rat}_Y$ with $(\mathfrak{Rat}_Y)_\Omega := \{\Theta \in \mathfrak{Rat}_X \mid \Theta \subseteq \Omega\}$.

g). Let $\Theta \in \mathfrak{Rat}_Y$ and $C = (\mathscr{O}_Y(\Theta), \mathscr{O}_Y^+(\Theta)), \Theta_X = \beta(\Theta)$ and $c \in \mathcal{F}_\Theta(C)$ the universal element. The composition $A \xrightarrow{b} B \xrightarrow{c} C$ is an element of F_{Θ_X} and has the universal property for that functor. The morphism,

$$(\mathscr{O}_X(\Theta_X), \mathscr{O}_{X^-}^+(\Theta_X)) \to (\mathscr{O}_Y(\Theta), \mathscr{O}_Y^+(\Theta)) = C$$

is an isomorphism.

The proof of the theorem occupies the ret? of this situation Applying c) in the situation of b), $(f_0 \bmod I) \in (A\langle X_1, \cdots, X_n\rangle / I)^\times$. Since $f_0 X_i = f_i$¡

$$f_0 A\langle X_1, \cdots, X_m\rangle / I = \langle f_0, \cdots, f_m\rangle_{A\langle X_1, \cdots, X_m\rangle / I} = A\langle X_1, \cdots, X_m\rangle / I.$$

Then b) follows from c). Also d) follows from c) and the remainder results in Proposition 1 - 3. c) $\Rightarrow$ a) is clear. c) $\Rightarrow$ e) Follows from the density of $A[X_1, \cdots, X_m]$ in $A\langle X_1, \cdots, X_m\rangle$ (Proposition 3.55). c) $\Rightarrow$ f) follows from the density of $A^+[X_1, \cdots, X_m]$ in $A^+\langle X_1, \cdots, X_m\rangle$ again by Proposition 3.55.

We first show c) (which takes a while). When this is done g) and h) ($\Omega \simeq \operatorname{Spa} B$) remains. Let

$$B^{\triangleleft} = (A\langle X_1, \cdots, X_m\rangle / J)[f_0^{-1}],$$

and B^+ be the integral closure of the image of $A^+\langle X_1, \cdots, X_m\rangle$ in $B^{\triangleleft}$ as in Equation 3.6. It follows from the results ? Proposition 3.55 that $B^{\triangleleft}$ is complete Huber, and that the following morphisms are open,

$$A^{\triangleleft}\langle X_1, \cdots, X_m\rangle \to A^{\triangleleft}\langle X_1, \cdots, X_m\rangle / J \to (A^{\triangleleft}\langle X_1, \cdots, X_m\rangle / J)_{f_0} = B^{\triangleleft}. \tag{3.7}$$

Since $A^+\langle X_1, \cdots, X_m\rangle$ is open in $A^{\triangleleft}\langle X_1, \cdots, X_m\rangle$, its follows that B^+ is an open subring of B. Since $A^+ \subseteq A^{\triangleleft\circ}$, it can be easily shown that

$$A^+\langle X_1, \cdots, X_n\rangle \subseteq A^{\triangleleft}\langle X_1, \cdots, X_m\rangle^\circ.$$

By Proposition 3.28, the images of elements of $A^+\langle X_1, \cdots, X_m\rangle$ is ? ? in Equation 3.7 is an open embedding. It easily follows from the definition of "bounded that the image of $A^+\langle X_1, \cdots, X_m\rangle$ in $B^{\triangleleft}$ is contained in $B^{\triangleleft\circ}$. By Proposition 3.41, this also holds for the integral closures of ? image. It follows that $B = (B^{\triangleleft}, B^+)$ is a Huber pair and the composition of morphisms in Equation 3.7 with $A^{\triangleleft} \to A^{\triangleleft}\langle X_1, \cdots, X_m\rangle$ defines a morphis $A \xrightarrow{b} B$. We now claim that $b \in \mathcal{F}_\Omega(B)$. Indeed, the image q_i of $X_i \in A^{\triangleleft}\langle X_1, \cdots, X_m\rangle$ in B is in B^+ by the definition. Thus $|q_i|_Y \leq 1$ for all $y \in Y := \operatorname{Spa} B$. Since $b(f_i) = q_i b(f_0)$ (as $f_0 X_i - f_i \in I \subseteq J$), we have

$$|b(f_i)|_y \leq |b(f_0)|_y.$$

Set $\beta = \operatorname{Spa}(b)$ then for $x = \beta(y)$,

$$|f_i|_x = |b(f_i)|_y \leq |b(f_0)|_y = |f_0|_x.$$

Also

$$|f_0|_x = |b(f_0)|_y \neq 0$$

as $b(f_0) \in B^{\triangleleft\times}$. Then $x \in \Omega$ and $b \in \mathcal{F}_\Omega(B)$ as claimed.

We want to check that b has the universal property having turning B a representing object for $\mathcal{F}_\Omega$

$$\operatorname{Hom}_{\mathrm{op}}(B, C) \overset{\varphi\mapsto\varphi b}{\to} \mathcal{F}_\Omega(C)$$

is bijective. For this, let C be a complete Huber pair and $A \overset{c}{\to} C$ a morphism of Huber pairs such that $\gamma(\operatorname{Spa} C) \subseteq \Omega$? $\operatorname{Spa}(c)$ (i.e. $c \in \mathcal{F}_\Omega(C)$). Let $Z = \operatorname{Spa} C$. For $z \in Z, |c(f_0)|_Z = |f_0|_{\gamma(z)} \neq 0$ as $\gamma(z) \in \Omega$. Thus, $c(f_0) \in C^{\triangleleft\times}$ by Theorem 3.1. Then $q_i = \frac{c(f_i)}{c(f_0)}$ exists in $C^{\triangle}$. For $z \in Z$, $|q_i|_z \leq 1$ as

$$|c(f_i)|_z = |f_i|_{\gamma(z)} \leq |f_0|_{\gamma(z)} = |c(f_0)|_z$$

as $\gamma(z) \in \Omega$. By Theorem 3.1, $q_i \in C^+ \subseteq C^{\triangleleft\circ}$. By Proposition 3.55, there is a unique morphism $A^{\triangleleft}\langle X_1, \cdots, X_m\rangle \overset{c_1}{\to} C^{\triangleleft}$ of nat-rings where composition with $A^{\triangleleft} \to A^{\triangleleft}\langle X_1, \cdots, X_n\rangle$ conincides with c and such that $c_1(X_i) = q_i$. Since $c(f_i) = q_i c(f_0)$ in $C^{\triangleleft}$, c_1 annhilates $f_0 X_i - f_i$. By its continuity, $c_1(I) = 0$. As $c_1(f_0) = c(f_0) \in C^{\triangle\times}$, c_1 annhilates $I : f_0$? By continuity $c_1(J) = 0$. Thus c_1 factors over $A^{\triangleleft} \to A^{\triangleleft}\langle X_1, \cdots, X_m\rangle/J$ and the first morphism in Equation (3.7). By the universal property of $(A\langle X_1, \cdots, X_m\rangle/J)_{f_0}$ from Proposition 3.54, c_1 factors uniquely as the second morphism in Equation (3.7) followed by $B^{\triangleleft} = (A\langle X_1, \cdots, X_m\rangle/J)_{f_0} \to C$.

$$c_1(A^+) = c(A^+) \subseteq C^+$$

as c is a morphism of Huber pairs and $q_i \in C^+$ (see the above construction). As $A^+[X_1, \cdots, X_n]$ is dense in $A^+\langle X_1, \cdots, X_m\rangle$, and C^+ open, $c_1(A^+\langle X_1, \cdots, X_m\rangle) \subseteq C^+$. Thus c_3(image of $A^+\langle X_1, \cdots, X_m\rangle$ in $B^{\triangleleft}$) $\subseteq C^+$. Since B^+ is integral over that image and C^+ integrally closed in $C^{\triangleleft}$, $c_3(B^+) \subseteq C^+$ and we have a morphism $B \overset{c_3}{\to} C$ of Huber pairs.

By construction $c = c_3 b$. That c is uniquely determined by this property follows from,

$$c_3\left(\frac{b(a)}{b(f_0)^l}\right) = \frac{c(a)}{c(f_0^l)}$$

for $a \in A^{\triangleleft}$ and $l \in \mathbb{N}$, together with the density of the image of $A_f^{\triangleleft}$ in $B^{\triangleleft}$ observed before. We have these checked the derivd uderived universal property for $A \overset{b}{\to} B$.

It remains to prove the last two assertions of Theorem 3.2. Bijectivity of $Y \overset{\beta}{\to} \Omega$, plus identification of topologies of $\mathfrak{Rat}$ and $(\mathscr{O}(\cdots), \mathscr{O}^+(\cdots))$. That $\beta(Y) \subseteq \Omega$ follows from $b \in \mathcal{F}_\Omega(B)$ which was checkd before. For the bijectivity of β, we need the following proposition.

Proposition 3.56. *Let A be the (completion or) sequential completion of a topological ring $\mathcal{A}$. Then* $\operatorname{Cont}(A) \to \operatorname{Cont}(\mathcal{A})$ *is bijective.*

Proof. Let u and v be ontinuous aluations on A. Let u_0, v_0 their compositions with $\mathcal{A} \to A$. Then $\Gamma_u = \Gamma_{u_0}$. Indeed, the image of $\mathcal{A}$ is dense in A. If $\gamma = u(a) \neq 0$, $W = K_{v<r}$ is open in A. Hence there is $\alpha \in \mathcal{A}$ whose image in A is in $a + W$. It follows that $v_0(\alpha) = v(a) = \gamma \in \Gamma_{u_0}$. Since these γ generates Γ_u, $\Gamma_u \subseteq \Gamma_{u_0}$. The other inclusion is obvious. Thus $\Gamma_u = \Gamma_{u_0}$ and $\Gamma_v = \Gamma_{v_0}$. For injectivity, we assume that u_0, v_0 are equivalent. Then without loss of generality,

$$\Gamma_u = \Gamma_{u_0} = \Gamma_{v_0} = \Gamma_v$$

and hence $u_0 = v_0$ and have to show $u = v$. But this is clear by the continuity of u and v.Indeed, if $a \in A$ then by the density of $\mathcal{A}$ in A, there is a sequence $(\alpha_i)_{i=1}^{\infty}$ such that

$$\lim_{i\to\infty} a_i = a$$

where a_i is the image of $\alpha_i \in \mathcal{A}$ in A. Since u and v are continuous,

$$u(a) = \lim_{i\to\infty} u(a_i) = \lim_{i\to\infty} u_0(\alpha_i) = \lim_{i\to\infty} v_0(\alpha_i) = \lim_{i\to\infty} v(a_i) = v(a).$$

Limits in $\Gamma \cup \{0\}$ exist as it is Hausdorff.

Thus $\mathrm{Cont}(A) \to \mathrm{Cont}(\mathcal{A})$ is injective. For the surjectiity, let $\mathcal{A} \xrightarrow{u_0} \Gamma = \Gamma_{u_0}$ be a continuos valuation which we must extend to A. Elements of A are equivalence classes of Cauchy sequences $(\alpha_i)_{i=1}^{\infty}$. Let $\gamma_i = v_0(\alpha_i)$. If $\lim_{i\to\infty} \gamma_i = 0$, then we put $v(a) = 0$. If $(\widetilde{\alpha}_i)$ is equivalent to the given one and $\delta \in \Gamma$, then $K = K_{v_0 < \delta}$ is open in $\mathcal{A}$, hence there is i such that $\alpha_i - \widetilde{\alpha}_i \in K$. Then $v_0(\widetilde{\alpha}_i) < \delta$ provided that $v_0(\alpha_i) < \delta$ which happens for sufficiently large i. Thus this case is independent of the choise of representatives of teh equivalence class of Cauchy sequences.

if $\lim_{i\to\infty} \gamma_i = 0$ does not hold, there is $\delta \in \Gamma$ such that $\gamma_i \geq \delta$ for sufficiently large i. Since $K = K_{v_0<\delta}$ is open and (α_i) is a Cauchy sequence, there is i' such that $\alpha_j - \alpha_k \in K$ when $(j, k) \geq i'$. By what was said before, there is $i > i'$ such that $\gamma_i \geq \delta$. If $j \geq i$, then $\alpha_i - \alpha_j \in K$ hence $\gamma_j = u_0(\alpha_j) = u_0(\alpha_i) = \gamma - i$ as $\alpha_i - \alpha_j \in K$. Thus (γ_i) stabilizes to some fixed $\gamma \in \Gamma$ as $j \to \infty$, and we put $v(a) = \gamma$. If $\widetilde{\alpha} \sim \alpha$ then $\widetilde{\alpha}_j - \alpha_j \in K$ for $j > i''$ hence $u_0(\widetilde{\alpha}_j) = \gamma$ when $j > \max\{i, i''\}$. hence this case too is independent of choice of α. One easily checks the rest. □

We now prove the bijectivity of $Y \xrightarrow{\beta} \Omega$. First this makes sense as $\beta(Y) \subseteq \Omega$ which follows from $b \in \mathcal{F}_{\Omega}(B)$. For the surjectivity, let $r \in \Omega$ be the equivalence class of the valuation u of $A\langle X_1, \cdots, X_n\rangle \xrightarrow{v} \Gamma \cup \{\infty\}$ where we assume $\Gamma = \Gamma_u$. We have to extend u to $B^{\triangleleft}$ which is done in several steps. We first show the existence and the uniquness of a continuous map,

$$A\langle X_1, \cdots, X_n\rangle \xrightarrow{v} \Gamma \cup \{0\}$$

such that $v(I) = 0$ and the composition,

$$A^{\triangleleft} \to A^{\triangleleft}\langle X_1, \cdots, X_m\rangle \xrightarrow{v} \Gamma \cup \{0\}$$

equals u. The uniqueness still holds for $v : \langle X_1, \cdots, X_m\rangle \xrightarrow{v} \Gamma' \cup \{0\}$ are allowed where Γ' is an ordered group containing Γ as an ordered subgroup. Since $x \in \Omega, |f_0|_x = v(x) \neq 0$, hence x extends in a unique way to the valuation below.

$$A^{\triangleleft}_{f_0} \xrightarrow{u_1} \Gamma \cup \{0\}.$$

The uniuqness holds in the same general sense as before, (i.e. values $\Gamma' \supseteq \Gamma$ are also true).

We define,

$$A[X_1, \cdots, X_n] \xrightarrow{v_0} \Gamma \cup \{0\}$$

by

$$(P) = u_1\left(P\left(\frac{f_1}{f_0}, \cdots, \frac{f_m}{f_0}\right)\right)$$

where

$$P \mapsto P\left(\frac{f_1}{f_0}, \cdots, \frac{f_m}{f_0}\right)$$

is the $A^{\triangleleft}$-algebra morphism $A^{\triangleleft}[X_1, \cdots, X_m] \to A^{\triangleleft}_{f_0}$. For

$$P = \sum_{\alpha\in\mathbb{N}^m} p_\alpha X^\alpha \in A[X_1, \cdots, X_m]$$

we have,

$$
\begin{aligned}
v_0(P) &= u_1 1\left(\sum_{\alpha\in\mathbb{N}^m} p_\alpha \prod_{i=1}^{m}\left(\frac{f_i}{f_0}\right)^{\alpha_i}\right),\\
&\leq \max_{\alpha\in\mathbb{N}^m} u_1\left(p_\alpha \prod_{i=1}^{m}\left(\frac{f_i}{f_0}\right)^{\alpha_i}\right),\\
&\leq \max_{\alpha\in\mathbb{N}^m} u(p_\alpha) \prod_{i=1}^{m}\left(\frac{v(f_i)}{v(f_0)}\right)^{\alpha_i},\\
&\leq \max_{\alpha\in\mathbb{N}^m} v(p_\alpha) \qquad \text{(b)}
\end{aligned}
$$

It follows that v_0 is continuous for the topology on $A^{\triangleleft}[X_1,\cdots,X_n]$ considered in Lemma 3.3 (as $v_(K_{v<r})[X_1,\cdots,X_m] \subseteq [0,r)_{\Gamma\cup\{0\}}$ by (b) and $K_{v<r}[X_1,\cdots,X_m]$ is open in $A^{\triangleleft}[X_1,\cdots,X_m]$).

By Proposition 3.56, there is a unique continuous extension,

$$A^{\triangleleft}\langle X_1,\cdots,X_m\rangle \xrightarrow{r} \Gamma,$$

satisfying our conditions. We have

$$v(f_0X_i - f_i) = v_0(f_0X_i - f_i) = v_0 = v_1\left(f_0\frac{f_i}{f_0} - f_i\right) = 1.$$

Hence r is annhilates the ideal generated by the $f_0X_i - f_i$. By the continuity, it also annhilates its closure I. Conversely, let $A^{\triangleleft}\langle X_1,\cdots,X_m\rangle \xrightarrow{v'} \Gamma'$ with $\Gamma \subseteq \Gamma'$ (ordered preserving inclusion of a subgroup) such that

$$A^{\triangleleft} \to A^{\triangleleft}\langle X_1,\cdots,X_m\rangle \xrightarrow{v'} \Gamma'\cup\{0\}$$

equals to v. Let $P \in A[X_1,\cdots,X_m]\backslash\{0\}$ be of degree at most d,

$$Q(X_0,\cdots,X_m) = X_0^d P\left(\frac{X_1}{X_0},\cdots\frac{X_m}{X_0}\right)$$

be the homogeneousation of P, and let $q = Q(f_0,\cdots,f_m) \in A$. One easily checks that

$$f_0^d P - q \in \langle f_0X_1 - X_1,\cdots,f_0X_m - f_m\rangle_{A^{\triangleleft}[X_1,\cdots,X_m]} \qquad \text{(c)}$$

Since v annhilates I, $v'(f_0X_i - f_i) = 0$ and it follows that

$$u(f_0)^d v'(P) = v'(f_0^d P) = v'(q) = u(q).$$

hence,

$$v'(P) = \frac{u(q)}{v(f_0^d)} = u_1\left(\frac{q}{f_0^d}\right) = u_1\left(P\left(\frac{f_1}{f_0},\cdots,\frac{f_m}{f_0}\right)\right) = v(P).$$

By the uniquness part of Proposition 3.56, $v' = v$.

Since $v(f_0) = u(f_0) \neq 0$, J is also annhilates by v. This is then a unique continuous valuation

$$A^{\triangleleft}\langle X_1,\cdots,X_m\rangle/J \xrightarrow{v_1} \Gamma\cup\{0\}$$

when compsosed with $A^{\triangle}\langle X_1,\cdots,X_m\rangle \to A^{\triangleleft}\langle X_1,\cdots,X_m\rangle/J$ equals v. As $v_1(f_0) = v(f_0) \neq 0$, this is a unique continuous,

$$B^{\triangleleft} \xrightarrow{w} \Gamma\cup\{0\}$$

where composition with $A\langle X_1,\cdots,X_m\rangle/J \to B^{\triangleleft}$ the second morphism in (a) equals v_1. Since this morphism is an open inclusion and v_1 is continuous, one easily check the continuity of w.

We claim that the equivalence class of w is an element of $Y = \mathrm{Spa}(B)$. By (b),

$$v(f) = v_0(f) \leq 1,$$

when $f \in A^+[X_1, \cdots, X_m]$. Since this ring is dense in $A\langle X_1, \cdots, X_m\rangle$, by the continuity of v,

$$v(f) \leq 1$$

when f belongs to the latter ring.

Thus, the image in $B^{\triangleleft}$ of $A^+\langle X_1, \cdots, X_m\rangle$ is contained in $V_{w\leq 1}$. Since $K_{w\leq 1}$ is the integrally closed subring of $B^{\triangleleft}$ of the image of $A^+\langle X_1, \cdots, X_m\rangle$. We obtain,

$$w(B^+) \subseteq [0,1]_{\Gamma\cup\{0\}}$$

as claimed.

Thus, the equivalence class y of w is an element of Y such that

$$\beta(y). = x.$$

If y' has the same property, then as $A^{\triangleleft} \to B^{\triangleleft, |\cdot|_{y'}} \to \Gamma' = \Gamma_{y'}$ equivalent to v and $\Gamma_x = \Gamma_u = \Gamma$, we may assume without loss of generality that $\Gamma \subseteq \Gamma'$ (with order preserving inclusion of a subgroup) and that the composition,

$$A^{\triangleleft} \to B^{\triangleleft} \overset{|\cdot|_{y'}}{\to} \Gamma'$$

actually equals to v. By the uniqueness assertion about v, the composition $A^{\triangleleft}\langle X_1, \cdots, X_m\rangle \to B^{\triangleleft} \overset{|\cdot|_{y'}}{\to} \Gamma'$ equals to v. Since $A^{\triangleleft}\langle X_1, \cdots, X_m\rangle \to A^{\triangleleft}\langle X_1, \cdots, X_m\rangle/J$ is injective, the composition $A^{\triangleleft}\langle X_1, \cdots, X_m\rangle/J \to B^{\triangleleft} \overset{|\cdot|_{y'}}{\to} \Gamma'$ and $A^{\triangleleft}\langle X_1, \cdots, X_m\rangle/J \overset{v_1}{\to} \Gamma \subseteq \Gamma'$ also coincide. Finally $w = |\cdot|_{y'}$ as it is uniquely determined by v_1. Thus $y = y'$ and the bijectivity of β is shown.

For the comparison of topological and of rational open subsets,it suffices to consider,

$$\Theta = \mathcal{R}_Y(g_0|g_1, \cdots, g_j),$$

where $g_i \in B^{\triangleleft}$ and $\langle g_1, \cdots, g_j\rangle_{B^{\triangleleft}}$ is open in $B^{\triangleleft}$, and show that $\beta(w) \in \mathfrak{Rat}_X$. This will identify the topology base as claimed when $(\Theta_X = \mathfrak{Rat}_X(f_0|f_1, \cdots, f_m))$, one can take $g_i =$ (image of f_i) in $B^{\triangleleft}$ which generate open ideal in $B^{\triangleleft}$ as $A^{\triangleleft} \to B^{\triangleleft}$ is adic.

By Proposition 3.3.3, this is an open subgroup $U \subseteq B^{\triangleleft}$ such that

$$\mathcal{R}_Y(g_0|g_1, \cdots, g_j) = \mathcal{R}_Y(r_0|r_1, \cdots, r_j, \cdots, r_m)$$

and

$$\langle g_0, \cdots, g_j\rangle_{B^{\triangleleft}} = \langle r_0, \cdots, r_m\rangle_{B^{\triangleleft}}$$

when $g_i - r_i \in U$ for $0 \leq i \leq j$ and $r_i in U$ when $j < i \leq m$. As $A^{\triangleleft}_{f_0}$ has a dense image in $B^{\triangleleft}$, we can take $r_i, 0 \leq i \leq j$, for this subring,

$$r_i = \left(\text{image in } B^{\triangleleft} \text{ of } \frac{\varphi_i}{f_0^{a_i}} \in A_{f_0}\right)$$

Since f_0 is a unit in $B^{\triangleleft}$, we can multiply all r_i and g_i by f_0^a with out changin Θ or $\langle g_0, \cdots, g_j\rangle$ or $\langle r_0, \cdots, r_j\rangle$. THus it can be assumed that r_i ins the image of $\varphi_i \in A^{\triangleleft}$ in $B^{\triangleleft}$. Then β maps Θ onto $\Omega \cap \mathcal{R}_X(\varphi_0, \cdots, \varphi_j)$ but $\langle \varphi_0, \cdots, \varphi_j\rangle_{A^{\triangleleft}}$. Let $V \subseteq A^{\triangleleft}$ be the preimage under $A^{\triangleleft} \to B^{\triangleleft}$ of the open subgropu $f_0^a U$. Then we still have $\beta(\Theta) = \Omega \cap \mathcal{R}_X(\varphi_0, \cdots, \varphi_0)$ if $\varphi_i \in V$ for $j < i \leq m$. As A is Huber it has (oifin) and (tn:0) and there a finite subset $S \subset A^{\triangleleft\circ\circ}$ generating an open ideal. By Fact 3.1.5, S is tn. Assume $S^d \subseteq V$, when V is sufficinetly large. As $A^{\triangleleft}$ has (0;fq), S^d still generates an open ideal. Selecting $\varphi_0, \cdots, \varphi_m$ such that they form a ? of S^d, we may assume that

$$\langle \varphi_0, \cdots, \varphi_m\rangle$$

is open in $A^{\triangleleft}$. Then $\mathcal{R}_X(\varphi_0|\varphi_1, \cdots, \varphi_m) \in \mathfrak{Rat}_X$.

Finally,

$$\beta(\Theta) = \Omega \cap \mathcal{R}_X(\varphi_0|\varphi_1, \cdots, \varphi_m) = \mathcal{R}_X(f_0\varphi_0| \text{ all other } f_i\varphi_j)$$

The topological claim then follows. If $\Theta_X = \beta(\Theta)$, one checks that

$$\mathcal{F}_\Theta(C) \to \mathcal{F}_{\Theta_X}(C),$$

$$B \xrightarrow{v} C \Rightarrow A \xrightarrow{\beta} B \xrightarrow{r} C$$

is bijective usig the definition and the bijectivity of β. Thus the composition of structure presheaves also holds.

Corollary 3.7.

$$\operatorname{Spa}(A) \cong \operatorname{Spa}(\hat{A})$$

4 Acyclicity proofs

The Acyclicity proofs usually consider a special ? of Čech cohomology to show that $\mathscr{O}_X$ extends from $\mathfrak{Rat}_X$ to $\mathscr{O}_X$. Acyclicity will typically follow using ? the follows.

Proposition 4.1 (Grothendieck Tohoku Paper Lemma 3.3.1)**.** *Let $\mathscr{A} \xrightarrow{F} \mathscr{B}$ be a left exact functor between abelian categories where $\mathscr{A}, \mathscr{B}$ contain sufficineltly many injective objects. Let $\mathscr{X}$ be a class of objects of $\mathscr{A}$ such that*

a). Every direct summand of (in particular, every object isomorphic to) an element of $\mathscr{X}$ is in $\mathscr{X}$.

b). Every objet A of $\mathscr{A}$ has a monomorphism $A \to X$ with $X \in \mathscr{X}$.

c). If

$$0 \to X' \to X \to X'' \to 0,$$

is a short exact sequence in $\mathscr{A}$ such that $X', X \in \mathscr{X}$, then $X' \in \mathscr{X}$ and

$$0 \to FX' \to FX \to FX'' \to 0.$$

Corollary 4.1. *All injective objects of $\mathscr{A}$ belongs to $\mathscr{X}$ and furthermore $X \in \mathscr{X}$ and $p > 0$, we have $R^pFX = 0$.*

4.1 Acyclicity Criteria for General Sheaves on $\operatorname{Spa} A$

In this subsection, we denote $X = \operatorname{Spa} A$ where A is a Huber pair.

Proposition 4.2. *Let $x \in X$ be such that $c\Gamma_x = \Gamma_x$ and U be an open subset of X such that $x \in U$. Then there is $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in (A^\triangleleft)^{m+1}$ such that*

$$\langle f_0, \cdots, f_m\rangle_{A^\triangleleft} = A^\triangleleft$$

and

$$x \in \mathcal{R}_X(f_0|f_1, \cdots, f_m) \subseteq U.$$

In particular, Fact 3.2.11, this holds when x is closed.

Proof. Since rational open subsets form a topology base, there are $(\varphi_i)_{i=0}^m$ such that

$$x \in \mathcal{R}_X(\varphi_0|\varphi_1, \cdots, \varphi_m) \subseteq U.$$

If $a \in A$ is such that $|a|_x \neq 0$, then this implies

$$x \in \mathcal{R}(a\varphi_0|\varphi_1, \cdots, \varphi_m) \subseteq \mathcal{R}(\varphi_0|\varphi_1, \cdots, \varphi_m) \subseteq U.$$

As $|\varphi_0|_x \in \Gamma_x = c\Gamma_x$, such a can be chosen such that

$$|\varphi_0 a|_x \geq 1.$$

Then

$$x \in \mathcal{R}(a\varphi_0|a\varphi_1, \cdots, a\varphi_m, 1) \subseteq \mathcal{R}(a\varphi_0|a\varphi_1, \cdots, a\varphi_m) \subseteq U.$$

Note that

$$\langle a\varphi_0, \cdots, a\varphi_m, 1\rangle_{A^\triangleleft} = A^\triangleleft.$$

$\square$

Remark 4.1.

1. *It also holds in the Tate case and the analytic case for arbitrary x.*

2. *Thus $\mathcal{O}_X(\mathcal{R}(f_0|f_1, \cdots, f_m))$ can be bounded by the ? Then 3b) (where ? f_0 is not necessary).*

Definition 4.1.

a). A rational covering of X is a covering $X = \bigcup_{i=1}^m \mathcal{R}_X(f_i|f_1, \cdots, f_m)$ such that

$$\langle f_1, \cdots, f_m\rangle_{A^\triangleleft} = A^\triangleleft.$$

b). The rational sieve defined by such $(f_i)_{i=1}^m$ is

$$\{U \in \mathscr{O}_X \mid U \subseteq \mathcal{R}(f_i|f_1, \cdots, f_m) \text{ for some } i \in \{1, \cdots, m\}\}.$$

i.e. the sieve generated by this covering.

Proposition 4.3.

1. *Every open covering $\mathcal{U}$ of X has a refinement which is a rational covering.*

2. *Eery coering sieve (i.e. $X = \bigcup_{U\in\mathscr{S}} U$) of X contains a rational sieve.*

Proof. Let $\mathscr{S}$ be a covering sieve of X. For $x \in X$, we have $U_x \in \S$ such that $x \in U_x$ and $(f_{x,i})_{i=1,\cdots,m_x}$ with

$$\langle f_{x,1}, \cdots, f_{x,m_x}\rangle_{A^\triangleleft} = A^\triangleleft$$

such that $x \in \mathcal{R}(f_{x,0}|f_{x,1}, \cdots, f_{x,m_x}) \subseteq U_x$. Then

$$X = \bigcup_{x\in X_{\mathrm{cl}}} \mathcal{R}_X(f_{x,0}|f_{x,1}, \cdots, f_{x,m_x}).$$

Indeed, the right hand side V contains all closed points. If $V \subsetneqq X$, then $X\backslash V$ would be a non-empty closed subset without closed points, a contradiction by Fact 1.2.7.

As X is spectral, it is quasi-compact, hence there are finitely many closed points $(x_i)_{i=1}^n$ such that

$$X = \bigcup_{j=1}^n \mathcal{R}(f_{x_j,0}|f_{x_j,1}, \cdots, f_{x_j,m_j}).$$

Let $f_{j_i} = f_{x_j,i}, n_j = m_{x_j}, U_j = \mathcal{R}(f_{j,0}|f_{j,1}, \cdots, f_{j,m_j})$. Let J be the set of fnctions ι defined on $\{1, \cdots, n\}$ such that

$$0 \le \iota(j) \le m_j,$$

and let $J_0 \subseteq J$ be the set of all $\iota \in J$ such that there is $j \in \{1, \cdots, n\}$ such that $\iota(j) = 0$. For all $\iota \in J$, let

$$f_\iota = \prod_{j=1}^n f_{j,\iota(1)}$$

then

$$\langle f_\iota \mid \iota \in J\rangle_{A^\triangleleft} = \prod_{j=1}^n \langle f_{j,0}, \cdots, f_{j,m}\rangle_{A^\triangleleft} = A^\triangleleft$$

by the assumption. Note that

a). $X = \bigcup_{i=1} 6n\mathcal{R}(f_{j,0}|f_{j,1}, \cdots, f_{j,m_j}) = \bigcup_{j=1}^m U_j \in \mathscr{S}$.

b). $\langle f_{j,0}, \cdots, f_{j,m_j}\rangle_{A^\triangleleft} = A^\triangleleft$.

c). $\max_{\iota\in J} |f_\iota|_x = \max_{\iota\in J_0} |f_\iota(x)|$ when $x \in X$. Note that when $x \in U_k$ and $\iota(j) = \begin{cases} \iota(j), j \ne k, \\ 0, j = k. \end{cases}$ then $\iota' \in J_0$ and $|f_\iota|_x \le |f_{\iota'}|_x$.

d). $\langle f_\iota \mid \iota \in J_0\rangle_{A^\triangleleft} = A^\triangleleft$. (Follows from b), c) and Theorem 3.1).

Then the $(f_\iota)_{\iota\in J_0}$ defines a rational sieve/covering. We claim that this rational sieve is contained in $\mathscr{S}$. This will finish the proof.

To prove this claim, it suffices to consider,

$$\Omega_\lambda = \mathcal{R}(f_\lambda | \text{ all } f_\iota \text{ with } \iota \in J_0).$$

with $\lambda \in J_0$ and to show that $\Omega_\lambda \in \mathscr{S}$, it suffices to find $j \in \{1, \cdots, n\}$ such that

$$\Omega_\lambda \subseteq U_j.$$

We claim that this holds when $\lambda(j) = 0$ (by Definition such j exists for $\lambda \in J_0$). Indeed, let $x \in \Omega_\lambda$ then

$$|f_\lambda|_x = \prod_{k=1}^{n} |f_{k,\lambda(k)}| \neq 0.$$

Hence

$$0 \neq |f_{j,\lambda(j)}|_x = |f_{j,0}|_x.$$

If $x \in U_j$, there would be $i \in \{1, \cdots, m_j\}$ such that

$$|f_{j,i}|_x > |f_{j,0}|.$$

Let

$$\lambda'(k) = \begin{cases} \lambda(k), & k \neq j, \\ i, & k = j. \end{cases}$$

Then

$$|f_{\lambda'}(x)| = |f_{j,i}(x)| \prod_{k\neq j} |f_{k,\lambda(k)}|_x > |f_{j,0}|_x \prod_{k\neq j} |f_{k,\lambda(k)}|_x = |f_\lambda(x)|$$

For the inequality in the middle use that this is not 0 as they are factors of $|f_\lambda|_x \neq 0$. By c), there is $\theta \in J_0$ such that $|f_\theta(x)| \geq |f_{\lambda'}|_x > |f_\lambda(x)|$, contradicting $x \in \mathcal{R}(f_\lambda | \text{ all } f_\iota \text{ for } \iota \in J_0) \subseteq \mathcal{R}(f_\lambda | f_\theta)$. $\square$

Proposition 4.4. *Let A be a Huber pair and $X = \operatorname{Spa} A$ and $\mathcal{G}$ be a presheaf of abelian groups on $\mathfrak{Rat}_X$ such that for every $\Omega \in \mathfrak{Rat}_X$ and every rational covering $\mathcal{U}$ of Ω, the map,*

$$\mathcal{G}(\Omega) \to \hat{H}^0(\mathcal{U}, \mathcal{G}) \tag{4.1}$$

is bijective. Then

$$\mathcal{G} \to (\underline{\operatorname{Sheaf}}(\mathcal{G}))_{\mathfrak{Rat}_X} \tag{4.2}$$

is an isomorphism in the category of presheaves on Ω.

Proposition 4.5. *Let $X = \operatorname{Spa} A$ where A is a Huber pair. The class $\mathscr{X}$ of sheaves $\mathcal{F}$ of abelian groups on X such that*

$$\hat{H}^p(\mathcal{U}, \mathcal{F}) = 0 \tag{4.3}$$

where $p > 0$ and $\mathcal{U}$ is a rational covering of $\Omega \in \mathfrak{Rat}_X$, together with the functor

$$\mathcal{F} \mapsto \mathcal{F}(\Omega) \tag{4.4}$$

where $\Omega \in \mathfrak{Rat}_X$ satisfy the assumption of Proposition 2.4.3. Then all injective sheaves are in $\mathscr{X}$ and $H^p(\Omega, \mathcal{F}) = 0$ when $\Omega \in \mathfrak{Rat}_X, \mathcal{F} \in \mathscr{X}$ and $p > 0$.

We have somewhat more convenient acyclicity criterion in the Tate case.

Definition 4.2. *Let $X = \operatorname{Spa} A$ where $(A^\triangleleft, A^+)$ is a Huber pair. We inductively define the property of a sieve $\mathscr{S}$ in $\Omega \in \mathfrak{Rat}_X$ to have a Laurent series $\leq o$ as follows.*

1. *A sieve has Laurent order 0 if it is the all sieve.*
2. *If $o > 0$ and Laurent order $\leq o - 1$ has already been defined then $\mathscr{S}$ has Laurent order $\leq o$ if there is $g \in \mathscr{O}_X(\Omega)$ such that $\mathscr{S}|_{\mathcal{R}_\Omega(g|1)}$ and $\mathscr{S}|_{\mathcal{R}_\Omega(1|g)}$ both have Laurent order $\leq o - 1$.*

The Laurent order of $\mathscr{S}$ is the smallest $o \in N$ such that $\mathscr{S}$ has Laurent order $\leq .o$ or ∞ if no such o exists. We denote $o_L(\mathscr{S})$ to be the Laurent order of $\mathscr{S}$.

Proposition 4.6.

a). $\mathscr{S} \subseteq \mathcal{T} \Rightarrow o_L(\mathcal{T}) \leq o_L(\mathscr{S})$.

b). $\Theta \subseteq \Omega \Rightarrow o_{L,\Theta}(\mathscr{S}|_\Theta) \leq o_{L,\Omega}(\mathscr{S})$

c). $o_L(\mathscr{S}) < \infty \Rightarrow \mathscr{S}$ is a covering sieve.

The following lemma does not yet need the Tate assumption.

Lemma 4.1. *Let $G = \{g_i \mid 1 \leq i \leq n\}$ be a non-empty finite subset of $A^{\triangleleft\times}$. Then the rational sieve defined by G has finite Laurent order.*

Proof. We write $g_i \preceq g_j$ if $\Omega = \mathcal{R}(g_j|g_i)$. Obviously, this is a partial order with allowed equivalence. We call (i,j) in order if at least one of $g_i \preceq g_j$ or $g_j \preceq g_i$ holds. We prove the claim by induction on the number of pairs not in order. If all pairs are in order then without loss of generality by renumbering G, we can assume $g_1 \preceq \cdots \preceq g_m$. Thus, all $g_i \leq g_m$. and the rational sieve contains $\Omega = \mathcal{R}_\Omega(g_m|g_0, \cdots, g_{m-1})$ hence is the all sieve thus has Laurent order 0.

Let (i,j) be not in order and the assertion shown for a smaller number of pairs not in order. Let $q = \frac{g_i}{g_j}$ and $\Theta_1 = \mathcal{R}_\Omega(q|1)$ and $\Theta_2 = \mathcal{R}_\Omega(1|q)$. Then the pairs in order stay in order after restricting to Θ_i and $g_j|_{\Theta_1} \preceq g_i|_{\Theta_1}$ and $g_i|_{\Theta_2} \preceq g_j|_{\Theta 2}$. Thus (i,j) becomes in order for $G_k = \{g_i|_{\Theta_k} \mid 1 \leq i \leq n\}$ where $k = 1, 2$.By the induction assumption and Definition 4.2, the rational sieve defined by G has finite Laurent order. □

Lemma 4.2. *Let A be a Huber pair such that $A^\triangleleft$ is Tate and there is $(f_i)_{i=0}^m \in A^{m+1}$ such that $\langle f_0, \cdots, f_m\rangle_{A^\triangleleft} = A^\triangleleft$. Then there is $\varepsilon \in A^{\triangleleft\times} \cap A^{\triangle\circ\circ}$ such that*

$$|\varepsilon|_x \leq \max_{0\leq i\leq n} |f_i|_x. \tag{4.5}$$

Proof. Let $s \in A^{\triangleleft\circ\circ} \cap A^{\triangleleft\times}$. By assumptino, there are $(\lambda_i)_{i=0,\cdots,m} \in (A^\triangleleft)^{m+1}$ such that

$$\sum_{i=0}^m \lambda_i f_i = 1.$$

Since A^+ is an open subring and $\lim_{j\to\infty} s^j\lambda_i = 0$, there is $1 \leq j \leq m-1$ such that all $\lambda_i s^j \in A^+$. Then one can define $\varepsilon = s^j$.

$$|\varepsilon|_x = \left|\sum_{i=0}^m (s^j\lambda_i)f_i\right|_x \leq \max_{0\leq i\leq m} |\lambda_i s^j|_x |f_i|_x \leq \max_{0\leq i\leq m} |f_i|_x.$$

since $|\lambda_i s^j| \leq 1$ since $x \in \operatorname{Spa} A$ and $\lambda_i s^j \in A^+$. □

Proposition 4.7. *In the situation of Definition 4.2, assume A (i.e $A^\triangleleft$) to be Tate. Then a sieve $\mathscr{S}$ in $\Omega \in \mathfrak{Rat}_X$ covers Ω if and only if $\omega_L(\mathscr{S}) < \infty$.*

Proof. The proof will work by showing that every rational sieve has finite o_L which proves the claim by Proposition 4.3 and 4.6 a).

As was said before it is sufficent to show that every rational sieve has finite Laurent order. Let $\mathscr{S}$ be the rational sieve defined by the finite subset of G of $\mathscr{O}_X(\Omega)$. We use induction on the number of elements of G which are not in $\mathcal{O}_X^\times(\Omega)$. If this number is 0, Lemma 4.1 can be applied. Let $g_1 \notin \mathcal{O}_X^\times(\Omega)$ and the assertion shown for a smaller number of non-elements. Since A is Tate, there is $\delta \in \mathcal{O}_X^\times(\Omega)$ such that $|\delta|_\omega < \max_{1\leq i\leq m} |g_i|_\omega$ holds for all $\omega \in \Omega$. Consider

$$\Theta_1 = \mathcal{R}_\Omega(g_1|\delta) = \mathcal{R}\left(\frac{g_1}{\delta}\middle|1\right), \Theta_2 = \mathcal{R}_\Omega(\delta|g_1) = \mathcal{R}\left(1\middle|\frac{g_1}{\delta}\right).$$

On Θ_1, g_1 becomes a unit as $|g_1|_\theta \geq |\delta|_\theta \neq 0$ when $\theta \in \Theta_1$ hence the induction assumption applies to $G|_{\Theta_1}$. But the induction assumption also applies to $G|_\Theta$ since

$$\mathcal{R}_{\Theta_2}(g_1|g_2, \cdots, g_m) = \emptyset$$

as for $\vartheta \in \Theta_2$, $|g_1|_\vartheta \leq |\delta|_\vartheta < \max_{1\leq i \leq m} |g_i|_\vartheta$. And,

$$\mathcal{R}_{\Theta_2}(g_i|g_1, g_2, \cdots, g_m) = \mathcal{R}(g_i|g_2, \cdots, g_m)$$

as for $\vartheta \in \Theta_2$,

$$|g_1|_\vartheta \leq |\delta|_\vartheta < \max_{2\leq i\leq m} |g_i|_\vartheta. \tag{4.6}$$

Hence $g_1|_{\Theta_2}$ can simply be removed from G_{Θ_2} without affection the rational sieve by (4.6) it follows that $g_2|_{\Theta_2}, \cdots, g_m|_{\Theta_2}$ have no common zero in Θ_2, hence define a rational sieve. ☐

Proposition 4.8. *Let $X = \operatorname{Spa} A$ where A is Tate, $\mathcal{G}$ be a presheaf on $\mathfrak{Rat}_X$ (of abelian groups) such that for all $\Omega \in \mathfrak{Rat}_X$ and all $g \in \mathcal{O}_X(\Omega)$, the sequence,*

$$0 \longrightarrow \mathcal{G}(\Omega) \xrightarrow{r\mapsto(r|_{\Omega_1}, r|_{\Omega_2})} \mathcal{G}(\Omega_1) \oplus \mathcal{G}(\Omega_2) \xrightarrow{(r_1,r_2)\mapsto(r_1|_{\Omega_{12}} - r_2|_{\Omega_{12}})} \mathcal{G}(\Omega_{12}) \longrightarrow 0 \tag{4.7}$$

is exact where $\Omega_1 = \mathcal{R}_\Omega(g|1), \Omega_2 = \mathcal{R}_\Omega(1|g)$ and $\Omega_{12} = \Omega_1 \cap \Omega_2$. Then $\mathcal{G} \to (\underline{\text{Sheaf}}(\mathcal{G}))|_{\mathfrak{Rat}_X}$ is an isomorphism in the category of pressheaves of abelian groups on X.

Proposition 4.9. *Let $X = \operatorname{Spa} A$ where the Huber pair A is Tate. Then the assumption in Proposition 2.4.3 hold for $\mathscr{A} =$ (sheaves of abeliangroups on X) and of the functor $\mathcal{F} \to \mathcal{F}(\Omega)$ and $\mathscr{X}$ the class of sheaves $\mathcal{F}$ on X such that*

$$0 \longrightarrow \mathcal{F}(\Omega) \xrightarrow{f\mapsto(f|_{\Omega_1}, f|_{\Omega_2})} \mathcal{F}(\Omega_1) \oplus \mathcal{F}(\Omega_2) \xrightarrow{(f_1,f_2)\mapsto(f_1|_{\Omega_{12}} - f_2|_{\Omega_{12}})} \mathcal{F}(\Omega_{12}) \longrightarrow 0 \tag{4.8}$$

is exact for all $\Omega \in \mathfrak{Rat}_X$ and all $g \in \mathcal{O}_X(\Omega)$ where Ω_1, Ω_2 and Ω_{12} are as in Proposition 4.8.

References

[1] Huber, R *Continuous valuations*, Mathematische Zeitschrift